$\hbar$, the Bekenstein-Hawking Entropy, and the Running-Vacuum Form of Dark Energy from the Cosmological Horizon

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Abstract

Applied to the cosmological horizon as a causal partition, the embedded-observation framework of [Main] determines the emergent action scale $\hbar = c^3\epsilon^2/(4G)$ from classical horizon temperature and thermal self-consistency, fixes the discreteness scale $\epsilon = 2\,l_p$ as the unique simultaneous solution, and obtains the Bekenstein-Hawking entropy $S = A/(4l_p^2)$ including the 1/4 coefficient by direct boundary-mode counting — the three conditional on the infrared detailed-balance condition H-slope (§3.2) together with the horizon and frame conditions of §2 and §8.5, the counting $S = A/\epsilon^2$ itself unconditional — and addresses the cosmological constant problem at Level G1 — this paper's register: horizon-normalization identities and conditional thermodynamic relations — by identifying the quantum vacuum energy and the effective cosmological constant as properties of logically distinct descriptions rather than commensurable quantities whose 10^{122} ratio requires cancellation; the decisive Level-G3 target, open, is a covariant equation — the dependency ladder in full: G1, horizon and mode-count identities (this paper's register); G2, weak-field and Newtonian consistency; G3, covariant field equations with conservation/Bianchi structure and the continuum limit, open; G4, the cosmological implications (cosmological constant, dark sector, Page curve, MOND and the Bullet Cluster), conditional on G3 and their listed assumptions — namely $E_{\mu\nu}[g, \mathrm{OI}] = 8\pi G\,(T_{\mu\nu} - \mathcal{F}[g,T]\,g_{\mu\nu})$ under which the distinct-descriptions argument becomes a theorem of the dynamics — field equations, stress-energy coupling, Bianchi structure, and the continuum limit being the named G3 opens. The recent LIGO/Virgo/KAGRA observation GW250114 (September 2025) confirms the black-hole area theorem at 99.999% confidence — a test of classical horizon dynamics, which the framework preserves; the 1/4 entropy coefficient itself has no direct empirical test. The framework carries dark energy in running-vacuum form, the functional form being the local expansion shared by every smooth model (§7.1) and the falsifiable content the Type II channel — a structural reading, not a derived law; matter conserved, compensation through running G — the exclusion of phantom crossing, and the magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$ (conditional on the §7.3 split and not established at theorem level, indistinguishable from zero) — consistent with the Bertini et

al. (2025) RG-corrected ΛCDM fit to DESI DR2 + DES-Y5 + Planck 2018, which reports $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ (consistent with zero at 2σ); a ~95% dark sector corollary in which the apparent invisible matter is a structural feature of the embedded observer's description rather than a new particle species; and a MOND-like acceleration scale $a_0 = cH/6$ from entropy displacement at the cosmological boundary (the dimensional factor imported from Verlinde), reproducing the baryonic Tully-Fisher relation and predicting Keplerian decline of galactic rotation curves at radii exceeding $r_M = \sqrt{6GM_B/(cH)}$ — the last confirmed by Jiao et al. (2023) in the Milky Way at ~ 17 kpc. The $H(z)$ dependence of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with Genzel et al.'s detection of declining outer velocities at $z = 0.9$–2.4. The results follow from partition geometry under the conditions named at each step and are independent of the specific substratum dynamics, treated in a separate companion paper [SM].

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1. Introduction

The cosmological observables that most resist explanation within standard physics — the value of $\hbar$, the Bekenstein-Hawking area law, the 10^{122} cosmological constant problem, the accelerated expansion, the ~95% dark sector, and the MOND-like behavior of galactic rotation curves — are treated here as consequences of a single structural fact: the cosmological horizon partitions the degrees of freedom of any sub-c observer into a visible sector and a causally inaccessible hidden sector.

Prerequisites. This paper takes as established the following results from a companion work [Main]: (i) an observer embedded in a deterministic bijection on a finite configuration space, coupled to a hidden sector that is non-trivial (C1), record-persistent (C2 — realized in our universe by the slow horizon, $\tau_S \ll \tau_B$), of sufficient memory capacity (C3 — with $N_H \gg N_V$ the comfortable regime), and read back (C4), has a reduced visible-sector description that is — in the memory-bearing sector those conditions characterize — non-Markovian, and that universally admits a unitarily evolving fixed-basis quantum representation — the representation universal over finite laws ($S \iff D \iff Q_{\mathrm{fb}}$), the necessity holding in the memory currency, with C1–C4 necessary as well as sufficient for accessible non-Markovianity (conditional on the eigenstate thermalization hypothesis for the C2 $\Leftarrow$ direction), and full operational quantum theory approached at every finite test with its unique selection open; (ii) the emergent description is uniquely determined by the partition, with any parameters of the emergent theory depending only on the partition boundary's geometric and thermodynamic properties (partition-relativity); (iii) the emergent transition probabilities $T_{ij}(t)$ determine the emergent Hamiltonian up to a diagonal-unitary D-gauge, an energy origin, and one global antiunitary conjugation — the two-branch form proved in §3.3 — via the phase-locking lemma applied to continuous-time sampling. The technical bridge from P-indivisible classical stochastic processes on finite configuration spaces to unitary quantum evolution on finite Hilbert spaces is Barandes' stochastic-quantum correspondence [35, 36], used by the companion paper for compression and interpretation — the fixed-basis representation itself is supplied internally there ($S \iff D \iff Q_{\mathrm{fb}}$). We treat these results as inputs and apply them to the cosmological horizon. The cosmological horizon independently verifies C1–C3: causal coupling via the Einstein constraint equations (C1), a slow-bath timescale of order the Hubble time $\tau_B \sim 10^{17}$ s (C2), and a hidden-sector capacity of order $S_{\mathrm{dS}} \sim 10^{122}$ modes (C3). C4, history readback, is a named realization condition at the cosmological cut but is not presently discharged (§2.2). Accordingly, application of the C1–C4 memory-bearing equivalence to our universe remains conditional on C4; the ETH condition for C2 necessity holds for the horizon complement (a generic chaotic many-body system).

The present paper reproves none of [Main]'s foundational results. Where a theorem below invokes a specific input — partition-relativity (§3.1), the emergent-description determination by transition probabilities (§3.2–§3.3), or the P-indivisibility/recurrence argument (§3.2 deep-sector lemma; §8 theorem requirements) — the invocation is stated as a known result and the reader is directed to [Main] for the proof.

The results are as follows. The emergent action scale $\hbar = c^3\epsilon^2/(4G)$ is determined by thermal self-consistency between the classical horizon temperature (computed from the Jacobson thermodynamic argument with no quantum input) and the Euclidean KMS period of the emergent QFT on the same horizon (§3). The discreteness scale $\epsilon = 2\,l_p$ is then fixed as the unique simultaneous solution of the action-scale relation and the definition of the Planck length (§4). The Bekenstein-Hawking entropy follows by mode counting, with the 1/4 coefficient fixed as the ratio $2\pi/8\pi$ between the Euclidean KMS period and the Jacobson prefactor (§5); the action scale, the discreteness scale and the coefficient share the conditions of the calibration — H-slope (§3.2) with the horizon and frame conditions of §2 and §8.5 — while the counting $S = A/\epsilon^2$ is unconditional. The recent LIGO/Virgo/KAGRA observation GW250114 confirms the area theorem at 99.999% confidence (the 1/4 coefficient itself has no direct test). The cosmological constant problem is dissolved by identifying the emergent quantum vacuum energy of order M_{Pl}^4 and the effective cosmological constant of order H^2/G as properties of logically distinct objects rather than commensurable quantities whose 10^{122} ratio requires cancellation (§6); the descriptive distinction is established and the dynamical form of the dissolution is conditional on the G3 coupling theorem (§6.2).

The framework's empirical content is developed in §7. Dark energy appears in running-vacuum form, the functional form being the local expansion shared by every smooth model (§7.1) and the content the Type II channel with matter conserved and compensation through running G — a structural reading of the boundary bookkeeping, not a derived law (§7.1); the magnitude of the H^2 coefficient is bounded at the emergent-QFT scale $|\nu| \sim (M_{\mathrm{SM}}/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ —

indistinguishable from zero — on either reading of the emergent layer's gravitational status (§7.1). This is consistent with the Bertini et al. (2025) direct RG-corrected ΛCDM fit to DESI DR2 + DES-Y5 + Planck 2018, which reports ν consistent with zero at 2σ. The dark-sector corollary predicts that ~95% of the gravitational budget of the universe is invisible to the emergent QFT without requiring new fundamental particles — a concordance rather than a puzzle. The acceleration scale $a_0 = cH/6$ emerges from entropy displacement at the cosmological boundary, with the dimensional factor 1/6 taken from Verlinde's emergent-gravity analysis, reproducing the MOND phenomenology in the deep-modified regime and predicting Keplerian decline at radii exceeding $r_M = \sqrt{6GM_B/(cH)}$; the latter is confirmed by the Jiao et al. (2023) detection of a Keplerian decline in the Milky Way rotation curve at ~ 17 kpc. The $H(z)$ dependence of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with the declining outer velocities observed at $z = 0.9$–2.4 by Genzel et al.

Three epistemic tiers are worth distinguishing before the technical development begins. **Tier 1** comprises what follows from partition geometry alone, for any finite deterministic system meeting the four conditions at its cosmological horizon: the energy-shell horizon temperature, the finite horizon counting, and the structural two-level distinction between the emergent vacuum energy and the effective cosmological constant. The results built on these carry named conditions and are stated with them throughout: $\hbar$, $\epsilon = 2\,l_p$ and the 1/4 coefficient are conditional on H-slope together with the horizon and frame conditions (§§2, 5); the descriptive-level distinction behind the cosmological-constant dissolution is established while the dynamical dissolution is conditional on the G3 coupling theorem (§6.2); and the Type II running-vacuum functional form is not presently derived, its local expansion being shared by every smooth model (§7.1). **Tier 2** comprises the dark-sector phenomenology: the ~95% invisible matter fraction, the MOND acceleration scale, and the rotation-curve crossover. These follow from the thermodynamic Clausius relation applied to the boundary reservoir and the geometric factor from spherical redistribution in de Sitter space; they inherit the conditions on the de Sitter temperature of §3.2, and the dimensional factor that sets the acceleration scale is imported from Verlinde's emergent-gravity analysis (§7.3). **Tier 3** comprises solution-specific quantities — the transition steepness between the MOND and Keplerian regimes, the greybody factor for Hawking radiation, and the specific microscopic realization of the substratum — which depend on the particular bijection φ and are addressed in the companion paper [SM].

The tier system maps to the framework's broader hierarchical structure articulated in [Structure §2]. *Tier 1* — the energy-shell horizon temperature, the finite horizon counting, and the two-level distinction — operates at Level C × Level G4: these results hold for any partial-trace observational system on a horizon-like causal partition, not specifically for OI's cubic-lattice substratum. They are universality-class-level structural results, and the conditional results built on them ($\hbar$, $\epsilon = 2\,l_p$, the 1/4 coefficient) are universality-class-level in the same sense once their named conditions are granted, since those conditions concern the horizon and the emergent dynamics rather than the lattice (§8.5). *Tier 2* — the dark-sector phenomenology — sits at Level D × Levels G1–G2, deriving from the thermodynamic Clausius relation applied to the specific boundary reservoir of OI's universality-class representative. *Tier 3* — solution-specific quantities — is at the bijection-specific level below Level D, depending on the particular φ within $[(S,\varphi)]/\mathcal{G}_{\rm sub}$. This stratification clarifies what a conditional universality-class result claims: not "OI predicts $\hbar$ uniquely" but "any horizon-bounded embedded-observer system meeting the named conditions gets this $\hbar$, and OI provides one specific realization."

The paper is organized as follows. §2 identifies the cosmological horizon as a causal partition and verifies the structural conditions. §3 derives the emergent action scale $\hbar$ in four steps: the classical horizon temperature, boundary-only dependence, dimensional determination, and thermal self-consistency. §4 fixes the discreteness scale $\epsilon = 2\,l_p$. §5 derives the Bekenstein-Hawking entropy and discusses the GW250114 confirmation. §6 dissolves the cosmological constant problem. §7 presents the quantitative predictions for dark energy, the dark sector, and dark matter, with a summary table in §7.4. §8 discusses scope, the status of the discreteness scale, vacuum energy and the Higgs potential, and the framework's falsifiability conditions. §9 concludes. The cubic-lattice realization of the substratum — which fixes the Standard Model gauge structure via the wave equation — is developed in a companion paper [SM] and is independent of the GR results presented here. The reconstruction theorem and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S,φ) are developed in [Substratum].

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2. The Cosmological Horizon as Causal Partition

2.1 The partition

The cosmological horizon — the boundary beyond which no signal propagating at or below c can reach the observer — implements the causal partition of Lemma 2: Γ_V is the interior, Γ_H everything beyond. This is a consequence of GR's causal structure, not a modeling choice. The universality of the observed quantum mechanics follows from all sub-c observers sharing the same causal structure.

Different observers have different horizon areas (hence different $S_{\rm dS}$), but $\hbar = c^3\epsilon^2/(4G)$ depends only on local geometric quantities — not on the observer's worldline or horizon area — so all observers share the same emergent action scale.

2.2 Verification of the conditions

(C1) In the ADM formulation, the bulk Hamiltonian is a sum of constraints that vanish on-shell; physical time evolution is generated by the boundary term at the horizon, which depends on data from both sides. The Hamiltonian and momentum constraints further correlate interior and exterior initial data on any Cauchy surface, so the physical phase space is a constraint surface within the kinematic product $\Gamma_V \times \Gamma_H$ (see §8.4 for implications). This is *stronger* than the $H_{\rm int} \neq 0$ required by Lemma 2: constraints enforce correlations that persist on all timescales and cannot be perturbatively removed. **Satisfied.**

(C2) $\tau_B \gtrsim 1/H \sim 10^{17}$ s; for laboratory processes, $\tau_S \sim 10^{-15}$ s. Ratio: $\tau_S/\tau_B \sim 10^{-32}$. **Satisfied.**

(C3) Hidden sector has A/ϵ^2 modes with $A \sim 10^{52}$ m^2; visible sector has $\sim 10^{80}$ baryons. No laboratory process saturates the hidden sector. **Satisfied.**

(C4) The horizon construction establishes bidirectional coupling (the constraint structure above), persistence (C2) and ample capacity (C3), but the present paper does not independently demonstrate that a visible-history record written into hidden boundary degrees is routed back into future visible conditionals within the accessible window — the read-write cycle that [Main §1.3] names as the property to be demonstrated in the cosmological realization. Bidirectional coupling is a strengthened form of (C1) and persistence is (C2); neither, nor their conjunction, supplies the routing that (C4) asserts, and since [Main §3.4] derives (C1) and (C3) from (C4), verifying those two does not supply it either. (C4) is therefore a named realization condition at the cosmological cut. **Not presently discharged.** Nothing in §3 consumes it: the $\hbar$ calibration is carried by H-slope together with the horizon and frame conditions.

2.3 Application

Which horizon, and why the calibration is stated on exact de Sitter. Two distinct surfaces are in play and they coincide only in exact de Sitter. The causal definition above is the cosmological event horizon, $R_{\rm event} = a(t)\int_t^\infty c\,dt'/a(t')$; the area $A = 4\pi c^2/H^2$ used below is that of the Hubble/apparent horizon $R_A = c/H$. Generic FLRW has $\dot H \neq 0$, no timelike Killing vector and hence no bifurcate Killing horizon, so exact Euclidean KMS periodicity — a stationary result — is not available there. For the flat-FLRW apparent horizon the Hayward–Kodama surface gravity gives $T_A = \frac{\hbar H}{2\pi k_B}\left|1 + \dot H/(2H^2)\right|$, and for flat ΛCDM with $\dot H/H^2 = -\frac{3}{2}\Omega_m$ this is $\approx 0.775\,\hbar H/(2\pi k_B)$ today — a 22% effect, not Planck suppressed, and larger than the lattice-dispersion correction $O((\epsilon H/c)^2) \sim 10^{-122}$ by 121 orders of magnitude. The two must not be conflated. The gap equation below is therefore stated as a calibration on an exact de Sitter horizon, where $R_{\rm event} = R_A = c/H$, the Killing structure exists, and the comoving frame is freely falling. Because κ cancels between $c^2\epsilon^2\kappa/(8\pi G)$ and $\hbar\kappa/(2\pi c)$, the resulting $\hbar = c^3\epsilon^2/(4G)$ does not depend on which de Sitter H calibrates it. Extending the argument to a genuinely dynamical FLRW horizon would require a Kodama–Hayward treatment showing the same factor enters both the classical and emergent thermal descriptions and cancels; that is not established here.

The classical horizon temperature is a counting derivative, not an assumption. Restrict the finite total system to a conserved energy shell and let visible configuration i carry energy e_i. If the boundary interaction is negligible at the shell scale, uniform counting gives $P(i) = \Omega_H(E - e_i)/\sum_j \Omega_H(E - e_j)$, so with $s_H = \ln \Omega_H$ and $\beta = s'_H(E)$, $P(i) \propto \exp[-\beta e_i + \frac{1}{2}s''_H(E)e_i^2 + O(e_i^3)]$. Gibbs is the leading finite-bath marginal of uniform counting, classically. Feeding in this paper's own relations $s_H = A/\epsilon^2$ and $dE = c^2\kappa\, dA/(8\pi G)$ gives $\partial s_H/\partial E = 8\pi G/(c^2\kappa\epsilon^2)$ and hence $k_B T_{\mathrm{micro}} = (\partial s_H/\partial E)^{-1} = c^2\kappa\epsilon^2/(8\pi G) = k_B T_{\mathrm{cl}}$ — with no quantum mechanics, no KMS condition, no vacuum choice and no infinite bath. On the exact de Sitter calibration $E = c^4 R/G$ and $s_H = 4\pi G^2 E^2/(c^8\epsilon^2)$, so $s''_H = \beta/E$ and the shell marginal is $P(i) \propto \exp[-\beta e_i + \beta e_i^2/(2E)]$: not exactly canonical, carrying the finite-reservoir correction $\delta_{\mathrm{bath}} \sim 1/(4s_H) \sim 10^{-123}$ at $s_H \sim 10^{122}$. This is a second correction of the same spectacular smallness as the lattice-dispersion term $(\epsilon H/c)^2$, arising for an unrelated reason; the two should not be conflated.

What this leaves. The chain above is classical and gives a Gibbs marginal in the classical subsystem energy. The horizon statement needs $\rho_{\mathrm{obs}} \propto e^{-\beta \hat{H}_{\mathrm{eff}}}$ in the emergent time-translation generator, which [Main] distinguishes from H_{tot} and does not prove survives the trace-out. H-state therefore splits into three narrower conditions: **H-shell** (the finite microcanonical factorization — $P(i) \propto \Omega_H(E - e_i)$ assumes weak boundary coupling, whereas C1 requires nonzero cross-boundary coupling and strong coupling would instead give a Hamiltonian of mean force), **H-energy** (the thermodynamic energy intertwines with $\hat{H}_{\mathrm{eff}}$), and **H-spectrum** (freely-falling UV regularity across the horizon). Only the third is a genuinely spectral condition.

H-balance: the minimal load-bearing state condition. Full H-spectrum is stronger than the calibration needs. For an accessible horizon probe with positive frequency gap ω, require only

$$\frac{\Gamma_\uparrow(\omega)}{\Gamma_\downarrow(\omega)} = e^{-\tau_K \omega}, \qquad \tau_K = \frac{2\pi c}{\kappa},$$

over the finite frequency window the observer can probe. KMS implies this; conversely, on a finite system, detailed balance across a transition graph connecting the relevant levels fixes the stationary populations to Gibbs form: from $W_{m\to n}/W_{n\to m} = p_m/p_n$ and $p_m/p_n = e^{-\tau(\omega_m - \omega_n)}$ on every accessible edge, $p_n = Ce^{-\tau\omega_n}$ on each connected component, and $\rho = Z^{-1}e^{-\tau\Omega}$ if the graph is connected. The finite-observer version tolerates a budget: edgewise error η integrates to $L\eta$ along a path of L accessible transitions, so no limitless observation is required. So the hierarchy is H-spectrum ⇒ H-balance ⇒ gap equation, with only the second arrow load-bearing.

Precision on microreversibility. With time-reversal involution Θ, $\Theta\varphi\Theta = \varphi^{-1}$ and $\Theta M_i = M_{\theta i}$, the exact trajectory-reversal map $x \mapsto \Theta\varphi(x)$ gives $N_{ij} = N_{\theta j,\theta i}$, the time-reversed detailed-balance relation. It reduces to $N_{ij} = N_{ji}$, and hence $N_i P_{ij} = N_j P_{ji}$, only when the coarse states are time-reversal even ($\theta i = i$). "Spatial partition" must therefore not be read as merely "Θ maps visible variables to visible variables": if visible configurations carry T-odd data such as momenta, only the $(\theta j, \theta i)$ form holds, and the symmetric form used downstream is unavailable.

Classical microreversibility already gives the classical half. With M_i the shell microstates compatible with visible macrostate i, $N_i = |M_i|$ and $N_{ij} = |\{x \in M_i : \varphi(x) \in M_j\}|$, time-reversal invariance gives a bijection between $i \to j$ and $j \to i$ trajectories, so $N_{ij} = N_{ji}$ and hence $N_i P_{ij} = N_j P_{ji}$ exactly. With $N_i = \Omega_H(E - e_i)$ this yields $\ln(P_{ij}/P_{ji}) = -\beta_E(e_j - e_i) + \frac{1}{2}s''_H(E)(e_j^2 - e_i^2) + \cdots$ — the thermal rate ratio from finite deterministic reversible counting, with no quantum mechanics. Equating the two descriptions of the same observable rate ratio gives $\beta_E \Delta E = \tau_K \Delta\omega$ up to the two controlled corrections, hence $dE/d\omega = \tau_K/\beta_E$ in the infrared, and on a connected accessible sector $E_n = \hbar\omega_n + E_0$. The relation naive H-energy tried to assume emerges instead from thermal rate consistency. The remaining qualification is an identification of observables — that the classical shell transitions and the frequency-resolved detector transitions are the same operational transitions — which, if it requires arbitrary detector couplings, touches the open coherent operational lift.

The infrared limit does not remove state dependence. Dispersive corrections vanish as $k \to 0$, but de Sitter admits non-Bunch-Davies α-vacua that are de Sitter invariant while failing detector detailed balance. With $r = e^{\alpha}$, the reverse/forward ratio $R_\alpha(\omega) = e^{-2\pi\omega/H}\big((1 + re^{\pi\omega/H})/(1 + re^{-\pi\omega/H})\big)^2$ has zero-frequency slope $-d\ln R_\alpha/d\omega|_0 = (2\pi/H)(1-r)/(1+r)$, which returns $2\pi/H$ only as $r \to 0$. So H-balance cannot be discharged by taking $\omega \to 0$.

H-slope: the minimal condition the calibration needs, and its status. Only the infrared slope is load-bearing. Writing the classical shell rate ratio R_E and the observer frequency response R_ω for the same operational transition with $\Delta E = E(\omega)$, $E(0) = 0$, and $-\partial_{\Delta E} \ln R_E|_0 = \beta_E$, the single requirement is $-\partial_\omega \ln R_\omega|_0 = \tau_K$ (**H-slope**), whence $E'(0) = \tau_K/\beta_E = c^3\epsilon^2/(4G) = \hbar$. Expanding $\ln R_E = -\beta_E E + BE^2 + \cdots$, $\ln R_\omega = -\tau_K\omega + C\omega^2 + \cdots$ and $E(\omega) = a\omega + d\omega^2 + \cdots$ gives $\beta_E a = \tau_K$ at first order exactly: the finite-bath and lattice-dispersion corrections alter d and higher coefficients, not the linear IR action scale. Full H-balance is still required for the global statement $E_n = \hbar\omega_n + E_0$ across a connected spectrum.

H-balance is a selection condition, not a representation theorem. [Main]'s intervention-dilation theorem realizes any finite action-labelled conditional family with one finite reversible dynamics, a fixed prior and permutation interventions. Take a two-gap probe with $\omega_2 = 2\omega_1$ and $R(\omega_1) = R(\omega_2) = \frac{1}{2}$: thermality at gap one forces $\tau = \ln 2/\omega_1$ and hence $R(\omega_2) = \frac{1}{4}$, so the family is nonthermal for every common inverse temperature — yet it is OI-realizable. So finite realization, reversibility and finite intervention statistics do not give H-balance, and by the α-vacuum slope above the failure already occurs at H-slope. Note also that a bare visible-space transition law read off a closed unitary would be bistochastic — for two levels the forward and reverse moduli coincide, giving ratio 1, the infinite-temperature value — which is one more reason that form is unavailable in general (Main §3.1); so H-slope requires the ancilla-carrying Q_{fb} form together with a detector gap, an interaction and an identification of excitation and de-excitation channels. Those are exactly the structures the composite and coherent instrument lift leaves open. **H-slope is therefore not an independent bridge to gravity; it is a horizon-specific selection test on that open lift.** Complete passivity is the natural candidate selector — it forces every transition to carry one common temperature and singles out Gibbs/KMS states — but it quantifies over composites, so invoking it relocates H-slope into the same open layer rather than discharging it.

The gap equation in its cleanest form: $\hbar = \tau_K/\beta_E$. Write the emergent evolution as $U_t = e^{-it\Omega_{\mathrm{eff}}}$ with $[\Omega_{\mathrm{eff}}] = \mathrm{time}^{-1}$. Before the action scale is fixed the physical energy generator can only be $H_{\mathrm{phys}} = a_*\Omega_{\mathrm{eff}}$ for some universal a_* with units of action; determining a_* is what this section does, so it must not be set to $\hbar$ in advance. Classical horizon counting supplies an inverse energy, $\beta_E = \partial s_H/\partial E = 8\pi G/(c^2\kappa\epsilon^2)$. The emergent horizon state, conditional on H-spectrum, supplies an inverse time, $\tau_K = 2\pi c/\kappa$. For a finite system a KMS state for evolution generated by Ω is Gibbs with exponent $-\tau_K\Omega$, so $e^{-\tau_K\Omega_{\mathrm{eff}}} = e^{-\beta_E a_* \Omega_{\mathrm{eff}}}$ up to normalization, giving $\tau_K = \beta_E a_*$ and

$$a_* = \frac{\tau_K}{\beta_E} = \frac{2\pi c/\kappa}{8\pi G/(c^2\kappa\epsilon^2)} = \frac{c^3\epsilon^2}{4G} = \hbar.$$

The action scale is exactly the conversion between an independently derived inverse-energy temperature and an independently derived inverse-time KMS period. No quantum mechanics enters the classical side and no $\hbar$ is inserted into τ_K beforehand.

Why the classical shell state must not be identified with $e^{-\beta\hat{H}_{\mathrm{eff}}}$. Suppose the configuration-diagonal shell state $\rho_{\mathrm{shell}} = \sum_i p_i|i\rangle\langle i|$ were required to be Gibbs for the emergent generator. Since $\rho_{\mathrm{shell}} > 0$ the matrix logarithm is unique, so $\Omega_{\mathrm{eff}} = -\tau^{-1}\log\rho_{\mathrm{shell}} + \mathrm{const}\, I$ is diagonal in the configuration basis, whence $|\langle j|e^{-it\Omega_{\mathrm{eff}}}|i\rangle|^2 = \delta_{ij}$ — no configuration dynamics at all. That contradicts the generic phase-locking sector of [Main §3.2], where the energy eigenbasis overlaps every configuration state and $V = I$ is recorded as a nongeneric degeneracy. The diagonal embedding $p(x) \mapsto \sum_x p(x)|x\rangle\langle x|$ is an embedding of classical probability, not a quantization map $E(x) \mapsto \hat{H}$, and using it as one is a no-go rather than an unproved bridge. A second, independent obstruction: $U_\varphi = e^{-i\Omega}$ has a non-unique Hermitian logarithm, $\omega_a = \theta_a + 2\pi n_a$ with n_a selectable per eigenmode, so the bare permutation does not determine continuous frequency gaps, while a conserved classical energy is merely constant on cycles.

Consequences for the named conditions. H-shell is therefore not load-bearing here: the calibration needs only the scalar $\beta_E = \partial s_H/\partial E$, which follows from the horizon density of states and the first law, not a visible/hidden product Gibbs marginal. That matters independently, since the shell conditional $\mu_H(h \mid i) \propto \mathbf{1}[E_H(h) = E - e_i]$ depends on i, whereas [Main]'s channel uses a fixed hidden prior $\mu_H(h)$; substituting one for the other would replace a product preparation by a correlated ensemble and a different projection theorem. H-shell is retained as an auxiliary result on classical probe equilibration and finite-bath corrections. H-energy in its naive form — configuration energies as eigenvalues of $\widehat{H}_{\text{eff}}$ — is false by the no-go above; its meaningful form is modular, that the static thermodynamic generator (the Hamiltonian of mean force at non-negligible coupling) agrees with the observer's dynamical generator in the infrared up to one universal action scale and an additive constant. That relation follows automatically once H-spectrum supplies the KMS property, so H-energy is redundant rather than open. **H-slope is the one load-bearing state condition: H-spectrum ⇒ H-balance ⇒ H-slope, and only the last is needed for the calibration.**

Status of $\hbar$ and the $1/4$. Both are conditional, not unconditional consequences of S1–S4: they require H-frame (the substrate preferred foliation is the cosmological comoving foliation, which S1–S4 do not supply and [SM] states is not derived) and H-state (the emergent cutoff-scale modes lie in the freely falling adiabatic/de Sitter vacuum class). H-state is not derivable from partition geometry: [Main]'s realization theorem admits different fixed hidden priors μ_H over the same $(\varphi, \text{partition})$, so vacuum-like and excited laws are both realizable, and the canonical uniform invariant measure is an equilibrium/typicality principle rather than an energy-minimization one. An additional state principle is therefore logically necessary.

Lemma 1 is now a consequence of partition geometry: the horizon has finite area $A = 4\pi c^2/H^2$, bounding coupled modes to $A/\epsilon^2 < \infty$. The Part I theorem applies: the observer's reduced description is P-indivisible and, by either route of [Main, §3], equivalent to unitary QM with $\hbar$ determined by the partition boundary.

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3. The Emergent Action Scale $\hbar$

3.1 The classical horizon temperature

Jacobson [1] showed $dE = T\,dS$ applied to local causal horizons yields Einstein's equations, with $dE = (c^2\kappa/8\pi G)\,dA$, where κ is the surface gravity of the horizon. The entropy density is $\eta = 1/\epsilon^2$ — one coupled mode per minimal cell. This is not an assumption about the number of states per cell but a consequence of two results: ϵ is defined as the minimal distinguishable scale (Lemma 1), so each cell of area ϵ^2 contributes exactly one boundary mode that couples across the partition; the number of internal states per mode (the alphabet size q) is a gauge freedom with no observable consequences (the q-gauge theorem: two systems with different q produce identical transition probabilities if they share the same coupling structure). Thus $dS = dA/\epsilon^2$. Equating:

$$T_{\rm cl} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B}$$

For de Sitter ($\kappa = cH$): $k_B T_{\rm cl} = c^3\epsilon^2 H/(8\pi G)$. No $\hbar$ appears.

Reconciliation with the internal component count. The counting $\eta = 1/\epsilon^2$ is one boundary mode per cell, whereas the substratum carries $K = 2d = 6$ internal components per site ([SM §4.5]). These are consistent, and the consistency is forced rather than assumed, once the components are read as what [SM §4.5] now conditionally identifies them as, under H-link: the $K = 2d$ components are the $2d$ link directions from a cell, with component a assigned to link direction $\hat{e}_a \in \{\pm\hat{e}_1, \dots, \pm\hat{e}_d\}$ and coupling only across that one link (the minimal-coupling factorization $\delta = 1$). A boundary cell on the horizon has exactly one outward-normal link crossing the partition; the other $2d - 1 = 5$ link directions are tangential to the boundary surface or inward, and carry no cross-partition coupling. The number of components whose coupling crosses the horizon per cell is therefore exactly one — the normal link — so $\kappa_m = 1$ and $\eta = 1/\epsilon^2$ follows from the link-direction structure, not from a separate mode-counting postulate. (The q-gauge argument settles the orthogonal question of states per mode; the component count is settled here by which links are normal.) This also confirms $\epsilon = 2\,l_p$ directly: with $\kappa_m = 1$ the relation $\epsilon = 2\sqrt{\kappa_m}\,l_p$ gives $\epsilon = 2\,l_p$, and the [SM §6] cutoff identification is unaffected. The Bekenstein-Hawking coefficient is independent of κ_m in any case — writing $\eta = \kappa_m/\epsilon^2$, the gap equation gives $\hbar = c^3/(4G\eta)$ and $S = \eta A = A/(4l_p^2)$ with κ_m cancelling — so the entropy law is a fixed point of the construction regardless, and the component count enters only the discreteness scale, where the link-direction structure fixes it at unity.

3.2 The emergent action scale

Step 1: Uniqueness. By partition-relativity (Prerequisites), $\hbar$ is derived, not free.

Step 2: $\hbar$ is structural, not volumetric.

Lemma (Boundary-only dependence). *Decompose the hidden sector as $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$, where $\mathcal{C}_B$ denotes boundary modes (within the interaction range of H_{int}) and $\mathcal{C}_D$ denotes deep modes (the remainder of the hidden sector). If the classical Hamiltonian is spatially local, then on timescales $t \ll \tau_B$:*

$$T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$$

where $T_{ij}^{(B)}$ depends only on the V-B dynamics.

Proof. (i) *Coupling structure.* Spatial locality of the classical Hamiltonian implies the interaction decomposes as $H_{\text{tot}} = H_V + H_B + H_D + H_{VB} + H_{BD}$, with no direct V-D coupling: a deep hidden-sector mode must propagate through the boundary layer before influencing the visible sector. The coupling chain is $V \leftrightarrow B \leftrightarrow D$, not $V \leftrightarrow D$.

(ii) *Frozen deep sector.* Let Δ_D be the spectral gap of $H_D + H_{BD}$ restricted to the deep sector conditioned on a fixed boundary configuration. The deep-sector evolution over time t displaces any initial configuration by $\mathcal{O}(\Delta_D\, t)$ in phase space. The spectral gap gives the inverse relaxation time of the slowest mode, so $\Delta_D \le 1/\tau_B$ (with equality when the slowest mode dominates the relaxation). For $t \ll 1/\Delta_D \le \tau_B$: $\|\varphi_t^D - \mathrm{id}\| = \mathcal{O}(\Delta_D\, t) = \mathcal{O}(t/\tau_B)$.

(iii) *Factorization of the Liouville integral.* The transition probability is:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{b\in\mathcal{C}_B} \sum_{d\in\mathcal{C}_D} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))]$$

By (i), $\pi_V(\varphi_t(x_i, b, d))$ depends on d only through the back-reaction $B \leftarrow D$, which by (ii) shifts b by $\mathcal{O}(t/\tau_B)$. Expanding:

$$\pi_V(\varphi_t(x_i, b, d)) = \pi_V(\varphi_t^{VB}(x_i, b)) + \mathcal{O}(t/\tau_B)$$

where φ_t^{VB} is the flow restricted to $V \times B$ with D frozen. The d-sum then contributes $|\mathcal{C}_D|$ identical terms at leading order:

$$T_{ij}(t) = \underbrace{\frac{1}{|\mathcal{C}_B|} \sum_{b\in\mathcal{C}_B} \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))]}_{T_{ij}^{(B)}(t)} + \mathcal{O}(t/\tau_B) \qquad \square$$

The correction is $\mathcal{O}(t/\tau_B) \sim 10^{-32}$ for laboratory processes. Since $T_{ij}(t)$ determines the emergent quantum description (Prerequisites; Barandes correspondence [35, 36]), and $T_{ij}^{(B)}(t)$ depends only on the V-B dynamics — which are characterized by the boundary geometry (A, ϵ, κ) and the constants c, G appearing in the classical Hamiltonian — the emergent action scale $\hbar$ inherits boundary-only dependence.

Corollary (deep-sector cardinality is gauge). *No observable of the emergent description — no transition probability, emergent Hamiltonian eigenvalue, or scattering amplitude — depends on the cardinality $|\mathcal{C}_D|$ of the deep hidden sector. Systems with the same $\mathcal{C}_V \times \mathcal{C}_B$ dynamics produce the same emergent physics to within $\mathcal{O}(\tau_S/\tau_B)$, whether $|\mathcal{C}_D|$ is finite, countably infinite, or uncountably infinite.*

Proof. $T_{ij}^{(B)}(t)$ is defined by a sum over $\mathcal{C}_B$ alone. The $\mathcal{C}_D$-dependent terms enter only through the $\mathcal{O}(t/\tau_B)$ back-reaction. Since the emergent Hamiltonian $\hat{H}_{\text{eff}}$ is uniquely determined by $\{T_{ij}(t)\}$ (phase-locking lemma, [Main, §3.1]; D-gauge theorem, §3.3), it inherits $\mathcal{C}_D$-independence at leading order. All observables — eigenvalues, transition amplitudes, scattering cross-sections — are functions of $\hat{H}_{\text{eff}}$ and therefore $\mathcal{C}_D$-independent. □

The cardinality of the deep hidden sector is gauge in the same technical sense as the alphabet size q: just as two systems with different q are physically equivalent if they share the same coupling structure, two systems with different $|\mathcal{C}_D|$ are physically equivalent if they share the same $\mathcal{C}_V \times \mathcal{C}_B$ dynamics. Lemma 1 requires S finite for the recurrence proof ([Main, §2.3]); the corollary establishes that the deep sector's contribution to this finiteness has no observable consequences. The question "is the universe finite or infinite?" has no empirical content within the framework — it is identified as gauge.

Step 3: Dimensional determination. Step 2 excludes volumetric (deep-sector) quantities, leaving boundary quantities. The boundary carries both *local* geometric data (ϵ, κ, and the constants c, G of the classical Hamiltonian) and a *global* quantity: the total area A, which forms the dimensionless ratio $A/\epsilon^2 = S_{\text{dS}}$. If $\hbar$ depended on S_{dS}, it would be observer-dependent — different observers have different horizon areas (§2.1) — contradicting the universality of the emergent action scale. (All sub-c observers share the same local physics and hence the same $\hbar$; §2.1.) This excludes A, leaving the local boundary quantities c, G, ϵ, and κ. The surface gravity κ is excluded because it varies between observers and epochs while $\hbar$ is observed to be universal across observers and constant in time: a laboratory experiment measures the same $\hbar$ regardless of the observer's local cosmological-horizon parameters, and for the cosmological horizon $\kappa = cH$ is itself time-dependent ($\dot{H} \neq 0$). The unique combination of c, G, ϵ with dimensions of action:

$$\hbar = \beta \frac{c^3 \epsilon^2}{G}$$

Step 4: Thermal self-consistency. The classical substratum assigns the partition boundary a temperature $k_B T_{\text{cl}} = c^2 \epsilon^2 \kappa/(8\pi G)$ (§3.1), containing no $\hbar$. The emergent QFT of Part I lives on this classical background. On the exact de Sitter calibration of §2 that background has a bifurcate Killing horizon with surface gravity κ, and regularity of the Wick-rotated metric requires Euclidean period $\beta = 2\pi c/\kappa$, giving a KMS state at $T_Q = \hbar\kappa/(2\pi c k_B)$ with $\hbar$ unknown. Two scope limits apply and are load-bearing. First, periodicity is not inherited by every QFT on this background: a preferred-frame dispersive lattice theory satisfies it only under the H-Hawking conditions of §8.5, and the dispersive-Hawking literature supplies counterexamples where they fail, so periodicity does not hold for every QFT on this background. Second, the KMS temperature parameter is fixed kinematically by κ; whether the emergent state belongs to that KMS class is state-dependent, and which class it occupies is H-spectrum, which §2 records as underivable from S1–S4. The equality below is therefore a conditional infrared calibration, not a theorem within the derived QFT. The two temperatures are computed independently — $T_{\rm cl}$ from the classical substratum alone (no QM), T_Q from the emergent QFT alone (no classical substratum details) — but they refer to the **same physical degrees of freedom**: the boundary modes $V \times B$ across which $H_{\rm int}$ couples the visible and hidden sectors. Step 2's boundary-only dependence shows that the emergent QFT is uniquely determined (at leading order in τ_S/τ_B) by these same $V \times B$ dynamics. Temperature is a state property; two complete descriptions of the same modes cannot assign different temperatures without logical contradiction. The matching $T_{\rm cl} = T_Q$ is therefore not an additional assumption but a consequence of the boundary modes being the same physical objects in both descriptions, with corrections at $\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$:

$$\frac{c^2\epsilon^2\kappa}{8\pi G} = \frac{\hbar\kappa}{2\pi c}$$

κ cancels — a non-trivial cross-check on Step 3, which excluded κ from $\hbar$ on observer-universality grounds. The thermal matching here recovers that exclusion from independent physics: κ enters both sides of the consistency equation, and only the combinations of c, G, ϵ predicted by Step 3 survive. Solving:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}}$$

The derivation is a gap equation: ϵ is the free geometric input, $\hbar$ is the output. The match is non-trivial: $T_{\rm cl}$ could have depended on volumetric hidden-sector data (excluded by Step 2), or T_Q could have depended on the state (it does not — the KMS temperature is kinematic in its parameter, conditional on H-balance in its class). That neither pathology obtains makes the gap equation a genuine determination. The non-circularity is structural: Part I establishes that a QFT emerges with *some* action scale $\hbar$; §3 determines *which* $\hbar$, using the independent classical temperature that Part I neither requires nor produces.

The counting $\eta = 1/\epsilon^2$ is definitional. Given that ϵ is the minimal distinguishable scale (Lemma 1), "one boundary mode per cell of area ϵ^2" is what "minimal" means: a cell finer than ϵ would subdivide the minimal scale, while counting fewer modes would require a distinction below ϵ to group them. The q-gauge theorem (alphabet size is observationally irrelevant) settles that what counts is the number of *cells*, not the number of internal states per cell. $\eta = 1/\epsilon^2$ is therefore not an independent physical ansatz about mode density but a consequence of the definition of ϵ combined with the irrelevance of sub-cell structure.

The predictive content lies not in the gap equation alone but in its consequences. The specific relationship $\hbar = c^3\epsilon^2/(4G)$ — rather than any other function of c,

G, ϵ — produces: the Bekenstein-Hawking formula with the exact factor 1/4 (§5), the CC dissolution with $S_{\rm dS}$ as the compression ratio (§6.3), and the Type II running-vacuum form of the dark energy (§7.1). Any alternative $\hbar(\epsilon)$ would fail at least one of these checks. The situation is analogous to deriving the Schwarzschild metric with M as a free parameter: the derivation has genuine content (the functional form) even though one input is not determined from within.

Jacobson's original derivation [1] uses $\hbar$-containing forms; this paper does not — it uses the classical identity $dE = (c^2\kappa/8\pi G)\,dA$ and the classical entropy density $\eta = 1/\epsilon^2$. The logical ordering differs from Jacobson's: his argument derives G from η and $\hbar$; the present argument takes G as a parameter of the classical Hamiltonian (Lemma 3) and derives $\hbar$ from G and ϵ. The two are consistent — substituting $\eta = 1/\epsilon^2$ and $\hbar = c^3\epsilon^2/(4G)$ into Jacobson's formula $G_{\rm eff} = c^4/(8\pi\eta\cdot\hbar c)$ recovers $G_{\rm eff} = G$ — but the roles of input and output are reversed. The Gibbons-Hawking temperature [2] $k_B T_{\rm GH} = \hbar\kappa/(2\pi c)$ is recovered as a prediction, not used as an input. The KMS condition used above is derived for continuum QFT on bifurcate Killing horizons; for the lattice-regularized theory emerging from Part I, the relevant result is that thermal periodicity of the Euclidean Green's functions is robust against UV modifications of the dispersion relation, with corrections at $\mathcal{O}((\epsilon\kappa/c^2)^2)$ [3, 4]. For the cosmological horizon, $\epsilon\kappa/c^2 = \epsilon H/c \sim 10^{-61}$, so lattice corrections are $\sim 10^{-122}$.

An independent necessity statement. The same conclusion is forced by an external theorem with a disjoint axiom base. For matter field equations whose principal polynomial is that of standard-model fields on a Lorentzian metric — the class in which the emergent sector of Part I falls, the single-cone structure being supported by the rotational-protection results of [SM §3.1] and by the symmetric-point universality of the block cones under H-blind ([SM §4.4]) — the gravitational closure theorem of Düll, Schuller, Stritzelberger and Wolz [Phys. Rev. D 97, 084036 (2018)] determines the gravitational action to be Einstein–Hilbert with cosmological constant, from canonical co-evolution and diffeomorphism invariance alone, with no thermodynamic input. The derivation above and the closure theorem therefore force the same dynamics from disjoint axiom bases; the closure route leaves G and Λ as undetermined constants, which is precisely where the present relation $\hbar = c^3\epsilon^2/(4G)$ and the analysis of §6–7 take over. The closure theorem, moreover, is one member of a family of independent necessity statements the emergent sector inherits once its inputs stand: Lovelock's uniqueness theorem [J. Math. Phys. 12, 498 (1971)] — second-order, metric-only, generally covariant dynamics in four dimensions is Einstein–Hilbert with cosmological constant — and the spin-2 self-coupling and soft-graviton results [Weinberg, Phys. Rev. 138, B988 (1965); Deser, Gen. Rel. Grav. 1, 9 (1970)], by which a massless spin-two mode with a complete matter sector is forced into universal coupling and the Einstein–Hilbert form at low energy. Four routes — thermodynamic, closure, Lovelock, spin-2 bootstrap — with disjoint axiom bases converge on the same dynamics; this is also the structural answer to the standard analog-gravity objection that emergent space-times deliver kinematics but not Einstein dynamics: given the emergent cone, matter content, and spin-2 mode, the dynamics is over-determined rather than underived. Each route's hypotheses name an internal obligation the framework must independently meet (the single cone, the massless spin-2 bound state, the absence of extra light gravitational modes), and the perturbative members of the family carry their own known subtleties in the literature. The finite-ϵ deformation of the principal polynomial, and its comparison against the closure of weakly birefringent electrodynamics [arXiv:1708.03870], is an open consistency check.

3.3 Gauge-fixing and the dimensional obstruction

Steps 1–2 establish that $\hbar$ depends only on boundary quantities; Step 4 will fix its value via thermal matching. A prior question must be addressed: does the transition-probability data $\{T_{ij}(t)\}$ determine the emergent Hamiltonian uniquely, or does residual gauge freedom leave $\hbar$ underdetermined? The following theorem settles what the argument needs: the transition data determine the full dynamical frequency content, and the dimensional obstruction then makes the determination of $\hbar$ a question no amount of such data can settle. The sharper question — how far the data pin the Hamiltonian itself — is settled by the two-branch theorem below.

Theorem (Bohr-frequency completeness). *Let $U(t) = e^{-iHt}$ with non-degenerate eigenvalues and non-vanishing configuration-basis overlaps. Then the transition probabilities determine the Bohr-frequency set $\{E_a - E_b\}$: each return probability expands as $|U_{ii}(t)|^2 = \sum_{a,b} |V_{ia}|^2 |V_{ib}|^2\, e^{-i(E_a - E_b)t}$, in which every gap value carries a strictly positive coefficient; every $|U_{ij}(t)|^2$ is a combination of the same frequencies; and consequently any U' of the same form with $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all i, j, t has $\{E'_a - E'_b\} = \{E_a - E_b\}$.*

Proof. Expanding $U_{ij}(t) = \sum_a V_{ia}\, e^{-iE_a t}\, V^*_{ja}$ gives $|U_{ij}(t)|^2 = \sum_{a,b} V_{ia} V^*_{ja} V^*_{ib} V_{jb}\, e^{-i(E_a - E_b)t}$; at $i = j$ the coefficients are $|V_{ia}|^2 |V_{ib}|^2 > 0$, and grouping terms by frequency value only adds positive quantities — degenerate gaps merge coefficients but cannot cancel them. Distinct characters $t \mapsto e^{-i\omega t}$ of $(\mathbb{R}, +)$ are linearly independent (Dedekind), so equality of two probability families forces equality of the per-frequency coefficients over the union of the two gap sets, and positivity makes the supports — the gap sets — coincide. □

The dimensional obstruction. This is what §3's argument consumes, and it is deliberately less than a reconstruction of H. The transition probabilities are dimensionless functions of t and determine temporal frequencies; converting a frequency to an energy requires the action scale $\hbar$ itself, so no amount of transition data fixes it. The residual ambiguities in passing from frequencies to a Hamiltonian carry no scale either: the energy origin $H \mapsto H + E_0$ contributes only the global phase $e^{-iE_0 t}$, the antiunitary reflection $H \mapsto -\bar{H}$ satisfies $e^{-i(-\bar{H})t} = \overline{e^{-iHt}}$ entrywise and so preserves every $|U_{ij}(t)|^2$ exactly, and the two-branch theorem below shows that nothing beyond these survives the coefficient matching. Step 4's thermal matching is therefore the mathematically obligatory entry point for the dimensionful input.

Theorem (D-gauge completeness, two-branch form). *Let* $U(t) = e^{-iHt}$ *with all energy gaps* $E_a - E_b$ *(*$a \neq b$*) distinct and all configuration-basis overlaps non-vanishing. If* $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ *for all* i, j, t*, then* $H' = DHD^\dagger + E_0$ *or* $H' = -D\bar{H}D^\dagger + E_0$*, for a time-independent diagonal unitary* D *and a real* E_0*.*

Proof. Equality of the probability families forces, by the character independence above, equality of every per-frequency coefficient of $|U_{ij}(t)|^2 = \sum_{a,b} V_{ia}V_{ja}^* V_{ib}^* V_{jb}\, e^{-i(E_b - E_a)t}$ over the union of the two gap sets; the gap sets coincide, and with all gaps distinct each frequency is carried by exactly one ordered mode pair on each side, so the matching forces a bijection ν of ordered pairs preserving gap values and, pair by pair, equality of the coefficient tensors $C_{ij}^{ab} = V_{ia}V_{ja}^* V_{ib}^* V_{jb}$ along ν. The classification of ν concerns the spectra alone, and it reduces to integer spectra: the realization constraints of a fixed correspondence — gap identities, orderings, distinctness of differences, failure of translation- and reflection-congruence — are homogeneous rational-linear equalities, strict inequalities, and inequations, so a real spectrum pair realizing a nontrivial correspondence yields an integer pair realizing the same one: the entries span a finite-dimensional $\mathbb{Q}$-subspace of $\mathbb{R}$, each basis-coordinate slice of the pair is itself a rational solution of the equality subsystem, rational combination coefficients near the real ones preserve the finitely many strict conditions, and a common denominator clears. For integer spectra the classification is the completed Piccard theorem: two rulers with all internal differences distinct and the same difference set are identical or mirror images, except for a single two-parameter six-mark family [A. Bekir and S. W. Golomb, IEEE Trans. Inform. Theory 53 (2007) 2864–2867]. In the congruent cases the coefficient lines give the multiplicative rigidity $W_{i,\tau a} = d_i \beta_a V_{ia}$ with d, β unimodular — the moduli by diagonal-line rigidity at $m \geq 3$, while at $m = 2$ the diagonal line factors as $(q-p)(1-p-q) = 0$ and the apparent hyperbola is exactly the reflection branch; the phases by the coboundary splitting on the pair lattice, the general solution of $\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl} = 0$ being $\delta_{ia} = \alpha_i + \beta_a$ — and the β's cancel in $H' = W \operatorname{diag}(E')W^\dagger$, giving the first branch; reflection congruence is the first branch applied to $(\bar{V}, -E)$, giving the second. The exceptional six-mark family induces, up to relabeling, the single frequency-preserving correspondence μ of the homometric pair $\{0, 1, 4, 10, 12, 17\}$ / $\{0, 1, 8, 11, 13, 17\}$ — the two-parameter family satisfies all fifteen gap identities of μ identically in its parameters — and μ admits no realization by unitary eigenbases with non-vanishing overlaps: its pulled-back four-cycle identities force flat moduli, the surviving monomial phase family confines the row parameters to the eight common roots-of-unity zeros of two mask polynomials, and the graph these eight generate has maximum clique three, short of the six mutually orthogonal rows required. The two printed branches are all that remains. □

The proof is kernel-verified end to end (`verification/lean-mathlib/OIBridge/`, composed statement `twoBranch_of_BGClassification`), with a single exception consumed as a cited external theorem: the integer classification of Bekir–Golomb, stated in the kernel as the explicit premise `BGIntegerClassification` — in the integer scope the cited text uses, the passage from integer to real spectra being proved, not assumed. The hypotheses are sharp: each relaxation fails on an explicit example. Omitting E_0 fails on $H + E_0$. Omitting the second branch fails on $-\bar{H}$, whose spectrum $\{-E_a\}$ is not a shift of $\{E_a\}$ for generic H; the second branch is an antiunitary ambiguity — time reversal composed with energy reflection — not a gauge transformation. Weakening the gap condition to eigenvalue non-degeneracy fails on the C_4 ring of [SM] Proposition 20: its spectrum $\{\pm 2a, \pm 2b\}$ has distinct eigenvalues but degenerate gaps, its overlaps are uniform Fourier moduli, its reflection symmetry gives $|U_{ij}(t)|^2 = |U_{ji}(t)|^2$ so that $\bar{H}$ reproduces every transition probability, and $\bar{H}$ is not diagonal-conjugate to H because conjugation preserves the loop product z^4, which is not real there — the same example that keeps weak-CP phases viable under reciprocity marks the exact boundary of Hamiltonian reconstruction. And the frequency set alone could not have decided the classification: the homometric pair above shares its fifteen pairwise differences without the rulers being translates or reflections of one another, which is why the proof runs on the coefficient-labelled frequency data, where the multiplicative relations $z^{ab} z^{bc} = |V_{\cdot b}|^2 z^{ac}$ of the vectors $z_i^{ab} = V_{ia} V_{ib}^*$ are available, and not on the difference set.

Corollary (operational antiunitary invariance). *Transposing the preparation, every operation, and the effect simultaneously —* $\rho \mapsto \rho^T$, $\mathcal{G} \mapsto T \circ \mathcal{G} \circ T$, $E \mapsto E^T$ *— preserves every finite circuit probability, for arbitrary operations and arbitrary finite sequences, hence for every fixed outcome branch of any adaptive instrument strategy; the transformation carries Kraus operations to Kraus operations with conjugated operators, preserves the trace normalization, and on unitary evolution is exactly the branch map* $H \mapsto -\bar{H}$.

Enlarging the data from transition probabilities to arbitrary preparations, instruments, adaptive finite circuits, and effects therefore cannot remove the second branch: at the purely operational level the two branches are not two empirically distinguishable theories but two antiunitarily related reconstructions of the same complete circuit statistics, and they become physically distinct only once an oriented structure is fixed. (Kernel: `circuit_invariance`, `string_invariance`, `transposeMap_kraus`, `kraus_normalization`, `unitary_channel_transpose` in `OIBridge/AntiunitaryInvariance.lean`.)

Corollary (thermodynamic orientation). *Under the theorem's hypotheses, the reflection branch inverts thermal orientation: the antiunitary image of the positive-temperature Gibbs state of* H *is the negative-temperature Gibbs state of* $H' = -D\bar{H}D^\dagger + E_0$*, and no population profile that is non-increasing in the energies without being uniform remains so under the reflected spectrum. One passive, non-maximally-mixed state fixed with respect to the emergent generator therefore excludes the antiunitary branch; positive-temperature Gibbs/KMS* *states are strictly passive, hence sufficient.*

The reconstruction's status is thus exactly split: circuit structure alone determines the dynamics up to ordinary gauge and one global antiunitary orientation; circuit structure together with one passive oriented state determines it up to ordinary gauge only, the excluded alternative collapsing the theorem's disjunction to $H' = DHD^{\dagger} + E_0$. Whether the substratum *derives* such a state is not claimed here; it is the H-orientation transport question, and it is narrower than the H-balance/KMS conditions of §3.2 — single-state passivity, not complete passivity and not the Gibbs law. Its classical layer is settled: exact finite-bath counting with $\beta_E = \partial S_H/\partial E > 0$ makes the counting-selected profile passive for the classical configuration energies, with no Gibbs approximation and no thermodynamic limit (kernel: `counting_passive`). The open steps are the stationarity of that state under the reconstructed evolution and the transport of the classical energy ordering to one of the two orientations of the reconstructed Bohr spectrum — a sign correspondence only, far weaker than H-spectrum. (Kernel: `transported_gibbs`, `gibbs_orientation`, `passivity_selector_nonuniform`, `orientation_excludes_reflection`, `gibbs_strictlyPassive` in `OIBridge/ThermalOrientation.lean`.)

The coherent-completion classification. The completion problem separates into three logically distinct questions — existence, classification, orientation — with three different statuses, recorded here exactly.

Existence. The classical fixed-basis operational theory admits a conservative quantum realization, but extension to a coherent, visible-local, multi-time representation is not automatic. For a stationary preparation, feasibility reduces to a convex condition determined by the reconstructed eigenvectors and the visible readout: `ShellRepresentationConsistency` holds precisely when the classical marginal is a mixture of the eigenvector population columns (kernel: `shell_representation_from_comb`, `comb_mixture_of_shell_representation`, with `uniform_overlap_obstruction` an explicit failure). Visible-local interventions impose further invariants and complete-positivity constraints, and an explicit finite carrier — preparation-feasible, with a globally reachable intervention table — admits no visible-local CPTP coherent lift at all (kernel: `twoByTwo_affine_rigidity`, `twoByTwo_nonCP`, `twoByTwo_no_local_lift`). Not every nominal carrier belongs to the coherent-completion class, and every statement below is about the completions that exist.

Classification. Conditional on existence, the remaining freedom is rigid. Two-slot operational contexts separate operators — one-slot contexts do not, even where the accessible algebra is already the full matrix algebra (kernel: `operational_separation`, `sameData_unique_state`) — so identical complete operational data induce a well-defined correspondence between two completions' accessible algebras, and the correspondence is bijective, unital, star-preserving, and positive in both directions by construction from the data, positivity transferring both ways through the self-duality of the positive cone (kernel: `sameData_orderIso`, `psd_iff_trace_nonneg`). It is consequently a Jordan $*$-isomorphism (kernel: `orderIso_jordan`, proved internally), and the finite-dimensional matrix-unit classification leaves exactly two possibilities.

Theorem (coherent-completion classification). *Let two OI-compatible coherent completions on the same finite visible dimension carry Hermitian accessible menus $\{G_i^{(1)}\}$, $\{G_i^{(2)}\}$, each spanning the full matrix algebra and containing the unit at a common index, and positive-semidefinite state families $\{\sigma_k^{(1)}\}$, $\{\sigma_k^{(2)}\}$, each separating ($\mathrm{Tr}(M\sigma_k) = 0$ for all k forces $M = 0$) and each generating the full positive cone by nonnegative finite combinations. If the two completions carry the same complete operational data, $\mathrm{Tr}(G_i^{(1)}\,\sigma_k^{(1)}) = \mathrm{Tr}(G_i^{(2)}\,\sigma_k^{(2)})$ for all i, k, then there are a linear map Φ with $\Phi(G_i^{(1)}) = G_i^{(2)}$ for every i and a unitary W with $\Phi(X) = WXW^\dagger$ for all X, or $\Phi(X) = WX^TW^\dagger$ for all X.* (Kernel: `sameData_unitary_or_transpose` in `OIBridge/JordanClassification.lean`, chaining `sameData_orderIso`, `orderIso_jordan`, and `matrixJordan_unitary_or_transpose`; the matrix-unit chain `orthogonal_resolution_rank_one`, `corner_form`, `corner_nilpotent`, `corner_unimodular`, `corner_cocycle`, `orientation_dichotomy` runs the same local-freedom, triple-consistency, phase-coboundary architecture as the Hamiltonian reconstruction of the theorem above.)

Every two existing OI-compatible coherent completions with identical complete operational data are therefore equivalent either by ordinary unitary gauge or by one global antiunitary orientation reversal. No larger coherent fibre survives these conditions.

Orientation. The residual transpose branch cannot be selected by the unoriented operational data themselves.

Theorem (operational orientation no-go). *Transposition carries every admissible completion to an admissible completion — Hermiticity, positivity, span, separation, the unit, and the full cone all transport — with exactly the same complete operational data, and realizes the transpose branch of the classification with $W = \mathbb{1}$ while being the conjugation action of no unitary. Consequently every selector that factors only through the operational data assigns the same value to the two members of an orientation pair; and, in general, no inhabited class of completions closed under transposition satisfies throughout a property P with $P(R) \Rightarrow \neg P(\Theta R)$.* (Kernel: `transpose_data_eq`, `transpose_completion_admissible`, `transpose_realizes_second_branch`, `transpose_not_inner`, `operational_orientation_noGo` in `OIBridge/OrientationSelection.lean`; `no_universal_oriented_property` in `OIBridge/OrientationClosure.lean`.)

The two named premises are accordingly not symmetric obligations. `ShellRepresentationConsistency` is an existence condition: it holds for a model precisely when it holds for the reflected partner $(\bar{V}, -E)$ (kernel: `shellRepresentation_transpose_stable`), so it cannot select an orientation. Orientation requires additional alignment — `OperationalTransitionIdentification`, or an `OrientedShellRepresentation` in which the represented spectral populations are non-uniform and passive with respect to the derived classical energy ordering. Those conditions are orientation-sensitive (kernel: `transitionIdentification_orientation_sensitive`, `orientedShellRepresentation_orientation_sensi` by the no-go they are not consequences of transpose-symmetric operational data alone, and either one, combined with the classification, excludes the transpose branch (kernel: `sameData_unitary_of_orientedReference`, `sameData_unitary_of_transitionIdentification`, `sameData_unitary_of_shellRepresentation`). The boundary is therefore

$$\boxed{\text{OI} + \text{OI-compatible coherent-completion conditions} \implies \text{QM}/\mathbb{Z}_2^{\text{anti}}}$$

for the coherent completions that exist, subject to the stated carrier, locality, genericity, and operational-completeness hypotheses; and if the positive classical thermodynamic orientation is additionally identified with the coherent energy ordering — through `OperationalTransitionIdentification` or the oriented shell condition —

$$\boxed{\text{OI} + \text{coherent completion} + \text{positive thermodynamic orientation} \implies \text{QM}}$$

up to ordinary unitary gauge. Two substantive questions accordingly remain: whether the actual OI substratum lies in the class admitting an OI-compatible coherent completion, and whether its derived classical thermodynamic arrow — $\beta_E = \partial S_H/\partial E > 0$, settled classically by exact counting (kernel: `counting_passive`) — enters that completion through an orientation-sensitive bridge. The classification of alternative coherent theories is otherwise closed: existing completions are ordinary complex quantum mechanics modulo one global antiunitary $\mathbb{Z}_2$, while some nominal carriers admit no coherent completion at all.

The operational-completion characterization. The classification above is a uniqueness statement conditional on existence. The existence side — which operational theories generated on a finite carrier are exactly quantum mechanics at the level of instruments and composites — is settled by a characterization with no external premise. Fix a nonempty finite visible system and an operational theory on it: a specification, for the system and for every finite ancilla level, of which finite outcome families of linear maps are available, closed under coarse-graining and outcome-conditioned composition, with a spectator-independent local readout of the ancilla. Five conditions are in play. (i) *Valid probabilities*: every available family sends positive states to positive states and its outcomes' traces sum to the input trace. (ii) *Trivial-ancilla consistency*: an operation available on the system is available on the system with a one-state ancilla adjoined. (iii) *Inert spectators*: an untouched independent system can be adjoined without changing an intervention. (iv) *Full reversible control*: every finite unitary on every composite is available. (v) *Iterated composition*: a composite may itself serve as the working system of a larger experiment — an operation obtained by attaching a fresh uniformly prepared ancilla to a composite, acting on the enlarged carrier, and discarding the ancilla is available on the composite.

Theorem (operational-completion characterization). *For every nonempty finite visible system, the available outcome families on the system and on every positive ancilla level are exactly the normalized finite Kraus instruments if and only if conditions (i)–(v) hold.* (Kernel: `exactAll_iff_physical_general` and `general_characterization` in `OIBridge/GeneralCarrier.lean`; necessity `physical_of_exactAll`; sufficiency through `krausSoundExt_of_validity_inert`, `compositeCompleteness`, and `exactAll_of_levelOne`. The one extension fact the sufficiency direction uses — every isometry into a composite extends to a unitary agreeing with it on the seed — is proved for every finite carrier from Mathlib's orthonormal-basis extension theorem, `finiteIsometryExtensionSF_discharged` in `OIBridge/IsometryExtension.lean`, so no external premise enters either direction.)

Conditions (i) and (ii) are well-formedness requirements of any operational theory; (iii)–(v) are the substantive selection principles, and for well-formed theories the characterization is by those three alone (kernel: `exactAll_iff_substantive`). Each of the five is independent of the other four and of the observation process itself: for each condition there is a qubit theory that realizes the sealed OI core — a finite C1–C4 embedded-observation process together with its actual visible readout — satisfies the other four, and fails that one (kernel: `five_way_minimality` in `OIBridge/RankGapTheory.lean`, with the witnesses `everywhereAvailable`, `countermodel`, `diagTheory`, `gapTheory`, `systemLoose`). In particular a theory that never creates coherence realizes the OI core and is not quantum (kernel: `oi_alone_not_qm`). Hence

$$\boxed{\text{bare finite OI} \;\not\Rightarrow\; \text{QM}}$$

while

$$\boxed{\text{OI-compatible operational theory} \;+\; \text{(i)–(v)} \;\iff\; \text{finite operational QM}}$$

for every nonempty finite system (kernel: `main_result`, `oi_compatible_classification` in `OIBridge/GeneralCarrier.lean`). The OI clause is not what does the work — every theory with full reversible control realizes the sealed core (kernel: `realizesSealedOICore_of_control`) — so the theorem is a classification of the completions compatible with embedded observation, not a derivation of quantum mechanics from it. The substantive physical question it isolates is whether the OI substratum supplies (iii)–(v); the determination of $\hbar$ in this section uses only the fixed-basis representation and does not wait on that question.

Quantum-complete OI (OI$^+$). The characterization suggests a layered extension rather than a redefinition of observation incompleteness. Let the OI core denote the original sealed finite observation-incompleteness condition, unchanged (kernel: `OICore`, which is `RealizesSealedOICore`, in `OIBridge/CompletedOI.lean`). Define OI$^+$ on a nonempty finite carrier by adjoining well-formedness — valid probabilities and consistency under a trivial ancilla, conditions (i) and (ii) — together with three substantive physical principles (kernel: `OIPlus` in `OIBridge/CarrierGeneralOIPlus.lean`; on the qubit carrier the OI-core conjunct is retained for its genealogical role, `OIHierarchy.OIPlus`, and the two definitions agree there, `oiPlus_qubit_iff`):

1. *Observational independence*: an available operation remains the same operation when an untouched system is adjoined. Equivalently, inert spectators do not change the operational possibilities; in particular, independently available operations on disjoint factors can be performed jointly (kernel: `observationalIndependence_iff_inert`, `parallel_of_observationalIndependence`).
2. *Reversible richness*: available reversible transformations admit operational inverses, and at every finite composite level there exists a passive drift and a finitely generated control family whose dynamical Lie algebra contains the special-unitary algebra. The finite reachability theorem then supplies exact unitary control (kernel: `control_of_reversibleRichness`; converse `reversibleRichness_of_control`), and the Lie-rank clause alone already does so, with no inverse of a control consumed (kernel: `control_of_lieRank`, from `universalReachability_of_lieRank_positive` in `OIBridge/PositiveReachability.lean`). Thus this is a certificate of sufficient reversible richness, rather than a direct postulate that every unitary transformation is available.
3. *Observer recursion*: a composite observable system is itself an admissible observable system, with its operational families inherited from the corresponding composite level. Equivalently, under the identity and relative-readout operations used by the construction, composites may themselves be used as systems in still larger experiments (kernel: `closure_of_observerRecursion`, `observerRecursion_of_closure`).

With these definitions, the kernel proves

$$\boxed{\mathrm{OI}^+ \iff \text{exact finite endomorphic operational QM}}$$

on every nonempty finite carrier (kernel: `carrier_general_oiPlus`). Equivalently, OI$^+$ is the physically rephrased form of completed OI, the core together with (i)–(v) (kernel: `CompletedOI`, `oiPlus_iff_completedOI`): observational independence corresponds to inert-spectator compositionality, reversible richness to sufficient composite unitary control, and observer recursion to iterated composition.

This equivalence is an axiomatic completion theorem, not a derivation of the additional principles from bare OI. None of observational independence, reversible richness, or observer recursion follows from the OI core: each can fail while the core, well-formedness, and the other two remain satisfied (kernel: `oiPlus_independence`). Nor is it claimed that these are the unique natural principles from which quantum mechanics may be selected. Indeed, because sufficient reversible control already realizes the sealed OI core, the OI-core conjunct is logically redundant once the full OI^+ conditions hold (kernel: `completedOI_iff_physical`). Its role in the definition is conceptual and genealogical: OI^+ identifies a quantum-complete extension of the original OI framework while preserving the distinction between what observation incompleteness itself entails and what additional physical structure selects the quantum sector. The five-condition characterization above and the OI^+ equivalence hold at the same scope, every nonempty finite system. The sealed OI core itself is a qubit process, so on a general carrier OI^+ consists of well-formedness and the three principles, and by the redundancy just noted nothing is lost by omitting the core conjunct there.

Primitive-source form of OI^+. The three physical principles of OI^+ admit a formulation one level deeper, in terms of physical implementations and embedded observers rather than of the available operations directly. On every nonempty finite carrier, exact finite endomorphic operational quantum mechanics is equivalent to three such principles (kernel: `carrier_general_oiPlusMin`, `oiPlusMin_iff_qm` in `OIBridge/MinimalRepertoire.lean`; with the elementary repertoire in place of the second principle, `carrier_general_oiPlusPos`, `oiPlusPos_iff_qm` in `OIBridge/PositiveReachability.lean`):

1. *Implementation locality.* Operational interventions are generated by finite families of admissible implementation operators with aggregate probability normalization; admissibility is invariant under relabelling and under adjoining an uncoupled spectator. This structure supplies complete positivity and observational independence (kernel: `observationalIndependence_of_implementationLocality`, `validity_of_implementationLocality` in `OIBridge/ImplementationLocality.lean`). No closure of the implementations under the adjoint is assumed, and no inverse of a control is assumed available: the Lie-rank condition supplied by the second principle gives full composite unitary control on its own, every unitary being, up to a global scalar phase that acts trivially on its conjugation channel, a positive word in the flows and the controls (kernel: `control_of_lieRank`, `universalReachability_of_lieRank_positive` in `OIBridge/PositiveReachability.lean`; the scalar phases are factors of the positive monoid there, not an operation of the repertoire), and on

a well-formed theory full control yields operational inverse accessibility (kernel: `inverseAccessibility_of_lieRank`).

2. *Phase-free richness.* At every finite composite level with two or more states, some pair of distinguishable states is continuously drivable, and every exchange of two distinguishable states is available. No phase operation enters. On three or more states one driven transition and the permutations generate the full special-unitary Lie algebra — permutation conjugation transports the drive to every pair, two drives sharing a state bracket to the imaginary transition on the third pair, and the real and imaginary transitions on a pair bracket to the population difference (kernel: `bracket_XX`, `hControl_perm` in `OIBridge/MinimalRepertoire.lean`) — and thereby yield the Lie-rank condition and hence exact finite unitary reachability at that level (kernel: `control_at_level`, through `control_of_lieRank`). On a two-state carrier the drive and the exchange commute and the generated algebra reaches no population difference (kernel: `not_hControl_two`); the levels with two or fewer states inherit control instead from the level with three times as many states, through the ancilla attachment and discard that embedded observation supplies (kernel: `descend`, `control_of_phaseFree`). The exchanges themselves are generated by one full cycle and one adjacent exchange (kernel: `phaseFree_of_cyclic`). Elementary transition richness — every pair of distinguishable states continuously drivable, every exchange available, and a quarter phase on every state, the quarter phase supplying the imaginary partner of a transition directly — is a stronger form of the same principle (kernel: `phaseFree_of_elementary`; `lieRank_of_elementary`, `hControl_star`, `bracket_XY`), and the two forms characterize the same theory (kernel: `oiPlusMin_iff_oiPlusPos`).

3. *Embedded observation.* One relabelling- and regrouping-invariant family of finite operational theories on all finite carriers has the given theory as its ambient member. This principle supplies observer recursion, iterated composition, and consistency under a trivial ancilla (kernel: `observerRecursion_of_embeddedObservation`, `systemToLevelOne_of_embeddedObservation` in `OIBridge/EmbeddedObservation.lean`).

The equivalence

$$\boxed{\begin{array}{l}\text{implementation locality} + \text{phase-free richness}\\ \qquad + \text{embedded observation} \iff \text{exact finite endomorphic operational QM}\end{array}}$$

holds on every nonempty finite carrier. It refines the OI^+ characterization, and it is not a claim that bare observation incompleteness entails these principles: the normalization half of operational validity is explicit in implementation generation, and neither principle is supplied by the others, the witnesses being stated for the Lie-rank condition that phase-free richness supplies (kernel:

redundancy_fails, lieRank_not_redundant). The nonclassical repertoire the equivalence consumes is one continuously driven transition together with the exchanges of distinguishable states; the quarter phase is dispensable at every level. The exchange clause is not reducible to the powers of a single cyclic permutation: on an even carrier one driven transition with one full cycle generates a conjugate of the real antisymmetric matrices, which reaches no population difference (kernel: not_hControl_evenCycle), so one full cycle together with one adjacent exchange is the two-element discrete form the kernel formalizes, sufficient for every exchange (kernel: phaseFree_of_cyclic), with no claim that a smaller discrete resource is excluded, and no stronger minimality of the one driven transition is claimed. Whether that driven transition can be supplied by a discrete resource is the question of the substratum-source form below. Inverse accessibility is not a hypothesis of the equivalence: Lie-rank richness gives full composite unitary control unconditionally, and on a well-formed theory full control yields inverse accessibility, well-formedness being supplied by implementation locality with embedded observation (kernel: inverseAccessibility_of_lieRank). An implementation class closed under the adjoint remains a property of exact quantum mechanics (kernel: reversibleImplementationLocality_of_qm in OIBridge/MicroscopicReversibility.lean), and, with the other two principles, the class with that closure and the class without it characterize the same theory (kernel: oiPlusPos_iff_oiPlusElem).

Substratum-source form. The three primitive-source principles are jointly supplied by one object: an implementation architecture — a class of admissible implementation operators closed under the operations of a finite operational theory — that is stable under an uncoupled spectator, invariant under relabelling of the carrier, closed under the adjoint, and drives the elementary transitions. A theory generated by such an architecture is exact finite endomorphic operational quantum mechanics on every nonempty finite carrier, and that quantum mechanics is generated by one (kernel: genTheory_qm_of_quantumArchitecture, qm_generated_by_quantumArchitecture in OIBridge/SubstratumSource.lean). The concrete OI substratum — finite states, bijective microscopic dynamics (Axiom 2), and the phase structure — supplies the structural part in full. Its interventions have monomial observable operators, a permutation of a diagonal (kernel: bijectiveOperator_monomial, phaseOperator_monomial in OIBridge/SubstratumInterface.lean), and that class, with nothing added, is closed under composition, coarse-graining, and ancilla blocks, stable under an uncoupled spectator, invariant under relabelling, and closed under the adjoint — the reversal of a bijective intervention is the inverse bijection and the reversal of a phase is its conjugate — with the exchanges of distinguishable states and the phases among its members (kernel: substratumClass_structurallyClosed, bijectiveOperator_conjTranspose, phaseOperator_conjTranspose in OIBridge/StructuralClosure.lean). What it does not supply, under finite bijective read-write dynamics, is the continuously tunable state-mixing operation: a monomial conjugation preserves diagonal states, whereas a continuously driven transition between two distinguishable states passes through operators with two nonzero entries in one row, which no bijection realizes. Finite reversible read-write dynamics, even with genuine hidden-memory and readback behavior, does not itself generate quantum state mixing: a selectable local coupling that is a bijection at every parameter value induces only permutation operators, a strict interpolation toward the exchange is not a bijection, and the memory-swap coupling moves information bidirectionally while remaining monomial (kernel: `readWriteSourced_not_qm`, `offDiagonal_interp_not_monomial`, `readWriteControl_independent` in `OIBridge/ReadWriteControl.lean`; `substratum_residual` in `OIBridge/StructuralClosure.lean`). Adding that one operation as an empirical controllability resource completes the quantum architecture: a structurally closed extension of the substratum class is a quantum architecture exactly when it drives the elementary transitions, it generates exact finite endomorphic operational quantum mechanics on every nonempty finite carrier when it does, and quantum mechanics is generated by such an extension (kernel: `substratum_extension_quantum_iff_drives`, `substratum_plus_control_qm`, `qm_generated_by_substratum_extension`):

$$\boxed{\begin{array}{l}\text{current OI substratum} + \text{continuous off-diagonal controllability}\\ \qquad\iff\ \text{exact finite endomorphic operational QM}\end{array}}$$

on every nonempty finite carrier. This is not a derivation of quantum mechanics from bare observation incompleteness. The controllability resource is not entailed by A1–A6; it is an empirical extension of the current substratum, entering as a hypothesis on the extended architecture. The scope is finite and endomorphic, on nonempty finite carriers, and the elementary control repertoire that enters is the one stated above, with no claim that it is mathematically minimal. Observation incompleteness alone admits many theories; the concrete substratum supplies every structural requirement of quantum mechanics but not continuous state-mixing controllability, and quantum mechanics is obtained exactly when that one physical resource is added.

Typed form. The statements above are endomorphic: every operation acts from a finite carrier back to the same carrier, the levels being $A \times \mathrm{Fin}\, n$. Finite-dimensional quantum mechanics also has operations between different carriers — preparations, isometric embeddings, partial traces, and instruments whose input and output dimensions differ. A typed finite operational theory has an availability predicate on finite outcome families of maps $M_S \to M_{S'}$ between any two finite carriers, with the closure rules already used above stated at their carrier-general type: identity, classical coarse-graining, feed-forward composition across carriers, relabelling along carrier bijections, attachment of a uniformly mixed fresh factor, discard of a factor, and a spectator-independent factor readout whose form is derived; no clause asserts a dilation (kernel: `TypedOperationalTheory` in `OIBridge/TypedCompletion.lean`). Restricting a typed theory to maps from a carrier to itself yields a finite operational theory on every carrier, its endomorphic shadow, which is an embedded-observation family without further assumption (kernel: `shadow_embeddedObservation`). The shadow is exact finite endomorphic operational quantum mechanics on every nonempty finite carrier — the conclusion above, carrier by carrier — exactly when, between any two nonempty finite carriers, an outcome family is typed-available precisely if it is a finite typed Kraus instrument, rectangular Kraus operators normalized on the input carrier (kernel: `typed_determined_iff`, `typed_determined_of_oiPlusMin`, `typed_determined_of_oiPlusPos` in `OIBridge/TypedPositive.lean`). Soundness wraps a typed operation into the register formed by the input and output carriers and compresses the square Kraus operators supplied by exactness; completeness realizes a typed Kraus instrument by attaching a uniform ancilla, relabelling, running a register instrument that places each Kraus operator on the output factor and records its index on the ancilla, and discarding; the converse restricts to one carrier. The interface itself carries no quantum content: the typed theory whose branches preserve diagonal matrices satisfies every rule and is not quantum mechanics (kernel: `typed_interface_not_quantum`). Hence

under the carrier-general typed operational interface,
current OI substratum + continuous off-diagonal controllability
$\iff$ exact finite-dimensional typed operational QM

for nonempty finite input and output carriers. Within the natural carrier-general extension of the operational rules already used above, "endomorphic" is a typing artifact rather than an additional physical restriction. The qualification stands beside the statement: uniform attachment and discard are not derived from the endomorphic formulation, which cannot express a map between different carriers; they are the carrier-general counterparts of the ancilla attachment and discard rules already present in the finite operational framework, and the determination theorem shows that adding them introduces no specifically quantum content. No pure-state preparation, no carrier-coherence condition, and no availability of all completely positive maps is assumed; the quantum content lies entirely in the endomorphic conclusion above. The scope of this statement is finite-dimensional.

Infinite-region completion. The finite theory admits a canonical infinite-region completion without a continuum limit and without a Hilbert-space representation. The substratum holds its fixed-spacing lattice fundamental, so the infinite-system construction it supplies is the directed system of finite spatial regions. For a finite region Λ the observable algebra is the full matrix algebra on the region's finite configuration space; inclusion of regions gives compatible injective unital $*$-homomorphisms, and observables supported on disjoint regions commute (kernel: `inclObs_mul`, `inclObs_injective` in `OIBridge/QuasilocalAlgebra.lean`, `emb_comm_of_disjoint` in `OIBridge/QuasilocalCharacterization.l`
The local algebra is the algebra of equivalence classes of finite-region observables, two of them agreeing exactly when they agree after inclusion into a common

region (kernel: `emb_eq_iff`); the inclusions are isometric for the operator norm, and the norm completion of the local algebra is a quasilocal C^*-algebra that is the closure of the union of the finite stages (kernel: `norm_inclObs`, `instCStarAlgebraQuasilocal`, `closure_iUnion_stage`). Every consistent family of finite-region density matrices extends uniquely to a state of that algebra, and the reversible finite-range substratum update extends uniquely to an isometric $*$-automorphism under which an observable of a region is an observable of an explicit finite region after each step (kernel: `quasiState_unique`, `quasiState_nonneg`, `heisQ_mul`, `norm_heisQ`, `heis_iterate_emb`).

The completion is characterized independently of the construction. A quasilocal system is a C^*-algebra carrying the same finite-region matrix algebras through compatible injective unital $*$-embeddings, with disjoint-region commutation and dense local stages; the definition names no completion, no state, and no representation (kernel: `QuasilocalSystem` in `OIBridge/QuasilocalCharacterization.lean`). Any such system is canonically and uniquely $*$-isomorphic to the region completion, by the universal property of the local algebra and of its completion, and a system carrying the same substratum-induced discrete dynamics receives a canonical isomorphism intertwining the automorphisms (kernel: `canonEquiv`, `canon_unique`, `systemEquiv_unique`, `canon_dyn`, `systemEquiv_dyn`). Hence

$$\boxed{\begin{array}{l}\text{the canonical infinite-region completion of } \mathrm{OI}_Q \\ \qquad \cong \ \text{the unique quasilocal fixed-lattice } C^*\text{-system carrying} \\ \qquad\quad \text{the substratum's local stages and OI-induced discrete dynamics}\end{array}}$$

where OI_Q denotes the quantum-completed substratum of the statements above — the current OI substratum together with continuous off-diagonal controllability — and not bare observation incompleteness. The statement removes the finite-region cutoff for the algebra, the state space, and the OI-induced discrete dynamics; it is not a characterization of all infinite-dimensional quantum instruments, nor of all locality-preserving quantum dynamics.

The restriction to the substratum-induced dynamics is a theorem rather than a convention. General locality-preserving automorphisms form a strictly larger class: conjugation by a phase unitary at one site preserves locality and is induced by no reversible finite-range substratum bijection, multiplying a single-site matrix unit by i where every transported observable carries entries that are sums of entries of the original (kernel: `phase_localityPreserving`, `phaseQ_ne_heisQ`).

Continuous time is separate. The discrete substratum dynamics does not determine a continuous one-parameter interpolation: two Hermitian generators give flows agreeing at every integer time and differing at $t = 1/2$ (kernel: `continuous_extension_not_unique` in `OIBridge/RegionLimit.lean`). What the substratum does determine is a path. The second-order update, in its phase-space form on the pair of the previous and current slices, factors exactly as a depth-two local circuit, a shear layer followed by a swap layer, each a product of commuting involutive gates, one per site, its reversibility a consequence of the factorization (kernel: `leap_eq_swap_shear`, `gate_comm`, `depth_two_circuit`, `leap_leap_symm` in `OIBridge/SecondOrderCircuit.lean`). Each layer is the time-one map of a strongly continuous one-parameter group of isometric $*$-automorphisms of the quasilocal algebra, driven gate by gate (kernel: `layerQ_add_time` in `OIBridge/SecondOrderLayer.lean`, `swapQ_add_time` in `OIBridge/SwapLayer.lean`, `layerQ_one_eq_heisQ`, `swapQ_one_eq_heisQ` in `OIBridge/SecondOrderDrive.lean`), and the two flows, run in the order the Heisenberg picture forces, the swap first, give a norm-continuous path of isometric $*$-automorphisms from the identity to the update's own Heisenberg action (kernel: `heisQ_of_comp`, `driveQ_isContinuousPath`, `driveQ_one_eq_heisQ` in `OIBridge/SecondOrderDrive.lean`). For the composite path no one-parameter-group law is established and no generator is exhibited: whether the path satisfies a one-parameter-group law, and whether one time-independent finite-range interaction has the update as its unit-time map, are open, and nothing here decides either. A continuous-time Hamiltonian law is additional structure where that formulation is wanted, and the equivalence above does not use it.

The lattice spacing stays fundamental. Nothing above asserts a spatial continuum limit or a refinement of the spacing; the continuum description used elsewhere in the framework is an effective calculational approximation rather than the object completed here. The uniqueness is among systems built from the substratum's finite-region matrix algebras rather than among all C^*-algebras, and no Hilbert-space representation and no superselection sector is selected at the level of the laws, a sector choice being an input of the initial-condition kind (kernel: `state_selection_audit` in `OIBridge/RegionTower.lean`).

Every member of the residual family is invisible to the transition data for a stated reason — diagonal rephasing moves no modulus, the energy origin is a global phase, the antiunitary reflection conjugates $U(t)$ entrywise — and none carries a scale, so §3.4's determination of $\hbar$ does not wait on the theorem. Cross-interval transition probabilities fix the relative gauge up to a single global phase; in the continuum limit this becomes an unobservable smooth function $e^{i\theta(t)}$.

Dimensional obstruction. The unitary $U(t)$ is dimensionless; $\hbar$ enters only via $\hat{H} = i\hbar\,\partial_t U \cdot U^\dagger$. No dimensionless data can fix a dimensionful constant — Step 4's thermal self-consistency provides the necessary dimensionful input.

Remark (phase recovery under discrete sampling). The spatial lattice discreteness does not discretize time: the continuous-time extension ([Main, §2.3]) provides $T_{ij}(t)$ as a genuinely continuous function of t via the Liouville marginalization of the Hamiltonian flow (Lemma 3). The D-gauge theorem therefore applies to the continuous object without aliasing corrections. Aliasing would arise only if the observer sampled T_{ij} at discrete intervals, in which case trans-Planckian frequencies could be misidentified — but these lie outside the emergent QFT's domain. The phase-recovery error is controlled by the same $\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$ suppression that governs the boundary-only dependence of §3.2.

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4. Self-Consistency and the Discreteness Scale

Rearranging $\hbar = c^3\epsilon^2/(4G)$:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\, l_p^2$$

The framework yields $\epsilon = 2\, l_p$ exactly. This is algebraically equivalent to the formula for $\hbar$: the framework contains one free geometric parameter (ϵ), and thermal self-consistency establishes its relationship to $\hbar$. The result is a self-consistency condition constraining ϵ to the Planck regime, not an independent determination. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would create a second trace-out, breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to physically identical configurations. The framework's predictive content comes from the relationship $\hbar = c^3\epsilon^2/(4G)$, the Bekenstein-Hawking formula, and the consequences that follow.

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5. The Bekenstein-Hawking Entropy

With $\epsilon = 2\, l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\, l_p^2} = S_{\text{dS}}$$

where $S_{\text{dS}} = A/(4\, l_p^2)$ is the Bekenstein-Hawking entropy of the de Sitter horizon. The factor of 1/4 in the Bekenstein-Hawking formula — historically introduced as a proportionality constant — is here fixed, under the conditions of the §3 calibration: each minimal cell of area $\epsilon^2 = 4\, l_p^2$ on the horizon contributes one unit of entropy, and $A/(4\, l_p^2) = A/\epsilon^2$ recovers the standard formula exactly. The factor of 4 in $\epsilon^2 = 4l_p^2$ traces directly to §3 Step 4's thermal matching: the ratio $2\pi/8\pi = 1/4$ between the Euclidean KMS period (numerator: thermal periodicity of QFT on a Killing horizon) and Jacobson's Einstein-equation prefactor (denominator: gravity's coupling in $dE = (c^2\kappa/8\pi G)dA$). The Bekenstein-Hawking 1/4 is therefore not an arbitrary constant but the consistency condition between gravitational and quantum thermal physics, conditional on H-slope (§3.2) and the H-Hawking conditions of §8.5. The same minimal-cell counting argument applies to any horizon with area A, giving $S = A/(4l_p^2)$ for black-hole horizons as well; the recent observational confirmation of the area law on GW250114 [5] is consistent with this extension.

Observational confirmation: GW250114. The Bekenstein-Hawking area law $dA \geq 0$ has long been regarded as a theoretical prediction of black-hole thermodynamics, but until 2025 the empirical evidence came primarily from agreement between classical horizon dynamics and semiclassical Hawking radiation predictions — none of it testing the 1/4 coefficient directly. The LIGO/Virgo/KAGRA observation of GW250114 (September 2025) provides the first direct test of the area law at the percent level: comparing the remnant horizon area inferred from the ringdown to the sum of initial horizon areas inferred from the inspiral, the area increase is consistent with the theorem at 99.999% confidence. The scope of this test is important to state precisely: GW250114 compares horizon areas and does not measure entropy, so it tests Hawking's area theorem — classical horizon dynamics, which the framework preserves — and not the 1/4 entropy coefficient. The coefficient, fixed here as the exact ratio $2\pi/8\pi$ between Euclidean KMS periodicity (thermal physics) and Jacobson's Einstein-equation prefactor (gravitational coupling), is a derived structural constant with no direct empirical test at present; any framework retaining classical general relativity passes the GW250114 test identically, so the observation is consistent with, but non-discriminating for, the framework. Untested is not untestable: the coefficient is equivalent to the Hawking-temperature normalization ($T = \hbar\kappa/2\pi c k_B$ plus the first law fixes it), so any measurement of a horizon's radiation would fix it directly. No known astrophysical source is accessible — a solar-mass black hole radiates at ~ 60 nK against the 2.7 K CMB — but one physically contingent channel exists: the final evaporation burst of a primordial black hole ($M_i \sim 5 \times 10^{14}$ g expiring today), whose burst light curve would measure $T(M)$ and hence the coefficient. Dedicated all-sky burst searches are active — Fermi-LAT [45], HAWC [46], and most recently LHAASO's all-sky search [47] — with only rate-density limits so far, and next-generation sensitivity projected for CTAO [48]; detection is contingent on PBHs existing in the relevant mass window. Analogue-gravity horizon experiments measure the kinematic (2π/KMS) side of the ratio but not the gravitational ($8\pi G$) side. The ongoing LVK observing runs will accumulate additional events testing the area theorem across a range of progenitor masses and spins.

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6. Dissolution of the Cosmological Constant Problem

6.1 The two levels of description

From finite-dimensional QM to QFT. Part I delivers QM on a finite-dimensional Hilbert space. The $N = A/\epsilon^2$ cells discretize the visible sector; spatial locality of the classical substratum restricts which cells interact on any given timescale. The emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$ — a lattice-regularized QFT with UV cutoff at $\epsilon = 2\,l_p$. The tensor product structure follows from the spatial product structure of the classical configuration space $\mathcal{C}_V = \mathcal{C}_1 \times \cdots \times \mathcal{C}_N$ (one factor per cell), which is not required by Lemma 1 but is a consequence of spatial locality in the classical substratum: each cell's configuration is a local degree of freedom.

Locality preservation — conditional (H-local-lift). A radius-1 reversible discrete map does not by itself fix a local continuous generator. Take four binary sites on a ring with the one-site shift $\varphi(x_1, x_2, x_3, x_4) = (x_4, x_1, x_2, x_3)$ — maximally local, each output site reading one nearest neighbour — and restrict U_φ to the one-particle orbit, where it is the 4-cycle. Writing $U_\varphi = e^{-iH}$, every Hermitian branch has $\lambda_k = \pi k/2 + 2\pi n_k$, so $H_{0,2} = \frac{1}{4}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) = -\pi/4 + (\pi/2)m$, which never vanishes ($m$ would have to be $\frac{1}{2}$); the minimum modulus is $\pi/4$. But $|1000\rangle$ and $|0010\rangle$ differ at opposite sites, and a Hamiltonian of onsite and nearest-neighbour terms has zero matrix element between them. So no branch of the logarithm is nearest-neighbour. The infinitesimal argument $T_{xx'}(dt) = |H_{x'x}|^2 dt^2 + \cdots$ presupposes observer-level locality rather than deriving it from locality of φ; [Main §3.2] already records that a finite permutation does not determine a continuous flow. The theorem below therefore holds under **H-local-lift**: the selected continuous observer generator is local or quasi-local. The normalized wave/Laplacian branch of [SM §4.1] supplies a plausible such generator, so this is a missing hypothesis, not an impossibility. Under H-local-lift: *If the classical Hamiltonian is spatially local (couples only neighbors), then the emergent quantum Hamiltonian inherits spatial locality.*

Proof. For infinitesimal dt and $x' \neq x$: $T_{x,x'}(dt) = |\langle x'|\hat{H}|x\rangle|^2\, dt^2 + O(dt^3)$. If x, x' differ at non-neighboring sites, spatial locality of the classical dynamics gives $T_{x,x'}(dt) = 0$ at leading order, hence $|\langle x'|\hat{H}|x\rangle|^2 = 0$. By the D-gauge theorem (§3.3), the residual freedom is a diagonal unitary D together with the global antiunitary conjugation, both of which preserve the vanishing of off-diagonal moduli, so $\langle x'|\hat{H}|x\rangle = 0$ in any gauge. The emergent Hamiltonian has the form $\hat{H} = \sum_k \hat{h}_k + \sum_{\langle k,l\rangle} \hat{h}_{kl}$. $\square$

Scope of the emergent QFT. The CC dissolution requires only that the emergent description assigns zero-point energy $\sim \hbar\omega/2$ per mode with UV cutoff at ϵ^{-1} — a property shared by any lattice QFT with local couplings, regardless of gauge group or matter content. The framework's predictions in Part II ($\hbar$, S_{dS}, the Type II RVM form of the dark energy) depend on partition geometry, not on the emergent field content.

Classical substratum (what geometric measurements probe): The horizon has classical thermal equilibrium. By the Friedmann equation, $\rho_{\mathrm{crit}} = 3H^2c^2/(8\pi G)$. No zero-point energy, no discrepancy. Total vacuum energy density: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3 — precisely observed.

Emergent QFT (what local quantum measurements probe): Zero-point energy

$\frac{1}{2}\hbar\omega$ per mode summed to the Planck cutoff gives $\rho_{\mathrm{QFT}} \sim 10^{113}$ J/m^3 — a $\sim 10^{122}$ discrepancy with the observed value [6, 7, 8].

6.2 Why gravity sees the classical value

Spacetime geometry is part of the classical substratum (Lemma 2): the metric evolves via Einstein's equations *before* the trace-out. The stress-energy sourcing gravity is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The semiclassical equation $G_{\mu\nu} = 8\pi G\langle\hat{T}_{\mu\nu}\rangle$ is itself an emergent approximation; at the classical level, the gravitational field equations never encounter the zero-point sum.

Ontological commitment. Classical spacetime is logically prior to QM; QM is emergent. This ordering is not a claim that spacetime is a substance-level primitive of the framework — the framework has no substance-level primitives, and under the reconstruction theorem, spacetime emerges from $[(S,\varphi)]/\mathcal{G}_{\mathrm{sub}}$ given the framework's irreducible starting commitments (the empirical fact that observation occurs, the empirical inputs E1–E4, and the structural assumptions A1–A6). It is a claim about the direction of the derivation chain — that the trace-out producing QM presupposes the classical geometric structure, not the reverse. This ordering is forced by the logical structure of the derivation through three independent requirements:

(i) Definiteness. The trace-out that produces the quantum-representable reduced description is defined by a *specific* partition $\Gamma = \Gamma_V \times \Gamma_H$. The partition must be definite — not in superposition, not subject to quantum uncertainty — for the marginalization integral ([Main, §1.4]) to be well-defined. A partition in superposition would yield an incoherent mixture of inequivalent quantum theories, not a single QM. The metric at the partition boundary must therefore be classical.

(ii) Non-circularity. The partition is defined by the causal structure: Γ_H is the set of degrees of freedom beyond the observer's causal reach (§2.1). Causal structure is determined by null geodesics of the metric. If the metric were derived from QM, then the derivation would require: QM $\to$ metric $\to$ causal structure $\to$ partition $\to$ QM — a circular dependency with no entry point. The circle breaks only if the metric exists prior to the partition.

(iii) Uniqueness of $\hbar$. The emergent action scale is determined by the boundary geometry (§3.2). If the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state — but $\hbar$ is observed to be kinematic in its parameter and state-dependent in its class. The geometry-first ordering is the only one consistent with a universal $\hbar$.

Descriptive distinction established; dynamical decoupling conditional on G3. That the zero-point sum belongs to the emergent description does not entail that it cannot source geometry: the two are different claims. Quantum transition data are invariant under $\Omega_{\mathrm{eff}} \to \Omega_{\mathrm{eff}} + CI$ (a global phase), so no ordinary observer statistic detects C; yet two gravitational completions, $\mathcal{G}_{\mu\nu} = 8\pi G T^{\mathrm{cl}}_{\mu\nu}$ and $\mathcal{G}_{\mu\nu} = 8\pi G(T^{\mathrm{cl}}_{\mu\nu} - \rho_C g_{\mu\nu})$, can share the same substratum, partition, reduced transition law, fixed-basis representation and action-scale calibration while differing in exactly that offset. Invisibility to quantum mechanics is not decoupling from gravity; the latter is a coupling theorem and is what G3 names as open. Accordingly the descriptive-level distinction is established and the dissolution is conditional. With that scope: any framework in which QM is logically prior to the metric must either (a) treat the zero-point sum as a gravitational source — producing the 10^{122} discrepancy requiring cancellation or fine-tuning — or (b) invoke an independent mechanism to decouple quantum vacuum energy from gravity. In the present framework, the zero-point sum is an artifact of the emergent description and never enters the stress-energy tensor; no decoupling mechanism is needed.

Cumulative evidence for the ordering. The geometry-first ordering produces a set of consequences that match observation, none of which were designed into the starting point. They are listed with their dependencies stated: items (2) and (3) are one algebraic fact, item (4) is partly definitional, and item (7) is a one-parameter derivation plus a three-parameter match, so eight of the entries are mutually independent: (1) the CC dissolution — the observed vacuum energy sits at $\rho \sim H^2/G$, the classical baseline, without fine-tuning (§6.3); (2)–(3) the relation $\hbar = c^3\epsilon^2/(4G)$ with its 1/4 coefficient and the Bekenstein-Hawking formula $S = A/(4l_p^2)$ — *one* consequence, since $\epsilon = 2\,l_p$ is algebraically equivalent to the gap equation (§4) and the entropy formula is its re-expression through the mode count $S = A/\epsilon^2$; the independent content is the single functional form and the KMS/Jacobson ratio behind the 1/4 (§5); (4) the dark-sector concordance — ~95% of ρ_{crit} is invisible to QFT; the *fraction* is $1-\Omega_B$ by construction (from $S_{\mathrm{struct}}/S_{\mathrm{total}} = \Omega_B$, §7.3), so it is partly definitional rather than an independent match — the non-trivial content is the matter-like vs dark-energy-like *split* within that 95%, tested separately (§7.1–7.3); (5) dark energy in running-vacuum form — the functional form is the local expansion shared by every smooth model (§7.1); the content is the Type II channel (a structural reading, not a derived law), the exclusion of phantom crossing, and the magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$ (conditional on the §7.3 split, not yet established at theorem level — §7.1; observationally indistinguishable from ΛCDM), consistent at 2σ with the Bertini et al. (2025) RG-corrected ΛCDM fit (itself consistent with zero); DESI DR2's preference for evolving dark energy favours the running-vacuum *form* over plain ΛCDM at the qualitative level, but its phantom-crossing *magnitude* is in mild tension (3–4σ) with the framework's $\nu \approx 0$ prediction (§7.1) [15]; (6) the MOND acceleration scale $a_0 = cH/6$ and the baryonic Tully-Fisher relation, derived from entropy displacement (§7.3); (7) the SM gauge group (a discrete structural output), together with the coupling sector's derived content — the induced $1/\alpha_0 = 23.25$ and the universality of the threshold (the coupling ambient for hypercharge pending H-observer-bundle and H-Y-vertex, [SM §6.5]) — the three coupling values at M_Z being reproduced through three matched parameters (δ_0, A, B), so that the $< 0.1\%$ figures are fits rather than consequences

[SM §6]; (8) declining rotation curves at high redshift, confirmed by Genzel et al. (§7.3); (9) deep-MOND cluster dynamics (e.g. Coma at the virial radius) matched by the simple interpolation function — though rich-cluster cores retain an open 1.0–1.5× residual not closable by baryons or by the framework's predicted light neutrinos (§7.3); (11) the Bullet Cluster lensing morphology reproduced by frozen boundary entropy (§7.3). Quantum-first orderings produce the 10^{122} discrepancy, have no natural account of the dark sector, and treat $\hbar$ and the 1/4 factor as unexplained inputs. The cumulative case is not a single tiebreaker but a pattern: the quantitative outputs of the geometry-first ordering are consistent with observation — with the DESI phantom-crossing magnitude (item 5) carried as the one live 3–4σ tension on its own dated schedule — while the central quantitative output of the quantum-first ordering is the worst prediction in physics.

The classical vacuum energy scale. The framework does not explain *why* $\rho \sim H^2/G$; this is set by initial conditions via the Friedmann equation. What it does is reclassify the CC puzzle: from quantum vacuum cancellation to classical initial conditions. The two problems are not equivalent. The QFT CC problem requires a cancellation between independent contributions — the bare cosmological constant and the zero-point energies of every field — accurate to 122 decimal places, re-tuned at each order in perturbation theory. The initial-conditions question asks why a single parameter (Λ, or equivalently H) takes its observed value; the scale $\rho \sim H^2/G$ is the natural energy density of classical cosmology, not a fine-tuned cancellation between unrelated quantities. The framework eliminates the 122-digit cancellation entirely; the remaining question — why H takes its value — is a standard cosmological problem addressable by inflationary dynamics or other initial-conditions mechanisms.

Quantum corrections to gravity. The emergent quantum description does feed back into gravitational dynamics through state-level quantities: §7.1 derives dark energy evolution because the emergent vacuum energy depends on H through the hidden sector's volume. What the framework excludes is the zero-point sum as a gravitational source. The zero-point energy is the ground-state eigenvalue of the emergent Hamiltonian $\hat{H}_{\text{eff}}$, which acts on $\mathcal{H}_V$. But gravity is sourced by the classical Hamiltonian H_{tot}, which acts on the full phase space Γ — and $\hat{H}_{\text{eff}}$ and H_{tot} are not the same operator. The eigenvalues of $\hat{H}_{\text{eff}}$ (including the zero-point energy) are properties of the *emergent description*; the energy that H_{tot} assigns to the corresponding classical microstates is the *classical vacuum energy* $\rho \sim H^2/G$. The Equivalence Principle applies to H_{tot}: everything that contributes to H_{tot} gravitates. The zero-point sum does not contribute to H_{tot}; it appears when the classical transition probabilities are reorganized into the Hilbert space formalism ([Main, §3.1]), not when the classical dynamics is evolved. The structural/volumetric distinction of §3.2 captures this: $\hbar$ (structural, a property of the correspondence) does not gravitate; the vacuum energy (volumetric, state-dependent, a property of H_{tot}) does, but at the classical scale $\rho \sim H^2/G$ rather than $\rho \sim M_{\text{Pl}}^4$.

6.3 The structural origin of the 10^{122} discrepancy

$\rho_{\mathrm{QFT}}/\rho_{\mathrm{classical}} \sim M_{\mathrm{Pl}}^4/(H^2/G) = S_{\mathrm{dS}} = A/(4\,l_p^2)$ — the Bekenstein-Hawking entropy, now identified as the information compression ratio of the trace-out. The "worst prediction in physics" is the entropy of the observer's blind spot — a category error, not a fine-tuning failure. This cannot be resolved from within, and not merely as a matter of present ignorance: any adjudication procedure available to an embedded observer is a function of its accessible data, and both descriptions are constructed from the same accessible data, so no criterion grounded in the accessible observations themselves can separate them — non-adjudicability follows from the equivalence itself, not from limited resources. (Rankings by theoretical virtue are not blocked by this factoring; but they order presentations of the same observational content rather than track fundamentality, and the framework's own answer to the fundamentality question is supplied at the substratum level, not adjudicated from within either description.) A future observation could break the equivalence, but that is falsification of the equivalence, not adjudication between the descriptions. Nor is this factoring limitation an artifact of the framework's particular observer architecture: Wolpert's limits of inference [9] and T. Breuer's self-measurement theorem [55] establish barriers of the same type for any embedded inference device under far weaker assumptions, corroborating that the limitation is device-general rather than model-specific (their results, unlike the factoring argument, do assume free availability of the device's setups). This is not a prediction awaiting future data — the observed vacuum energy sits at the classical geometric scale $\rho \sim H^2/G$, the value the framework expects.

Convergence with Maldacena's de Sitter observer cancellation. Maldacena [38] has recently shown that the unphysical phase of the de Sitter sphere partition function — long understood to obstruct a clean dS thermodynamic interpretation — cancels exactly when one incorporates an observer with a clock. The framework presented here arrives at the same cancellation by a different route: the substratum-count formulation makes the partition trace-out the *defining* operation rather than an added ingredient, so the cancellation Maldacena requires an observer-with-clock to achieve is structurally forced. CLPW [39] establish that the von Neumann algebra of QFT observables in a gravitational subregion promotes from type III to type II once an observer is included, with the resulting entropy pinned to the substratum's $\log|G_{\mathrm{sub}}|$ in the present construction. HUZ [40] argue that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom; the present framework provides a microscopic realization in which that dimension is computed directly from the partition geometry. The post-2022 mainstream observer-essentiality literature thus converges with the present construction on substantive content while differing in mechanism — a triangulation that strengthens the empirical case for both programs.

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7. Predictions and Tests

What the classical tests test. A scope observation frames this section. The classic precision tests of general relativity — perihelion advance, light deflection, Shapiro delay, gravitational redshift, frame dragging — test the vacuum Kerr and Schwarzschild geometries: Ricci-flat *solutions*, not the field equations that select them. The tested solution set radically underdetermines the equations: any theory whose vacuum sector contains those geometries in the tested regimes passes identically, and theories with different field equations exploit exactly this freedom. The radiative sector (binary-pulsar decay, gravitational waveforms) probes the vacuum *dynamics* and discriminates more; the *sourced* sector — cosmology, where the Friedmann geometry is nothing like Ricci-flat — is where the equations themselves, their coupling structure and their response to sources, are tested directly. This framework's relationship to that division is specific on both sides. Because the emergent field equations *are* the Einstein equations (§5), the framework inherits every vacuum-solution and radiative confirmation exactly — its derivation route is invisible to Ricci-flat tests, as an emergence claim should be. And because the framework's distinctive content is at the *equation level* — the thermodynamic origin of the equations, the gauge status of the uniform vacuum offset (§6), the running-vacuum channel below — its novel exposure concentrates precisely in the sourced sector, which is where the predictions of this section live. The classical record's silence about the equations is therefore not a loophole the framework hides in but the reason its forward tests are cosmological.

7.1 Dark energy evolution in running vacuum form

$\hbar$ is H-independent (§3.2), but the emergent vacuum energy is a state-level quantity depending on the hidden sector's volume. **Not a distinctive prediction.** The expansion below is correct but universal: for any smooth $f(H)$, setting $g(X) = f(\sqrt{X})$ with $X = H^2$ gives $f = g(X_0) + g'(X_0)(H^2 - H_0^2) + O((H^2 - H_0^2)^2)$ around any $H_0 > 0$. Every smooth dark-energy model has that local form, and the counterfamily $\rho_\Lambda = \rho_0 + A(H^2 - H_0^2) + B(H^2 - H_0^2)^2$ with arbitrary B satisfies the same boundary H-dependence with matter conserved and the Bianchi relation carried by running G. Nothing in the architectural commitments forces $B = 0$, and the running-vacuum literature carries even powers of H, H^4 in other regimes, and $\dot{H}$ terms. So: if "Type II form" means the first local Taylor term it is non-predictive; if it means a global $c_0 + \nu H^2$ law, it is not derived here. The defensible statement is that boundary architecture requires some state dependence on cosmological geometry, whose smooth late-time normal form contains an $H^2 - H_0^2$ term. With that said: any smooth $\rho_\Lambda(H)$ admits the Taylor expansion

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2)$$

This is the Running Vacuum Model [10, 11]. The framework forces this structural form through three architectural commitments: (i) the boundary entropy $S_{\rm dS} = A/\epsilon^2$ depends on H via $A = 4\pi c^2/H^2$, so the emergent vacuum energy inherits an H-dependence; (ii) matter is conserved in the trace-out ontology (§7.2), which forbids direct vacuum–matter energy exchange; (iii) Bianchi consistency with matter conservation then forces $G(H)$ to run in tandem with $\rho_\Lambda(H)$ [11]. The structural commitment is therefore the Type II channel — matter conserved, compensation through running G — while the functional form itself is the local expansion shared by every smooth model, as stated above. A consistency requirement accompanies the identification: the entropy-displacement dynamics of §7.3 redistributes boundary entropy between the uniform and structured budgets, and the Type II classification requires that this redistribution not constitute vacuum–matter energy exchange in the emergent conservation equations. This holds at the level of the budget identities used here; the demonstration at the level of the completed split is deferred to the §7.3 coefficient analysis.

Type II identification. The architecture just described places the framework in the *Type II* class of RVM implementations [11]. Matter is locally conserved, $G(H)$ runs mildly with cosmic expansion, and the Bianchi identity is preserved by the combined running of $\rho_{\rm vac}(H)$ and $G(H)$. By contrast, *Type I* RVM implementations assume $G = G_N$ fixed and introduce a direct vacuum–DM exchange term $Q = 3\nu H\rho_{\rm dm}$; these are physically distinct models sharing the same functional form for $\rho_{\rm vac}(H)$.

What the framework determines about ν. The architecture above fixes the functional form but not the coefficient. A rigorous derivation of ν requires completing §7.3's dark-energy / dark-matter split analysis — specifically, the H-dependence of the uniform-vs-structured partition of boundary entropy — which has not been done at the level of a theorem. What the foundations *do* determine are:

- $\rho_s(H) = 3c^2H^2/(8\pi G) = \rho_{\rm crit}(H)$ exactly, from the gap equation combined with classical-horizon thermodynamics (§7.2). The total gravitating boundary energy tracks the critical density at every H as an identity, not a prediction.
- $S_{\rm struct}/S_{\rm total} = \Omega_B(H)$, from the Clausius-relation entropy-displacement argument of §7.3. The structured (matter-associated) fraction of boundary entropy equals the baryon fraction at each epoch.
- What is *not* fixed: how the non-baryonic $(1-\Omega_B)$ fraction further subdivides into the cosmological-constant-like (uniform) and dark-matter-like (structured) components as a function of H. That subdivision determines ν.

Magnitude from the emergent-QFT channel. The self-consistent emergent-QFT calculation bounds the H^2 coefficient directly. In OI the matter fields and the curved-spacetime background are both emergent from the classical discrete substratum; the Shapiro-Solà β-function formula [33, 34] for the vacuum-energy running coefficient

$$\nu = \frac{1}{48\pi^2} \sum_i a_i \frac{M_i^2}{M_{\rm Pl}^2}$$

is derived from one-loop QFT on a curved semiclassical background and applies rigorously within the emergent QFT sector, with M_i the physical masses of the emergent fields. Applied to SM content ($M_i \leq m_t \approx 173$ GeV), the formula gives

$$|\nu_{\rm QFT}| \sim (M_{\rm SM}/M_{\rm Pl})^2 \sim 10^{-32}$$

— observationally indistinguishable from zero.

A potential additional contribution from the substratum mode count is constrained by the boundary architecture established in §3.1 and §5: boundary modes are catalogued by spatial location alone (one degree of freedom per cell of area ϵ^2), with no inherent frequency assignment. Under the natural 2D dispersion $dN/d\omega \propto \omega$, modes concentrate near the UV cutoff and contribute a fraction $(H\epsilon)^2 \sim 10^{-122}$ at horizon frequency, well below the emergent-QFT magnitude. Absent additional substratum structure giving log-uniform boundary-mode distribution at the cosmological horizon, the framework's prediction reduces to $|\nu| \approx |\nu_{\rm QFT}|$ — observationally indistinguishable from zero at all foreseeable precision, including DESI Year 5. That absence is now a computed result rather than a modeling assumption: the direct extraction of the boundary spectral measure from the substratum dynamics (the surface-Green's-function computation of §7.4, Remark) finds a power-suppressed local distribution — $dN/d\omega \propto \omega^4$, with the 2D $\propto \omega$ used here an upper bound on the infrared weight — and no log-uniform component, at any slow-bath ratio; the interacting kernel is the remaining residual.

Remark (gravitational status of the emergent running — two readings, one forecast). The magnitude assignment above imports the emergent-QFT running of $\rho_{\rm vac}$, while §8.3 denies that the emergent zero-point sum gravitates absolutely and the emergent-QFT scope statement assigns the Part II predictions to partition geometry rather than emergent field content. These commitments are compatible under two readings, which we state explicitly rather than leave implicit. *Reading 1 (curvature-response sources):* the substratum budget fixes the constant part of ρ_Λ, while the emergent layer's curvature response — the $\sim m^2H^2$ piece that survives adiabatic renormalization precisely because it is an epoch-differential (relative) effect in the sense of §8.3 — feeds back as a source. On this reading $\nu = \nu_{\rm QFT} + \nu_{\rm sub}$, and evaluating the adiabatic running of [34, 49] over the emergent SM spectrum gives a definite sign: $\nu_{\rm QFT} \approx -(4–8) \times 10^{-33}$ — negative for all natural curvature couplings ($|\xi - 1/6| \leq 1/6$), fermion-dominated (top-quark share $\approx 99.9\%$), with sign reversal requiring boson-sector couplings $|\xi - 1/6| \gtrsim 0.4$. Heavy states at M_X, if they belong to the emergent QFT, would dominate the M^2-weighted sum and the sign would follow their statistics — an input the present construction does not fix. *Reading 2 (strict geometry-first):* only partition geometry sources gravity; the emergent running is the observer's effective attribution rather than a source, and $\nu = \nu_{\text{sub}}$ alone — smaller still under the local spectral measure, with sign open pending the boundary-measure computation (see the locality remark of §7.4 and [SM §3.1]). The framework does not adjudicate between the readings here, and for observational purposes it does not need to: both give $w(z) = -1$ to all achievable precision and a null at DESI Year 5, differing only in the attribution and sign of a residual no foreseeable instrument resolves. (This non-adjudication is resource-relative, and thereby unlike the in-principle non-adjudicability of §6.3: the two readings are not observationally equivalent — they assign different sign and attribution to a real residual — so the factoring argument does not apply, and a sufficiently sensitive probe would in principle separate them. The §6.3 limitation protects only pairs of descriptions faithful to the same accessible data.) What both readings exclude is an observable running in the DESI-favored quadrant ($w_0 > -1$, $w_a < 0$) attributable to the emergent-QFT channel: under Reading 1 that channel is negative (phantom-side), and under Reading 2 it is not a source. Any detected positive-quadrant residual therefore requires the substratum channel on the non-local spectral measure, with its correlated Lorentz-violation signature.

Observational status. DESI data [12, 13] report evolving dark energy at 2.8σ–4.2σ. The 2026 multi-variant analysis [14] of DESI DR2 + Planck PR4 + supernovae tests several Type I RVM variants (vacuum–DM exchange), finding the "flipped RVM" and "RVM with threshold" variants preferred over ΛCDM at a level comparable to the more flexible $w_0 w_a$CDM parameterization — the running-vacuum functional form those variants instantiate is the local expansion the framework shares with every smooth model (§7.1), so the preference is qualitative support for the form, not for the framework, and the Type I variants tested are not a direct test of OI's Type II prediction.

A direct Type-II-equivalent test is provided by Bertini, Novaes, von Marttens, and Shapiro (arXiv:2509.24026) [15], who fit a renormalization-group-corrected ΛCDM model — structurally equivalent to Type II RVM at leading order in ν, with both $G(\mu)$ and $\Lambda(\mu)$ running and matter conserved — against Planck 2018 + DESI DR2 + DES-Y5, reporting

$$\nu^{\text{Bertini}} = -(2.5 \pm 1.3) \times 10^{-4}$$

with the ΛCDM limit at the 2σ boundary of the posterior. This is consistent at 2σ with the framework's expectation ($\nu \approx 0$, conditional on the Type II reading). An earlier pre-DESI canonical RVM fit by Gómez-Valent and Solà [16] returned $\nu = (1.34 \pm 0.38) \times 10^{-3}$ with 3.5σ preference over ΛCDM; the migration from this value to Bertini's as the data improved from pre-DESI to DESI DR2 is itself informative. DESI Year 5 ($\sim 1\%$ precision on ν) provides the decisive test: a null at that precision confirms the prediction; a high-confidence detection of $|\nu|$ at the 10^{-4} level would require additional substratum structure not currently derivable from §3-§5.

Coupled neutrino test. The same dataset that constrains ν also constrains the neutrino sector. The DESI DR2 + CMB analysis [17] finds $\Sigma m_\nu < 0.0642$ eV (95% CL) assuming ΛCDM, prefers normal mass ordering, and bounds the lightest neutrino mass at $m_l < 0.023$ eV — all three results match the OI predictions of normal ordering with Σm_ν near the oscillation minimum of 0.059 eV. The same analysis finds a 3σ tension with the lower oscillation limit assuming ΛCDM, interpreted as "a hint of new physics not necessarily related to neutrinos" — a mismatch the framework's RVM dark energy would *relax*: an evolving-DE analysis loosens the ΛCDM Σm_ν bound, comfortably accommodating the framework's ~ 0.059 eV prediction, though the RVM-corrected bound has not yet been computed. JUNO first results [18, 19] confirm the world-leading precision on $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ and $\delta m^2 = (7.48 \pm 0.10) \times 10^{-5}$ eV^2, with the solar/reactor 1.5σ tension persisting. The neutrino sector predictions and the RVM dark energy prediction are coupled in the framework — both follow from the same lattice structure — and the joint observational support is therefore not independent confirmation of separate effects but a single empirical pattern consistent with the framework's central mechanism.

Falsification gradient and decisive test structure. On its Type II reading the framework expects $w(z) \approx -1$ — the running of the cosmological constant in Type II RVM is H^2-suppressed relative to the leading Λ contribution, so the effective equation of state stays at -1 to within deviations of order ν. DESI DR2's mild preference for phantom crossing ($w_0 > -1$ at present, $w_a < 0$ with $w(z > z_{\rm cross}) < -1$) is therefore in mild tension (3-4σ) with that expectation. The Bertini et al. [15] RG-corrected ΛCDM fit, which is the closest published Type II RVM equivalent, sits at the 2σ boundary of the posterior — consistent with the framework but not yet a confirmation. DESI Year 5 ($\sim 1\%$ precision on ν, available 2028-2029) provides the decisive three-way discrimination: (i) a null result at that precision supports the framework's expectation $\nu \approx 0$ and disconfirms the DESI DR2 phantom-crossing preference; (ii) a high-confidence detection of $|\nu| \sim 10^{-4}$ confirms running-vacuum dynamics consistent with both Type II RVM structure and DESI DR2 trends, but at a magnitude requiring substratum structure beyond §3-§5; (iii) a high-confidence detection of phantom crossing falsifies the framework's prediction at structural level. The emergent-channel sign computation (§7.1 remark) sharpens outcome (ii): a detected running in the DESI-favored quadrant cannot be attributed to the emergent-QFT channel on either reading of its gravitational status, leaving the non-local substratum branch — with its correlated Lorentz-violation signature — as the only in-framework attribution. The current mild tension with DESI DR2 is therefore the kind of healthy empirical exposure — a falsification gradient with concrete decisive test scheduled — that distinguishes a framework engaged with data from one tuned to fit it.

7.2 The dark-sector corollary

The trace-out that produces the quantum-representable reduced description has an automatic gravitational consequence.

Corollary (Invisible gravitational budget). *Under Lemmas 1–3 and conditions (C1)–(C4), the cosmological trace-out that produces the memory-bearing dynamics quantum mechanics represents simultaneously renders ~95% of the universe's gravitational budget invisible to the emergent QFT.*

Proof. Three steps, each using only results already established.

(i) Total effective density. The horizon carries $S_{\mathrm{dS}} = A/\epsilon^2$ modes (§5). The classical temperature is $T_{\mathrm{cl}} = c^3\epsilon^2 H/(8\pi G k_B)$ (§3.1), computed entirely within the classical substratum with no reference to $\hbar$ or the emergent quantum description. The total thermal energy is $E_s = S_{\mathrm{dS}} \cdot k_B T_{\mathrm{cl}}$. By the shell theorem, distributing over $V_H = 4\pi c^3/(3H^3)$:

$$\rho_s = \frac{E_s}{V_H} = \frac{A}{\epsilon^2} \cdot \frac{c^3\epsilon^2 H}{8\pi G} \cdot \frac{3H^3}{4\pi c^3} = \frac{3c^2H^2}{8\pi G} = \rho_{\mathrm{crit}}$$

The identity $\rho_s = \rho_{\mathrm{crit}}$ is a consequence of Jacobson's Clausius relation applied to the de Sitter horizon. Verlinde [20] takes the identity as a premise; the present framework grounds it in the Clausius relation of the underlying classical substratum, using the boundary-mode counting fixed in §3.1. The factor ϵ^2 cancels between $S_{\mathrm{dS}} \propto 1/\epsilon^2$ and $T_{\mathrm{cl}} \propto \epsilon^2$, so the result is independent of the specific value of ϵ: what the identity actually uses is Jacobson's $dE = (c^2\kappa/8\pi G)\, dA$ plus an area-law entropy plus the Friedmann volume.

Three clarifications are essential for what follows.

Classical, not emergent. The calculation above uses only pre-trace-out quantities: the number of boundary modes (A/ϵ^2), the classical temperature (T_{cl}), and the Hubble volume (V_H). No step invokes the emergent quantum description. This is what distinguishes ρ_s from the QFT zero-point energy $\rho_{\mathrm{QFT}} \sim M_{\mathrm{Pl}}^4$: the latter is computed as $\sum \frac{1}{2}\hbar\omega$ over modes of the *emergent* quantum field theory — a quantity that exists only after the trace-out. The framework denies that ρ_{QFT} gravitates (§6.2), because it is an artifact of the emergent description. The boundary entropy's thermal energy, by contrast, is a classical quantity that predates the trace-out and gravitates at the classical level. That both involve $\hbar$ numerically (because $T_{\mathrm{cl}} = T_{\mathrm{GH}}$) is a consequence of the gap equation's self-consistency, not a dependence on QM. The ratio $\rho_{\mathrm{QFT}}/\rho_s \sim S_{\mathrm{dS}} \sim 10^{122}$ — the same compression ratio identified in §6.3 — confirming that the two quantities occupy different levels of the framework's hierarchy.

Physical locus. The entropy modes are physically localized on the horizon — they are the hidden-sector boundary degrees of freedom of §3.2. The uniform-medium description is a gravitational equivalence (shell theorem), not a claim about physical delocalization. No entropy occupies the bulk.

Why the equivalence is useful. When matter is present, it locally deforms the horizon geometry, redistributing the boundary entropy. The gravitational effect of this redistribution on bulk observers is most easily computed using the uniform-medium equivalence: the deformed boundary is equivalent to a non-uniform effective bulk source.

General d. Running the same construction in d spatial dimensions: the horizon is a $(d-1)$-sphere of area $A_{d-1} = [2\pi^{d/2}/\Gamma(d/2)]r_H^{d-1}$; the Hubble volume is $V_d = [\pi^{d/2}/\Gamma(d/2+1)]r_H^d = A_{d-1}r_H/d$; the classical horizon temperature $T_{\rm cl} \propto H$ is dimension-independent (Gibbons–Hawking); and the Friedmann equation in d spatial dimensions is $H^2 = [16\pi G_d/(d(d-1))]\rho$. The boundary entropy is $S = A_{d-1}/\epsilon^{d-1}$ (one mode per ϵ^{d-1} cell) and the gap equation fixes $\hbar = c^3\epsilon^{d-1}/[2(d-1)G_d]$ by the same thermal self-consistency argument as at $d = 3$. Collecting the geometric factors in $\rho_s = S\,k_B T_{\rm cl}/V_d$ and dividing by $\rho_{\rm crit}$ gives

$$\frac{\rho_s}{\rho_{\rm crit}} = \frac{2}{d-1}$$

— unity only at $d = 3$. This is the general-d formula cited in [SM §3.2] as the boundary-entropy filter selecting $d = 3$.

This boundary-entropy filter is one of three independent $d = 3$ selectors in the framework: (i) substratum link counting under H-link (SM §4.5, $K = 2dm$ with $m = 1$ giving $K = 6$; coupling-degree minimization does not establish it); (ii) Bravais-lattice uniqueness for the SM gauge group (SM Theorem 7b); (iii) the present boundary-entropy filter. The three derive $d = 3$ from very different physical considerations (substrate dynamics, crystallographic representation theory, cosmological boundary thermodynamics). The convergence is structurally significant.

(ii) Invisibility. The argument has two logically distinct components.

(ii-a) Construction-level invisibility (from Part I alone). The boundary modes comprising $S_{\rm dS}$ are hidden-sector degrees of freedom — the variables traced out to produce the quantum description. By the trace-out structure, no hidden-sector degree of freedom maps to an operator on $\mathcal{H}_V$. This is a theorem of Part I, independent of any gravitational content: the hidden sector is invisible to the emergent description by construction.

(ii-b) Gravitational consequence (from the ontological ordering of §6.2). The physical import — that the boundary entropy's gravitational influence appears as additional gravity with no source in $\langle\hat{T}_{\mu\nu}\rangle$ — requires the geometry-first ordering established in §6.2: the metric exists at the pre-trace-out level, and the stress-energy sourcing gravity is the classical stress-energy, not the emergent quantum expectation value. Under this ordering, the gravitational interaction between bulk matter and boundary entropy is mediated by the metric at the classical level, modifying the geometry but contributing no term to the emergent

$\langle \hat{T}_{\mu\nu} \rangle$. To the embedded observer, this appears as additional gravity with no particle-sector source.

The corollary's predictions are therefore conditional on the same geometry-first ordering that dissolves the CC problem in §6.2. This is not an additional assumption: the ordering is already required by Part II's logic (the partition is defined by null geodesics, so the metric must exist prior to the partition). What the corollary provides is a second, qualitatively distinct *consequence* of that ordering.

(iii) Budget. The emergent QFT accounts for the baryonic sector (~5% of ρ_{crit}). The remaining ~95% is boundary entropy — gravitationally active (because ρ_s is a classical quantity computed before the trace-out, unlike $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4$ which is an artifact of the emergent description) but invisible to the emergent QFT. This matches the observed composition, in which ~95% of the gravitational content has no source in particle physics. □

Observational status of particle dark matter searches. The corollary's prediction is that no particle dark matter exists — the dark sector is the trace-out, not a missing particle species. The strongest current constraint is the LZ Collaboration's 417-day analysis (December 2025) [21], which is the most sensitive WIMP search ever conducted and reports null results across the 3–9 GeV range, extending world-leading constraints from XENONnT and PandaX-4T to lower mass. The continued failure to detect particle dark matter across multiple decades of increasingly sensitive experiments — summarized in the literature as "the waning of the WIMP" — is consistent with the corollary's prediction. Definitive falsification of the corollary requires positive detection of a particle species accounting for the structured dark sector, which has not occurred.

Memory effects and P-indivisibility. Matter locally deforms the horizon geometry, redistributing the boundary entropy. This displacement persists because $\tau_B \gtrsim 1/H$ (condition C2): the hidden-sector modes carrying the redistributed entropy do not thermalize back to the unperturbed configuration on sub-Hubble timescales. Verlinde [20] identifies this as a "memory effect" from the non-thermalization of de Sitter states at sub-Hubble scales. In the present framework, this non-thermalization is not assumed but derived: it is P-indivisibility operating in the gravitational sector, with (C2) ensuring that entropy displacements persist without thermal overwriting.

The corollary is a consistency check for the total dark sector budget, not a derivation of its internal structure. The uniform component of the boundary entropy corresponds to dark energy (§6); the structured component — the matter-induced redistribution of boundary entropy producing local gravitational effects resembling dark matter — is derived in §7.3. The ratio $\rho_{\text{QFT}}/\rho_s \sim S_{\text{dS}} \sim 10^{122}$ confirms the parallel with §6.3: the information compression ratio and the gravitational occlusion fraction are two aspects of a single phenomenon.

Falsifiability. The corollary would be falsified by detection of dark matter particles accounting for the full ~27% structured dark sector, or by evidence that the dark sector's gravitational effects are unrelated to the boundary geometry.

7.3 Dark matter from entropy displacement

The total dark sector budget is established by §7.2. This section derives the structured component from the entropy displacement induced by baryonic matter.

Inherited conditions. The chain below rests on the de Sitter temperature of §3.2, which is conditional: T_{dS} requires H-slope (the infrared detailed-balance slope), H-frame (the preferred foliation is the comoving one) and the H-Hawking conditions of §8.5, and the action-scale calibration is stated on an exact de Sitter horizon. Everything in §7.3 therefore carries those conditions. A second assumption is local to this section: writing $Q = M_B c^2$ takes the entire rest energy of a static configuration to be delivered as heat to the boundary. That is a statement about how the emergent sector couples to the boundary — the same coupling map G3 names as open — and does not follow from (C1)'s requirement of nonzero cross-boundary interaction alone. A third input is imported: the dimensional factor converting the Hubble acceleration cH into the critical acceleration is Verlinde's, not derived here (Step 2).

Step 1: Entropy displacement (conditional on the above). Baryonic matter M_B is coupled to the boundary entropy through H_{VB} (condition C1). The boundary is a thermal reservoir at $T_{\mathrm{dS}} = \hbar H/(2\pi c k_B)$ (§3.2, conditional). For a static matter configuration in gravitational equilibrium — a quasi-static process — the Clausius relation gives the entropy displacement:

$$\Delta S = \frac{M_B c^2}{k_B T_{\mathrm{dS}}} = \frac{2\pi M_B c^3}{\hbar H}$$

This follows from the generalized second law (Theorem A.6) applied to the matter-boundary system: the boundary's entropy reduction is Q/T for a reversible process, where $Q = M_B c^2$ and $T = T_{\mathrm{dS}}$. For dynamic configurations, the Clausius inequality $\Delta S \geq Q/T$ only strengthens the displacement.

Step 2: The critical acceleration (conditional; imported). The displaced entropy redistributes over the volume between the matter and the horizon. The Jacobson mechanism converts the entropy gradient into additional radial acceleration. The conversion is dimensional: in d spacetime dimensions, the factor relating the natural Hubble acceleration scale cH to the empirical MOND acceleration is $(d-3)/[(d-2)(d-1)]$. This derivation is due to Verlinde [20] (eq. 1.7), and follows from the interplay between volume-law entropy distributed over $(d-1)$ spatial dimensions and area-law entropy scaling with $(d-2)$ surface dimensions, together with the standard surface-density-to-acceleration relation via Gauss's law. In our universe ($d = 4$):

$$a_0 = \frac{(d-3)}{(d-2)(d-1)} \cdot cH = \frac{1}{2 \cdot 3} \cdot cH = \frac{cH}{6} = 1.09 \times 10^{-10}\ \mathrm{m/s}^2 \quad (H_0 = 67.4\ \mathrm{km\ s}^{-1}\mathrm{Mpc}^{-1})$$

This is the MOND critical acceleration with no fitted parameter: $cH/6$ runs from 1.09×10^{-10} m/s^2 at the Planck value of H_0 to 1.18×10^{-10} m/s^2 at $H_0 = 73$, against the radial-acceleration-relation fit $g_\dagger = (1.20 \pm 0.02_{\mathrm{rand}} \pm 0.24_{\mathrm{sys}}) \times 10^{-10}$ m/s^2 [22], a ratio of 0.91–0.98 that lies inside the stellar mass-to-light systematic. The factor 1/6 has dimensional origin — it is fixed by $d = 4$ together with the volume-vs-area entropy interplay — and is taken over from Verlinde's analysis rather than rederived from the substratum.

The OI framework's contribution beyond Verlinde's derivation is *grounding the volume-law entropy mechanism in derivable structure rather than postulate.* In Verlinde's framework, the de Sitter states' non-thermalization ("memory effect") is hypothesized; in OI, it is derived from condition C2 ($\tau_B \gg \tau_S$) operating in the gravitational sector. This addresses the consistency objection of [37] by replacing the postulated elasticity with a structural consequence of P-indivisibility at the cosmological boundary. The published Verlinde-MOND consistency debate is consequently relevant to the strength of the elasticity assumption in [20], not to the OI derivation, where the relevant content is structurally grounded.

Step 3: The deep-MOND regime. For baryonic acceleration $\ll a_0$, the displaced entropy dominates. The Jacobson mechanism converts the entropy gradient into curvature, giving $g_{\mathrm{total}} = \sqrt{g_B \cdot a_0}$ and rotation velocity:

$$v^2 = r \cdot g_{\mathrm{total}} = \sqrt{GM_B \cdot cH/6} = \text{constant} \qquad \Longrightarrow \qquad v^4 = GM_B \cdot cH/6$$

This is the baryonic Tully-Fisher relation. Flat rotation curves emerge with no dark matter particles and no free parameters beyond the baryonic mass.

Step 4: Interpolation constraints. The full interpolation between the Newtonian regime ($g_B \gg a_0$) and the deep-MOND regime ($g_B \ll a_0$) is constrained by the framework but not uniquely determined. The entropy displacement defines a potential Ψ_D with $g_D = -\nabla\Psi_D$, and the framework requires:

(i) *No free parameters.* g_D depends only on g_B and $a_0 = cH/6$ — no halo-specific parameters, no fitting constants.

(ii) *Conservative.* $\nabla \times g_D = 0$, since g_D derives from the entropy potential.

(iii) *Monotonic.* $\partial g_D/\partial g_B > 0$ — more baryonic mass produces more entropy displacement, hence more dark matter effect.

(iv) *Correct limits.* $g_D \to 0$ for $g_B \gg a_0$ (Newtonian); $g_D \to \sqrt{g_B \cdot a_0}$ for $g_B \ll a_0$ (MOND).

(v) *Analytic.* The entropy redistribution is a smooth function of the coupling structure — no discontinuities or phase transitions in the interpolation.

These constraints narrow the interpolation to a one-parameter family indexed by the transition steepness — a property of the particular bijection φ, in the same category as fermion masses (solution-specific rather than structural). The simplest member satisfying all five constraints:

$$g_D = \frac{g_B}{2}\left(\sqrt{1+\frac{4a_0}{g_B}}-1\right)$$

which is equivalent to $g_{\text{total}} = g_B \cdot \nu(g_B/a_0)$ with $\nu(y) = (1+\sqrt{1+4/y})/2$ — the "simple" MOND interpolation function corresponding to $\mu(x) = x/(1+x)$. This form matches the empirical radial acceleration relation (McGaugh et al. [22]) to < 0.1 dex across all galaxy-relevant accelerations.

Epistemic status. Steps 1–3 are Tier 2 and conditional: the displacement formula follows from the Clausius relation applied to the boundary reservoir (§3.2, A.6) and therefore inherits the conditions on T_{dS} together with the local assumption $Q = M_B c^2$; the dimensional factor is imported (Step 2); given these, $a_0 = cH/6$ and $v^4 = GM_B \cdot cH/6$ contain no fitted parameter. The interpolation constraints (i)–(v) are Tier 2 on the same conditions; the transition steepness is Tier 3 (solution-specific).

Redshift evolution. Since $a_0(z) = cH(z)/6$ and $H(z) = H_0\sqrt{\Omega_m(1+z)^3+\Omega_\Lambda}$, the critical acceleration increases with redshift: $a_0(z=1) = 1.79\,a_0(0)$, $a_0(z=2) = 3.03\,a_0(0)$, $a_0(z=3) = 4.57\,a_0(0)$. The MOND crossover radius $r_M = \sqrt{6GM_B/(cH)}$ correspondingly shrinks — for a Milky-Way-like baryon budget ($M_B \approx 6\times10^{10}\,M_\odot$ at the stellar-plus-cold-gas level), $r_M = 9.0$ kpc at $z=0$ but 5.2 kpc at $z=2$ and 3.1 kpc at $z=5$. Galaxies with effective radii comparable to or exceeding r_M show flat rotation curves (the familiar local behavior); galaxies with $r_{\text{eff}} < r_M$ show declining, Keplerian-like rotation curves. Since r_M shrinks with increasing z, high-redshift galaxies are systematically more baryon-dominated — their rotation curves decline at smaller radii.

This prediction is borne out by the data. Genzel et al. [23, 24] report stacked rotation curves for massive star-forming galaxies at $z = 0.9$–2.4 showing a declining outer velocity at $> 3\sigma$ significance relative to local spirals — consistent with the OI prediction. In ΛCDM, flat rotation curves from NFW halos are expected at all redshifts; explaining the Genzel data requires anomalously low halo concentrations or baryon-dominated inner regions. The cleanest local test is Jiao et al. [25], who detect for the first time a Keplerian decline in the Milky Way's own rotation curve from ~ 19 to ~ 26.5 kpc using Gaia DR3 kinematics, with the flat rotation curve hypothesis rejected at 3σ. Our own galaxy is the strongest single piece of evidence for the $H(z)$-dependent crossover prediction at $z=0$, where the OI framework predicts the crossover at $r_M \sim 17$ kpc for the Milky Way's full baryon budget including the hot circumgalactic gas halo ($M_B \sim 2 \times 10^{11}\, M_\odot$) — essentially where Jiao et al. observe the transition.

Tully-Fisher tension and resolution. McGaugh et al. (2024), analyzing Nestor Shachar et al. (2023) JWST kinematics, report no evolution in the Tully-Fisher relation out to $z \sim 2.5$, seemingly contradicting $v^4 \propto H(z)$. However, this comparison uses *stellar* mass M_*, not baryonic mass $M_B = M_*/(1 - f_{\rm gas})$. At high redshift, gas fractions are large ($f_{\rm gas} \sim$ 50–70% at $z \sim 2$; Tacconi et al. 2018). The OI-predicted shift in the baryonic TF ($\Delta \log M_B = \log(H_0/H(z)) = -0.48$ dex at $z = 2$) is almost exactly compensated by the gas mass not included in M_*: the cancellation requires $f_{\rm gas} = 1 - H_0/H(z)$, which gives 44% at $z = 1$, 58% at $z = 1.5$, and 67% at $z = 2$ — squarely in the observed range. The apparent "no evolution" in the stellar TF is therefore *predicted* by the OI framework: the dynamical shift (higher a_0) and the compositional shift (higher gas fraction) cancel when only stellar mass is measured. The definitive test is the *baryonic* TF at $z > 1$ with reliable ALMA gas masses: the OI framework predicts a normalization shift of -0.48 dex at $z = 2$ at fixed $v_{\rm flat}$.

Observational predictions. Three parameter-free, quantitative tests:

(i) *Rotation curve evolution.* The outer rotation curve evolves with z through $a_0(z) = cH(z)/6$. For a $10^{10.8}\, M_\odot$ exponential disk with $r_{\rm eff} = 5$ kpc, the outer slope $\Delta v/v$ between $r_{\rm turn}$ and $2r_{\rm turn}$ is -0.08 at $z = 0$, evolving to -0.06 at $z = 2$ and -0.05 at $z = 5$ — flattening as the disk increasingly sits in the deep-MOND regime, while the asymptotic $v_{\rm flat}(z) = (GM_B \cdot cH(z)/6)^{1/4}$ shifts upward by $E(z)^{1/4}$ (+32% at $z = 2$). The Genzel et al. declining $z = 1$–2 rotation curves are consistent with this picture: their observed declines reflect rotation curves entering MOND-flat regimes at higher $v_{\rm flat}$ than at $z = 0$, not steeper Keplerian declines. Particle dark matter predicts no evolution in either the outer slope or in $v_{\rm flat}$ at fixed M_B.

(ii) *Tully-Fisher evolution.* The baryonic Tully-Fisher relation evolves: $v_{\rm flat}^4 = GM_B \cdot cH(z)/6$. At fixed baryonic mass, $v_{\rm flat} \propto H(z)^{1/4}$: 7% higher at $z = 0.5$, 16% higher at $z = 1$, 32% higher at $z = 2$. Equivalently, at fixed $v_{\rm flat}$, the inferred baryonic mass at $z = 2$ is $H_0/H(z = 2) = 0.33\times$ the local value — galaxies appear to need less dark matter at high z. Testable with JWST and ALMA kinematics combined with photometric baryonic mass estimates.

(iii) *No halo-specific parameters.* In particle DM, each galaxy has a dark matter halo characterized by at least two free parameters (concentration and mass). In the OI framework, the dark matter effect at radius r depends only on the enclosed baryonic mass $M_B(< r)$ and $H(z)$ — no halo-specific parameters. This predicts: the radial acceleration relation (RAR) at fixed g_B has zero *intrinsic* scatter (at fixed z) — observational scatter is set by measurement uncertainties — and the *intrinsic* scatter does not correlate with halo properties. Current SPARC data [22] show remarkably small RAR scatter (~ 0.13 dex), consistent with OI but requiring fine-tuning in ΛCDM.

Falsifiability. The entropy displacement account would be falsified by: (a) detection of dark matter particles accounting for $\sim 27\%$ of $\rho_{\rm crit}$; (b) high-redshift rotation curves inconsistent with the predicted $H(z)$ dependence — specifically, if galaxies at $z > 2$ show flat rotation curves at $r > r_M(z)$; (c) baryonic Tully-Fisher violations exceeding baryonic mass uncertainties; (d) RAR scatter correlating with halo properties at fixed g_B and z.

Cluster-scale status. Galaxy clusters present the most challenging regime for MOND-like theories. For the Coma cluster ($M_B \approx 1.4 \times 10^{14}\, M_\odot$, $R_{200} \approx 1.9$ Mpc), the acceleration at the virial radius is $g_B/a_0 \approx 0.05$ — deep in the MOND regime. With the simple interpolation function above, the predicted velocity is $v = \sqrt{g_B \cdot \nu(g_B/a_0) \cdot R_{200}} \approx 1260$ km/s; the observed velocity dispersion implies ~ 1270 km/s — a match to better than 1%. For other rich clusters (Perseus, A2029, A1689), the simple interpolation reduces the standard MOND mass shortfall from a factor ~ 2 to ~ 1.0–1.5; the residual is **not** closable by undetected cluster baryons. Modern censuses find massive clusters ($M_{500} \gtrsim 2 \times 10^{14}\, M_\odot$) baryon-complete to within $\sim 7\%$ of the cosmic fraction inside r_{500} (Planck cosmology), and consistent with the cosmic value at the virial radius; the late-time missing baryons localized by FRB dispersion measures (Connor et al. 2025) reside in the diffuse IGM and cosmic filaments — *between* clusters, not inside them. The earlier appeal to WHIM adding 30–100% conflated the *cosmic* missing-baryon fraction (filamentary) with a cluster-internal reservoir that the data do not support. Nor is the residual absorbable by massive neutrinos: the framework predicts $\Sigma m_\nu \approx 0.059$ eV (§ above), two orders of magnitude below the ~ 2 eV that would be required. The rich-cluster residual is therefore an **open problem**. It most plausibly reflects the unresolved transition between the *local* MOND-like interpolation used here (galaxies, cluster dynamics) and the *CDM-like* linear behavior of the same entropy displacement established for the CMB and Bullet Cluster: a single field cannot be MOND-like in cluster cores and CDM-like in linear growth without a specified crossover, and deriving that crossover from the non-local relaxation kernel is deferred to the substratum-level analysis (the same uncomputed kernel as §7.3). The deep-MOND limit $v^4 = GM_B \cdot a_0$ (Tier 2) is identical for all interpolation functions; only the intermediate-acceleration behavior (Tier 3) differs. Galaxy-scale predictions are unaffected: at $g_B/a_0 = 0.5$ (typical outer disk), different interpolation functions agree to < 0.07 dex — well within the RAR scatter of 0.13 dex.

Colliding clusters (Bullet Cluster). The Bullet Cluster (1E 0657-56) presents the most-cited evidence for particle dark matter: after the collision, gravitational lensing peaks coincide with the galaxy concentrations ($\sim 15\%$ of baryonic mass), not with the X-ray gas ($\sim 85\%$ of baryonic mass) that was ram-pressure stripped to the center. In a local MOND picture where g_D tracks the instantaneous baryonic distribution, the lensing should peak where the gas is — contradicting observations. The OI framework resolves this through the non-local character of entropy displacement. The dark gravity arises from boundary entropy redistribution at the cosmological horizon, and the horizon's response time is $\tau_{\text{relax}} \sim H^{-1} \approx 14$ Gyr. The relaxation timescale identification ($\tau_{\text{relax}} \sim 1/H$) is itself a Tier 2 derived quantity from the de Sitter boundary's response to perturbations; specific calculations of mode-by-mode relaxation are deferred to a substratum-level analysis. The collision crossing time is $t_{\text{cross}} \sim 720\,\text{kpc}/4700\,\text{km/s} \approx 0.15$ Gyr — only 1% of the relaxation time. The entropy displacement is therefore *frozen* at the pre-collision configuration, centered on the galaxy concentrations (which defined the gravitational potential wells for gigayears before the collision). The gas moved during the collision; the boundary entropy did not. This reproduces the observed morphology: lensing peaks at the galaxy positions, gas peak between them — without invoking collisionless particles. The mechanism makes a testable prediction: for very old post-collision systems ($t_{\text{since}} \gg 1$ Gyr), the dark gravity should gradually relax toward the gas distribution on timescale $\sim H^{-1}$. Particle dark matter predicts no such relaxation.

CMB acoustic peaks. The CMB power spectrum provides the most precise test of the dark matter paradigm: the odd/even peak height asymmetry requires a non-oscillating gravitational source (CDM in ΛCDM) that provides potential wells for the photon-baryon fluid to oscillate in. The entropy displacement satisfies this requirement through the same thermodynamic averaging that resolves the Bullet Cluster. The Clausius relation $\Delta S = \delta Q/T_H$ involves the *net* heat transfer to the horizon boundary. For an oscillating perturbation (compression then rarefaction), the net entropy displacement per acoustic cycle is zero — the oscillating component averages out. Only the secular (growing) component produces net displacement. The entropy displacement perturbation therefore does not oscillate with the photon-baryon fluid; it tracks the growing mode exclusively.

The entropy displacement perturbation then evolves identically to CDM in the linear regime: (i) it has the same effective density ($\rho_E \approx \Omega_{\text{CDM}} \times \rho_{\text{crit}}$, from the dark-sector corollary §7.2); (ii) it does not couple to photons (it is a gravitational effect at the horizon, not a fluid component); (iii) it grows under its own gravity from adiabatic initial conditions set by inflation; (iv) its growth equation $\ddot{\delta}_E + 2H\dot{\delta}_E = 4\pi G\rho_{\text{eff}}\delta_E$ has the same source term as CDM. The CMB power spectrum — which depends on Ω_b/Ω_m (peak ratios), the growth of perturbations before recombination (peak heights), the sound horizon r_s (peak positions), and the damping scale (high-ℓ suppression) — is therefore predicted to match ΛCDM at linear order. A direct CLASS comparison (2026-04-26) confirms this: with the framework's Type II RVM at $\nu = 10^{-4}$ (the Bertini 2025 1σ high end) encoded via CLASS's CLP fluid dark-energy module, the four acoustic peak amplitudes (TT) match ΛCDM to $\lesssim 0.02\%$, full-range D_ℓ^{TT} residuals have median 0.006% and maximum 0.08%, and corresponding D_ℓ^{EE} residuals are similar — all 20–300× below cosmic variance and 5–50× below Planck 2018 binned precision at every ℓ from 2 to 2500. *Epistemic status: Tier 2, inheriting the conditions of*

*§7.2–7.3. The CLASS run verifies the residual RVM dark-**energy** effect on the spectrum (the dark matter is modeled as standard CDM, the dark-energy sector as the CLP-fluid RVM); it does not by itself test the displacement$\equiv$ CDM step, which is the analytic result (i)–(iv) above. The temporal half of that equivalence — that a single relaxation kernel reproduces CDM-like growth at recombination while keeping the Bullet frozen — is demonstrated below; a direct Boltzmann run implementing the displacement kernel explicitly, including its k-dependence, is the remaining numerical check.*

Relaxation kernel: Bullet–CMB consistency. The frozen-displacement account of the Bullet Cluster (slow τ_{relax}) and the growing-mode tracking the CMB requires (fast response at recombination) are reconciled by a single kernel once τ_{relax} is read as the *epoch-local* horizon time $\tau_{\text{relax}}(z) \sim 1/H(z)$ — the ≈ 14 Gyr quoted above is its $z \approx 0$ value, not a fixed constant. Because the linear growing mode also evolves on $\sim 1/H(z)$, the displacement relaxation equation $d\delta_E/dt = (\delta_m - \delta_E)/\tau_{\text{relax}}$ becomes, in scale-factor form, $d\delta_E/da = (\delta_m - \delta_E)/a$ — *epoch-independent*, with solution $\delta_E = \frac{1}{2}\,\delta_m + \mathcal{O}(a_{\text{rec}}^2/a)$. Thus $\delta_E \propto a$ tracks the CDM growing mode at every epoch with no lag at recombination and no temporal cutoff; the constant ratio is absorbed into the calibration $\rho_E \approx \Omega_{\text{CDM}}\rho_{\text{crit}}$ (§7.2). A *fixed* $\tau_{\text{relax}} = 1/H_0$ would instead suppress the response by $H_0/H(z) \sim 1/10$ at $z = 100$ and distort the peaks — confirming that the epoch-local reading is the correct one (numerically verified: δ_E/δ_m constant to $< 0.1\%$ across $z = 1100 \to 0$ for $\tau = 1/H(z)$, versus an order-of-magnitude lag for $\tau = 1/H_0$). The Bullet remains frozen for the independent reason that a cluster collision is anomalously fast on the cosmic clock: $t_{\text{cross}}/\tau_{\text{relax}}(z{=}0.3) \approx 0.15/12 \approx 0.013 \ll 1$. The two regimes are therefore controlled by *different* ratios — $\tau_{\text{relax}}/t_{\text{growth}} \sim \mathcal{O}(1)$ at all epochs (tracking) versus $\tau_{\text{relax}}/t_{\text{collision}} \gg 1$ (freezing) — and do not conflict. Since the Jacobson sourcing is spatially local, the non-locality is *temporal* (a memory over τ_{relax}), not spatial, and imposes no k-cutoff on the linear growth. *Open computation (not an inconsistency):* the k-dependence of the relaxation rate, required for a full mode-by-mode kernel. *Epistemic status: Tier 2, inheriting the conditions of §7.2–7.3 (temporal reconciliation verified numerically, relaxation_kernel_check.py 2026-06-29; k-dependence deferred).*

7.4 Summary of GR-side predictions

The GR-side quantitative predictions of the framework are collected in the table below. Each follows from partition geometry at the cosmological horizon; no predictions are fit to the quantities they predict.

Prediction	Formula / value	Observed / status	Match
Emergent action scale	$\hbar = c^3(2l_p)^2/(4G)$	$\hbar_{\text{obs}} = 1.055 \times 10^{-34}$ J $\cdot$ s	structural
Discreteness scale	$\epsilon = 2\,l_p$	Planck length	structural

Prediction	Formula / value	Observed / status	Match
BH entropy coefficient	$S_{BH} = A/(4l_p^2)$	derived ($2\pi/8\pi$); no direct test — GW250114 confirms the classical area theorem, which the framework preserves	consistent (non-discriminating)
Cosmological constant	$\rho_\Lambda \sim H^2/G$	observed value	no discrepancy
Dark energy form	Type II channel; functional form the local expansion shared by every smooth model (§7.1); magnitude $\approx \lvert\nu_{\mathrm{QFT}}\rvert \sim 10^{-32}$, indistinguishable from zero	Bertini 2025 $\nu = -(2.5{\pm}1.3){\times}10^{-4}$	consistent at 2σ (Bertini consistent with zero)
Dark sector fraction	$\Omega_{\mathrm{dark}} \approx 95\%$	$\Omega_{\mathrm{CDM}} + \Omega_\Lambda \approx 95\%$	consistent
MOND acceleration scale	$a_0 = cH/6 = 1.09\text{–}1.18 \times 10^{-10}$ m/s^2 ($H_0 = 67.4\text{–}73$)	$g_\dagger = (1.20 \pm 0.02 \pm 0.24) \times 10^{-10}$ m/s^2	within systematic; factor imported
Baryonic Tully-Fisher	$v^4 = GM_B \cdot cH/6$	SPARC data	within 0.13 dex RAR scatter
Rotation curve crossover	$r_M = \sqrt{6GM_B/(cH)}$	Milky Way: 9–17 kpc depending on the baryon budget (6×10^{10}–$2 \times 10^{11}\,M_\odot$; §7.3 uses the former)	Jiao et al. 2023, 3σ
High-z rotation curves	More baryon-dominated with z	Genzel et al. $z = 0.9$–2.4	consistent at $> 3\sigma$

The gravitational sector's dark-energy content is geometry-dependent bookkeeping whose smooth late-time expansion contains an $H^2 - H_0^2$ term — the local form shared by every smooth model (§7.1) — read in the Type II running-vacuum channel; the framework does not presently derive a distinctive Type II law, the channel being selected by architectural commitments: the Planck-scale

UV cutoff is forced by the gap equation (§3.2, §4), and matter conservation plus Bianchi consistency fixes the coupled running of $G(H)$ and $\rho_{\mathrm{vac}}(H)$. To state the distinctiveness precisely: a $c_0 + \nu H^2$ expansion at low energy is generic for any covariant analytic $\rho_{\mathrm{vac}}(H)$; the framework-specific content is the Type II channel (matter conserved, compensation through G), the coefficient class, and the exclusion of phantom crossing. The *magnitude* of ν is bounded by the self-consistent emergent-QFT calculation (Reading 1 of the §7.1 remark; smaller still on Reading 2): $|\nu_{\mathrm{QFT}}| \sim (M_{\mathrm{SM}}/M_{\mathrm{Pl}})^2 \sim 10^{-32}$, indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3.1 and §5, which catalogues modes by spatial location with no inherent frequency assignment; the natural 2D field-theoretic distribution concentrates modes near the UV cutoff and contributes $|\nu|_{\mathrm{substratum}}$ well below $|\nu_{\mathrm{QFT}}|$. The framework's commitment is therefore the Type II reading with magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}|$ — the magnitude conditional on the §7.3 split and not yet established at theorem level (§7.1) — observationally indistinguishable from zero. The Bertini et al. (2025) measurement $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ is consistent with this expectation at 2σ, where Bertini is itself consistent with zero. DESI Year 5 ($\sim 1\%$ precision on ν) provides the decisive empirical test of the reading: a null supports the Type II reading and its near-null magnitude; a high-confidence detection at the 10^{-4} level would require additional substratum structure (specifically, log-uniform mode-per-decade distribution at the cosmological horizon) not currently derivable from §3-§5.

Remark (the magnitude question reduces to emergent locality). The choice between the two boundary-mode spectral measures is not an independent free parameter: it is the same question as the radiative emergent-Lorentz problem ([SM §3.1, point iii]). The natural 2D field-theoretic measure ($dN/d\omega \propto \omega$, giving $|\nu| \approx |\nu_{\mathrm{QFT}}|$) is the one induced when the coarse-grained boundary dynamics inherits the locality of the range-1 substratum; the log-uniform measure ($dN/d\ln\omega = \text{const}$, giving $|\nu| \sim 10^{-4}$) is not the spectrum of a local field theory in any dimension, and would require the coarse-graining to generate scale-invariant (non-local) boundary dynamics — precisely the failure mode that would also generate a relevant Lorentz-violating operator. The two open questions therefore share one premise: whether the substratum-to-observer coarse-graining preserves locality. A consequence is that the DESI Year 5 measurement is effectively a test of emergent horizon locality, with a built-in cross-check: a detection of $|\nu|$ at the 10^{-4} level would, under this analysis, require emergent non-locality, which must independently manifest as Lorentz violation — so a positive ν cannot be absorbed as a free parameter without a correlated Lorentz-violation signature. This reduces the framework's count of independent open questions; the magnitude of ν is forced to ≈ 0 conditional on the same emergent locality the framework requires for Lorentz invariance, rather than being independently open. (The reduction is supported by a stretched-horizon Clausius computation on the de Sitter static patch: the near-horizon redshift generates a log-uniform *observed* boundary spectrum ('t Hooft brick-wall kinematics), but the running coefficient is fixed by $d\rho_\Lambda/dH^2$, whose integrand is near-wall-dominated, so the redshift band reconstructs the area-law baseline and contributes to the running only at $O((H\epsilon)^2)$ — the local value. A log-uniform $\nu \sim 10^{-4}$ would instead require the *intrinsic* boundary theory to be scale-invariant (an RG fixed point the range-1 blocking does not supply), not merely a redshifted local spectrum. This defeats the one mechanism that could generate log-uniform weighting from local dynamics. The direct boundary-lattice computation confirms this independently: the density of states of any bounded-range (range-1) 2D boundary theory is $dN/d\omega \propto \omega$ (2D-local), with correlation length $\xi \sim 1/\Delta$ finite off criticality; the log-uniform measure is the density of states of a scale-invariant (critical, $\xi \to \infty$) theory only, reachable at the lattice level solely by tuning the gap to $\Delta \sim H\epsilon \sim 10^{-61}$ — the same horizon hierarchy the cosmological-constant dissolution removes, so a log-uniform ν would be self-undermining. That terminal residual is now discharged at the linear-map level: an exact surface-Green's-function extraction of the boundary measure from the substratum dynamics itself — the semi-infinite half-space trace-out of the reconstruction-selected wave equation, with no Gaussian effective model — gives a boundary-projected spectral measure $dN/d\omega \propto \omega^4$ (analytic edge count +4; estimator validated on the known +1/+2 controls), flat across thirty orders of magnitude in the slow-bath ratio τ_S/τ_B and stable under generic quenched deformation, with the log-uniform ω^{-1} measure appearing nowhere. The effective one-degree-of-freedom-per-cell 2D measure used in the running-vacuum bookkeeping above ($dN/d\omega \propto \omega$) is therefore an upper bound on the infrared boundary weight of the microscopic trace-out, so $|\nu| \approx |\nu_{\mathrm{QFT}}|$ holds a fortiori; which of the two local measures the ν integral runs over affects the prefactor only, not the fork. The remaining residual is the interacting (condensate-dressed) kernel, shared with [SM §3.1].)

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8. Discussion

The foundational emergent QM result is treated as established in [Main] and not re-derived here. The cubic-lattice realization of the substratum — which determines the Standard Model gauge structure via the wave equation — is independent of the present results and is developed in [SM]. The reconstruction theorem, the substratum gauge group, and the synthesis claim that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same (S, φ) are developed in [Substratum].

8.1 Scope

The characterization theorem's equivalence is (ii) $\iff$ (iii) under (C1)–(C4) — both directions proved in the foundational paper — with the quantum representation following for the source's finite-configuration class, the C2 $\Leftarrow$ direction conditional on the hidden sector's mixing hypothesis (ETH is its physical motivation; see [Main, §3.4]). The C1, C3, and C4 necessity directions are unconditional — C4's is immediate from observed non-Markovianity. For the cosmological horizon application the mixing hypothesis holds for any generic chaotic hidden sector — the horizon complement is such a system, and its coupling reads the boundary degrees it writes, which is (C4)'s realization — so the equivalence applies in our universe. The theorem does not identify which physical systems satisfy the conditions (empirical) nor derive $\hbar$ for specific systems (requires Part II). Universality in our universe follows from the shared causal partition defined by null geodesics.

8.2 The status of the discreteness scale

ϵ does not smuggle in a quantum assumption. Lemma 1 requires finite-dimensionality, motivated by the classical fact that finite-area boundaries admit finitely many modes. The result $\epsilon = 2\,l_p$ is a *consequence* of self-consistency (§4). Holographic entropy bounds [26] provide independent, non-quantum motivation.

8.3 Vacuum energy: relative effects and the Higgs potential

Relative effects. Casimir forces and Lamb shifts are predictions *within* the emergent QFT — relative effects depending on energy differences between configurations. The framework denies that the *absolute* zero-point sum gravitates, because gravity operates at the logically prior classical level (§6.2).

The Higgs potential. The strongest CC objection is not the zero-point sum but the electroweak Higgs potential: symmetry breaking shifts $V(\phi)$ by $\sim$ (200 GeV)^4, exceeding the observed Λ by $\sim$ 55 orders of magnitude. The framework's response follows from its ontological ordering and the structural/volumetric distinction of §3.2, applied as a precise principle:

What gravitates: Relative energy differences between configurations of the emergent fields. An electron at rest vs. no electron differs by $m_e c^2$ in the classical substratum — there is a classical microstate difference corresponding to this energy difference. Relative differences appear in the classical stress-energy tensor and gravitate. Casimir forces, Lamb shifts, and particle masses all fall in this category.

What does not gravitate: Absolute energy levels assigned to configurations by the emergent description. The Higgs vacuum energy shift $V(\phi_{\min})$ is the energy the emergent QFT assigns to its ground state — the configuration obtained after symmetry breaking. But the "ground state" in the emergent description corresponds to a class of classical microstates whose energy in the classical substratum is $\rho \sim H^2/G$, not $\rho \sim (200\ \text{GeV})^4$. The shift is a property of how the emergent description organizes the microstates (the correspondence), not a property of the microstates themselves (the classical dynamics).

The reconstruction-gauge form of this claim. The distinction between absolute and relative energy is not merely a characterization of the correspondence; it is a theorem of the reconstruction. The marginal phase-locking result ([Main

§3.4]) determines the emergent Hamiltonian from the complete visible transition data only *up to an additive energy shift*: the uniform offset of $\hat{H}$ is a gauge parameter of the marginalization, present in no visible transition statistics at any precision. The embedded observer's data therefore contains no fact about the absolute zero-point offset — the quantity whose Planck-scale magnitude constitutes the cosmological constant problem is, at the emergent level, unobservable in principle rather than merely unmeasured, and the problem's premise (that this offset is a physical energy awaiting gravitational response) attributes physical content to a gauge degree of freedom. The scope of this argument is exactly the scope of the split above: it covers the *uniform* offset only; mode structure that produces relative differences — Casimir forces, Lamb shifts, particle masses — corresponds to classical microstate differences, is determined by the visible data, and gravitates.

The principle is uniform: the zero-point sum, the Higgs vacuum energy, and the QCD condensate energy are all absolute properties of the emergent ground state. The dissolution applies to each for the same reason — gravity operates at the classical level (§6.2), where none of these quantities appear.

Why QCD binding energy gravitates but the quantum vacuum does not. The proton mass is ~99% QCD binding energy — a purely quantum effect. If gravity couples to the classical substratum, how does it "know" about masses generated by emergent quantum dynamics? The answer is that QCD binding energy and vacuum zero-point energy occupy different categories in the structural/volumetric distinction of §3.2. Relative energies within the emergent QFT — energy differences between configurations — correspond to differences in the coupling structure of φ, which the Jacobson derivation reads as curvature. A proton vs. three free quarks corresponds to two distinct classes of classical microstates with different energies in the classical substratum; the binding energy $\sim$ 938 MeV is a *relative* quantity that appears in H_{tot} and gravitates. The absolute baseline — the zero-point energy summed over all modes — corresponds to the total information capacity of the hidden sector (S_{dS}), which is a property of the partition geometry, not of the dynamics, and therefore does not source curvature.

8.4 Assumptions, limitations, and falsifiability

The theorem requires Lemma 1 (independent in Part I; consequence of geometry in Part II), the stochastic-quantum bridge ([Main, §3.1] and Appendix A; two routes reach its transition-statistics layer, neither reaching the operational instrument algebra), and genericity conditions (non-degenerate spectrum, non-vanishing overlaps), which hold for all but a measure-zero set. The P-indivisibility proof uses finiteness via the recurrence argument ($\varphi^N = \text{id}$). The deep-sector cardinality corollary (§3.2) establishes that only $\mathcal{C}_V$ and $\mathcal{C}_B$ need be finite — both guaranteed by holographic entropy bounds [26] independently of the de Sitter finite-dimensionality conjecture — while the deep sector $\mathcal{C}_D$ may be infinite. The accessible-timescale backflow lemma ([Main, §2.3]) provides a second route to P-indivisibility that is independent of $|\mathcal{C}_H|$. The characterization theorem becomes an equivalence on observable timescales ($T_{\mathrm{obs}} \ll \tau_B \sim 10^{17}$ s) — physically indistinguishable from a timeless identity for all experiments within the age of the universe.

The sharp partition and constraint structure. Lemma 2's product decomposition $\Gamma = \Gamma_V \times \Gamma_H$ is the kinematic phase space. In GR, the Hamiltonian and momentum constraints restrict the physical phase space to a submanifold $\Gamma_{\mathrm{phys}} \subset \Gamma_V \times \Gamma_H$ on which interior and exterior data are correlated. P-indivisibility survives on a constraint surface because restoring P-divisibility would require decoupling the sectors entirely, violating the constraints; the constraint surface thus enforces (C1) non-perturbatively. For systems other than the cosmological horizon, the fidelity of the product approximation determines the fidelity of the emergent quantum description.

Planck-scale ordering. The classical-prior ontology may break down at l_p, where "geometric" and "quantum" lose operational distinction. If a deeper theory unifies both, this framework describes the effective macroscopic regime.

Falsifiability. The *theorem* would be falsified by a failure of both the stochastic-quantum correspondence and the Stinespring construction for the class of processes generated here — a possibility excluded by the independent proofs in [Main, §3.1] and Appendix A. The *cosmological application* (the Type II channel with magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}|$) would be falsified by definitive observational demonstration that dark energy deviates from the ν-parameterized Type II form — e.g., a high-confidence detection of strong $w(z)$ variation incompatible with $w = -1 + \mathcal{O}(\nu)$. The magnitude prediction $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$ (conditional on the §7.3 split, not yet established at theorem level) admits a stratified falsification structure: a measurement $|\nu| \gtrsim 10^{-4}$ at high significance would require log-uniform boundary-mode spectral structure beyond §3-§5; a measurement $|\nu| > 10^{-3}$ at high significance would constitute a structural challenge to the magnitude prediction without falsifying the Type II functional form itself. The *dark-sector concordance* (§7.2) would be falsified by detection of dark matter particles accounting for the full structured dark sector, or by evidence that the dark sector's gravitational effects are unrelated to the cosmological boundary geometry.

The falsifiability content is stratified by hierarchy level (per [Structure §2.4]). The Type II running-vacuum functional form is not a discriminating prediction at any level: its local expansion is shared by every smooth $\rho_\Lambda(H)$ (§7.1), so a measured departure from the Type II form would not by itself refute the universality-class claim. The dark-sector concordance is a Tier 2 / Level D $\times$ Levels G1–G2 prediction, conditional on §3.2 — falsification would refute OI's specific universality-class representative without necessarily refuting the Tier 1 partition-geometry results. The magnitude prediction $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$, conditional on the §7.3 split, is the falsifiable content of the dark-energy sector: a measured $|\nu| \sim 10^{-3}$ would refute it at the specific-representative level.

A near-term mechanism-side test. A more tractable falsification target is provided by the recently developed operational distinction between classical and quantum memory in non-Markovian processes [27, 28]. Some non-Markovian quantum dynamics can be simulated using only classical stored information, while others require genuinely quantum memory in the environment, with the dividing line set by entanglement structure of the channel's Choi state. The framework predicts that all *fundamental* non-Markovian dynamics in nature — those arising from the cosmological trace-out rather than from engineered quantum baths — should be classical-memory simulable, because the hidden sublattice is by construction a classical substrate. A physically realized fundamental open-system process whose memory provably exceeds the classical bound would falsify the framework's foundational ontology even without cosmological data. This is a near-term tractable test in a literature where the experimental and theoretical tools are advancing rapidly.

Black-hole horizons under partition-relativity. The framework's mechanism is grounded in the cosmological horizon as the primary partition for any sub-c observer in this universe (§2.1). A black-hole horizon within the cosmological visible sector is a *local* causal boundary: it prevents an external observer from probing the black-hole interior by direct geometric access, but it does not redefine that observer's epistemic partition in the sense of [Main, §1.4]. Matter falling into a distant black hole was already inside the external observer's Γ_V before the merger and remains inside it after; the trace-out that defines the external observer's emergent quantum description is over the cosmological horizon, not over the black-hole horizon. The framework therefore predicts that gravitational wave signatures from binary black-hole mergers follow standard general-relativistic ringdown morphology, with no OI-specific echo, comb, or wall-reflection features. A four-event coherent matched-filter stack on LVK O1–O4 strain data using a dedicated discreteness-scale echo template (with the predicted detector-frame delay $\Delta t = (r_+/c)\ln(r_+/2l_p)$ at the remnant mass and spin) returned a clean null: stacked statistic -0.45, empirical $p = 0.66$, with the previously hypothesized echo amplitude excluded at 5.9σ. The framework's BH physics for external observers is identical to standard GR plus the cosmological-scale dark-sector effects of §7.2–§7.3.

Nested and time-dependent partitions. The primary partition for any sub-c observer in this universe is the cosmological horizon (§2.1). When additional causal boundaries are present — most naturally, a black hole horizon within the cosmological patch — the trace-out machinery extends to a nested formalism, addressed in Appendix A. The nested formalism is a self-consistency exercise: it shows that the framework's emergent description is well-defined when multiple boundaries coexist, and that quantities like the generalized second law and the Page curve emerge naturally. It is *not* a claim that external observers' emergent quantum descriptions acquire BH-horizon-specific features; under partition-relativity (above), the cosmological horizon remains the primary partition. Appendix A resolves the formal nested-partition machinery in three layers.

(i) Consistency of nested trace-outs (Appendix A, Theorems A.1–A.4). Two procedures — tracing out the trans-cosmological region and the black hole interior simultaneously (direct) or sequentially (first cosmological, then black hole) — produce the same CPTP channel, the same emergent Hamiltonian (up to D-gauge), and the same $\hbar$. The key result (Theorem A.2) is that the quantum partial trace within the emergent description agrees with the classical marginalization on the substratum: the emergent QM is self-consistent.

(ii) The generalized second law (Appendix A, Theorem A.6). The sum $S_{\text{matter}} + S_{\text{BH}} + S_{\text{dS}}$ is non-decreasing, proved from strong subadditivity and CPTP monotonicity of the emergent quantum description. Any process that shrinks the black hole (decreasing S_{BH}) transfers information to the visible sector and cosmological boundary, compensating the decrease.

(iii) The Page curve (Appendix A, Theorems A.7–A.9). For an evaporating black hole, the entanglement entropy of the radiation follows $S_{\text{rad}}(t) = \min(n_R(t), n_B(t)) \cdot \ln q$ plus exponentially small corrections, where $n_B(t)$ is the shrinking boundary mode count and $n_R(t) = n_B(0) - n_B(t)$. Unitarity is preserved (bijectivity of φ), the turnover occurs at the Page time $t_P \approx 0.646\, t_{\text{evap}}$ (structural, independent of mass and greybody factors), and the Page value is recovered for all but an exponentially negligible fraction ($\sim e^{-10^{77}}$ for $30M_\odot$) of initial conditions — from bijectivity (cycle decomposition on the finite energy shell) and Popescu-Short-Winter typicality [29], conditional on the scrambling condition H-scramble of Appendix A, which is not established there: uniformity over classical microstates is not uniformity over quantum pure states. The greybody factor is solution-specific.

Laboratory tests of the characterization theorem may be possible in analogue gravity systems where the hidden sector capacity is tunable.

8.5 Universality-class character of the partition-geometry results

§1 separates what follows from partition geometry alone — the energy-shell horizon temperature, the finite horizon counting, and the two-level distinction between the emergent vacuum energy and the effective cosmological constant — from the results that additionally require a named condition: the action scale $\hbar$, the discreteness scale $\epsilon = 2\, l_p$ and the 1/4 coefficient (H-slope with the horizon and frame conditions of §§2–5), the dynamical dissolution of the cosmological-constant problem (the G3 coupling theorem, §6.2), and the running-vacuum form (not presently derived, §7.1). This section states which of these are universality-class results — holding for any substratum meeting structural conditions S1–S4 below — and in what form.

The universality claim, in three parts. Let $\mathcal{S}$ be any finite deterministic substratum (not necessarily OI's cubic lattice) admitting a partition into a visible sector V and hidden sector H at a cosmological-horizon-like causal boundary, satisfying [Main]'s structural conditions C1 (non-trivial coupling), C2 (record persistence — slow bath $\tau_S \ll \tau_B$ the realization), C3 (sufficient memory capacity — $N_H \gg N_V$ the comfortable regime), and C4 (history readback).

(a) Unconditional. The following follow from partition geometry alone:

(U1) *Energy-shell horizon temperature.* The classical inverse-energy temperature $\beta_E = \partial s_H/\partial E$ of the horizon, from the horizon density of states (§3.1).

(U2) *Finite horizon counting.* The boundary entropy is the finite mode count $S = A/\epsilon^2$, one mode per cell of area ϵ^2 (§5).

(U3) *Two-level distinction.* The emergent quantum vacuum energy ($\sim M_{\rm Pl}^4$) and the effective cosmological constant ($\sim H^2/G$) refer to logically distinct objects across two levels of description; the 10^{122} ratio is a compression ratio, not a failure of cancellation (§6.1).

(b) Conditional universality. If, in addition, H-slope, H-frame and the H-Hawking conditions hold at the horizon of $\mathcal{S}$, then with the same numerical content as for OI's substratum:

(C1′) *Emergent action scale.* $\hbar = c^3\epsilon^2/(4G)$, where ϵ is the partition's discreteness scale (§3.2).

(C2′) *Discreteness scale.* $\epsilon = 2\,l_p$, fixed by the simultaneous solution of the action-scale relation and the definition of the Planck length (§4).

(C3′) *Area law.* $S = A/(4\,l_p^2)$, with the 1/4 coefficient the ratio $2\pi/8\pi$ between the Euclidean KMS period and Jacobson's Einstein-equation prefactor (§5).

The conditions are statements about the horizon and the emergent dynamics, not about the cubic lattice, so the conditional is itself a universality-class statement: any substratum in the class that meets them obtains (C1′)–(C3′).

(c) Not universality-class results in the present state. The dynamical dissolution of the cosmological-constant problem is conditional on the G3 coupling theorem, whose form S1–S4 do not fix (§6.2). The Type II running-vacuum functional form is not derived; its magnitude $|\nu| \approx |\nu_{\rm QFT}| \sim (M_{\rm SM}/M_{\rm Pl})^2 \sim 10^{-32}$ is conditional on the §7.3 split (§7.1). The dark-sector phenomenology is Tier 2: it inherits the conditions of §3.2 and the dimensional factor imported in §7.3.

What the derivations actually depend on. Reviewing §3–§7:

- The *ℏ derivation* (§3) uses: the infrared detailed-balance slope H-slope, which identifies the classical inverse-energy temperature with the emergent KMS period (§3.2); the classical horizon temperature from the Jacobson thermodynamic argument (depends only on classical horizon geometry); the Euclidean KMS period of the emergent QFT on the horizon (depends only on the partial-trace structure of the channel and the horizon's discreteness scale); thermal self-consistency requiring agreement between the two (depends on the consistency of emergent QFT with classical thermodynamics at the horizon). None of these depends on OI's specific cubic-lattice structure; the calibration depends on the cosmological-horizon partition, the partial-trace/Stinespring machinery [Main §3], H-slope, and the stated horizon, frame and KMS conditions. C1–C4 are realization diagnostics tracked at the concrete cut ([Main §1.3], §2.2): C1–C3 are verified there, while C4 remains not presently discharged and is not a hypothesis of the calibration. One further dependency is carried by the KMS step of Step 4 and should be stated: the Euclidean-regularity argument presupposes the emergent theory has local boost structure at the horizon (a bifurcate Killing horizon with a well-defined imaginary-time circle). That prerequisite is the local-boost-invariance of the emergent description — the same object whose protection is the framework's Lorentz-sector item in the boost channel ([Main §3.5]): the symmetric-point matching under H-blind ([SM §4.4]) and, conditional on it, the custodial running, which rests on a reported one-loop calculation that is not independently verified ([SM §3.1]). The $\hbar$ derivation is therefore not parallel to the Lorentz-violation accounting but partially downstream of its boost sector: a failure of emergent local boost invariance at the horizon would propagate to the gap equation, whereas the rotational sector does not enter here (spatial anisotropy does not touch the imaginary-time periodicity). The UV-dispersion robustness handled above ($\mathcal{O}((\epsilon\kappa/c^2)^2)$ corrections) is a separate and weaker condition; the boost-structure prerequisite is the load-bearing one and is inherited, not established, in this paper.

Free-level status, and the narrowed form of the conditional. Three results reduce what the boost prerequisite requires. First, the free-level cone is not exact per channel on the normalized $d = 3$ observer branch, and the direction dependence runs opposite to what an axis-based reading suggests. With $\cos\omega = \frac{1}{3}\sum_i \cos(ak_i)$ and the clock calibrated by $\Delta t = a/(c\sqrt{3})$,

$$\frac{\Omega}{ck} = 1 + \frac{a^2k^2}{24}\Big(\tfrac{1}{3} - \sum_i n_i^4\Big) + O(a^4k^4), \qquad n_i = k_i/k,$$

and since $\frac{1}{3} \le \sum_i n_i^4 \le 1$ the leading deviation lies in $[-a^2k^2/36,\, 0]$. It vanishes on the body diagonals, where all three arguments coincide and $\cos\omega = \cos(ak/\sqrt{3})$ exactly, and is maximal along the lattice axes. The (1+1) reduction does not rescue it: zeroing the transverse momenta leaves their two $+1$ contributions in the three-dimensional operator, giving $\cos\omega = (2+\cos ak)/3$. So the KMS prerequisite is not met exactly along axes, and an exact all-momentum cone is available only on a measure-zero set of directions — not on a generic horizon normal.

What replaces it is dispersive robustness rather than exactness. Under the Schützhold–Unruh result the Hawking temperature depends on the product of the low-energy phase and group velocities; writing $\Omega = ck[1 + q(\hat{n})a^2k^2 + \cdots]$ gives $v_{\rm ph}v_g/c^2 = 1{+}4q\,a^2k^2{+}O(a^4k^4)$ with $q = \frac{1}{24}(\frac{1}{3}{-}\sum_i n_i^4)$, hence $-a^2k^2/9 \le v_{\rm ph}v_g/c^2 - 1 \le 0$ and

$$\Big|\frac{\Delta T_H}{T_H}\Big| \;\lesssim\; \frac{1}{9}\Big(\frac{\epsilon\kappa}{c^2}\Big)^2.$$

This holds under named conditions, not universally — call them **H-Hawking**: (i) the preferred microscopic frame becomes the local freely falling horizon frame; (ii) cutoff-scale outgoing modes are in their freely falling ground state; (iii) cutoff dynamics is adiabatic relative to κ; (iv) the observer dispersion is the normalized stable infrared branch; (v) the horizon geometry varies smoothly on scales $\gg \epsilon$. Conditions (iv) and (v) are discharged by §4.1 and the scale separation; (i) and (ii) are open. The dispersive corrections vanish as $k \to 0$, so $\hbar = c^3\epsilon^2/(4G)$ and the 1/4 coefficient survive as an infrared thermal-calibration statement, with Planck-suppressed departures at finite frequency — not as an exact microscopic KMS theorem. Second, the off-axis and isotropization corrections belong to the $\mathcal{O}((\epsilon\kappa/c^2)^2)$ class already carried above; numerically, $(\epsilon\kappa/c^2)^2 \approx 6 \times 10^{-122}$ at the cosmological horizon (the same dimensionless ratio $(H\,l_p)^2$ that constitutes the §6 discrepancy — a scale identity, recorded as such) and $\approx 3 \times 10^{-77}$ for a solar-mass horizon. Third, the conditional accordingly narrows to its interacting core: radiative velocity-splitting between emergent species — the standard protection problem of emergent Lorentz invariance, in which loop corrections on a discrete substrate generically split species velocities at $\mathcal{O}(\alpha)$ unless a mechanism forbids it. This core is *identical* to the open Lorentz-sector boost item ([Main §3.5]; [Substratum §6.4]) — one open item, not two. The protection has a matching part and a running part. At the symmetric point the six carrier components are dynamically identical, so the substratum dynamics and a component-blind trace-out commute with the signed permutations of the components, whose six-dimensional representation is irreducible; under H-blind the complete quadratic response is custodially scalar and every massless block shares one cone at the matching scale ([SM §4.4], Proposition 4b). Conditional on that matching, the running is the custodial normalization of the Lorentz-sector accounting. The running part rests on a calculation reported in the companion paper ([SM §3.1]) whose driver and numerical output are not available for verification: a one-loop lattice calculation of the boost-projected self-energy curvature reported that, with the universal tree-level cone absorbed, the sector-differential coefficient obeys a quadratic-with-logarithm decoupling law $\delta c \sim \Delta C_2\,(\alpha_0/4\pi)(M_X/M_{\mathrm{Pl}})^2[\,c_L \ln(M_{\mathrm{Pl}}/M_X) + c_0\,]$ in the cross-charged condensate mass M_X, rather than the non-decoupling constant of the generic fine-tuning problem. What is certified is the symmetry statement — the signed-permutation symmetry of the carrier components forbids a condensate-independent differential counterterm at the regulator level — not the reported form. What remains of the conditional, under H-blind, is therefore not the existence of a protection but its quantitative closure: the reported decoupling form itself, pending an independent reproduction; the vertex-weighted one-loop point value of the normalization (structurally bounded, $\Delta C_2 \in [0.58, 2.08]$); the non-logarithmic constant of that loop (the substratum's own nonlinearity is component-local and contributes only to the universal cone, [SM §3.1]); and the condensate-gap determination of M_X. On the reported form and constants the allowed window would be $M_X \lesssim 10^8$ GeV on the worst-case ΔC_2, $\sim 4 \times 10^7$ GeV with the reported kernel constants, against the tightest multi-species bounds, with a phenomenological floor at 10^3–10^6 GeV set by leptoquark bounds — the cross-charged sector is baryon-safe at the gauge level, so proton decay does not close the window from below ([SM §3.1]) — an illustrative one-and-a-half-to-four-and-a-half-decade window carrying the provenance of that calculation. The $\hbar$ chain's boost conditional thus has the same two parts the Standard-Model sector carries: a certified matching theorem and a reported running calculation, the latter the one item shared across both papers and the one still to be verified.

- The *discreteness scale fixing* (§4) is purely algebraic given the $\hbar$ relation: rearranging it gives $\epsilon^2 = 4\hbar G/c^3 = 4l_p^2$. It inherits that relation's conditions and nothing else.
- The *area law derivation* (§5) uses: the boundary mode count $N_{\text{modes}} = A/\epsilon^2$ (depends only on the boundary's geometric area and the discreteness scale; unconditional); the identification of mode count with entropy via the partial-trace structure; and, for the 1/4 normalization, the calibration of §3.2, whose conditions it inherits.
- The *CC dissolution* (§6) uses: the structural distinction between the emergent quantum vacuum energy (a property of the emergent QFT description) and the effective cosmological constant (a property of the classical substratum's gravitational dynamics). The distinction is structural — it follows from the two-level description architecture of any embedded-observer system, not from OI's specific substratum. The dynamical dissolution — that the observer's inability to see a vacuum offset entails its gravitational decoupling — is a coupling statement conditional on the G3 theorem (§6.2).
- The *running-vacuum form* (§7.1) uses: boundary entropy as bookkeeping degree of freedom in the gap equation; the Clausius relation $\delta Q = T\delta S$ at the horizon; the fact that boundary entropy is finite and depends on H via $A \propto H^{-2}$. What these yield is state dependence on cosmological geometry whose smooth late-time normal form contains an $H^2 - H_0^2$ term; they do not single out the Type II law, whose local expansion every smooth $\rho_\Lambda(H)$ shares.

Sufficient structural conditions. The unconditional results (U1)–(U3) require the following structural conditions on the substratum:

(*S1*) *Cosmological-horizon partition.* The substratum admits a partition V at a causal horizon with timescale $\tau_B \sim H^{-1}$ and information capacity $S_{\text{dS}} \sim 10^{122}$.

(*S2*) *C1–C3 satisfied.* The partition has substantial coupling (C1), slow-bath structure (C2), and high capacity (C3), per [Main §3].

(*S3*) *Finite discreteness.* The substratum has a finite discreteness scale ϵ that admits identification with the boundary's mode-counting unit. [Main §3]'s Lemma 1 establishes finite-dimensionality from C1–C4; here we additionally require that the discreteness scale appears in the boundary mode count.

(*S4*) *Stinespring/partial-trace structure.* The visible-sector dynamics is generated by partial trace of substratum-level dynamics over the hidden sector, with the trace producing a CPTP channel admitting Stinespring dilation [Main §3.2].

These conditions are weaker than OI's full structural conditions A1–A6: A1 (finiteness) implies S3 with finite ϵ; A6 (background independence) is required for the SM gauge-group derivation but not for the partition-geometry results; A2-A5 (determinism, bounded coupling, center independence, linearity) are required for OI's specific universality-class representative but not for the universality-class-level claims. A substratum satisfying S1-S4 but violating some of A2-A5 (e.g., a substratum with stochastic dynamics, or with nonlinear evolution) would still produce (U1)–(U3), and — under H-slope, H-frame and H-Hawking at its horizon — $\hbar = c^3\epsilon^2/(4G)$, $\epsilon = 2\,l_p$ and the area law with its 1/4 coefficient, provided the partial-trace machinery and thermal self-consistency hold.

Theorem-level statement. The universality claim can be stated at the level of the framework's hierarchical structure (per [Structure §2]):

Theorem (universality of the partition-geometry results). For any substratum $\mathcal{S}$ satisfying S1–S4, (U1)–(U3) follow with the same content as derived in this paper for OI's specific substratum; if in addition H-slope, H-frame and the H-Hawking conditions hold at its horizon, so do (C1′)–(C3′) with the same numerical content.

The proof is the §3–§6 derivations themselves, with the observation that each step depends only on S1–S4 and the named conditions, not on OI's specific structural conditions A2–A5. A formal proof at the level of explicit theorem-style presentation has not been written out in this paper; the present section identifies the structural conditions required at each step, and the claim is established at that level.

Empirical implication. The universality-class results have limited direct empirical content. GW250114 confirms Hawking's area theorem — classical horizon dynamics that any framework retaining general relativity, at any hierarchy level, preserves; it does not measure entropy and therefore does not test the 1/4 coefficient at any level. Conversely, the dark-sector concordance (§7.2), the MOND-like phenomenology (§7.3), and the rotation-curve crossover (§7.3) depend on OI's specific gap-equation solution — inheriting its conditions and the dimensional factor imported in §7.3 — and are Level D $\times$ Levels G1–G2 predictions; their empirical confirmation would support OI's specific universality-class representative.

The stratification clarifies what would falsify what: a measured violation of the area theorem would challenge the classical horizon dynamics the framework presupposes; a direct empirical test of the 1/4 coefficient, if one becomes available, would test the universality-class claim; a measured deviation from the dark-sector phenomenology would falsify OI's specific representative without necessarily refuting the universality class. The framework's empirical content thus operates at multiple hierarchy levels, with falsifiability stratified accordingly.

8.6 Relation to other entropic and emergent-gravity programs

The thermodynamic derivation of §3–§5 places this framework within a broader program that treats gravity as entropic or emergent rather than fundamental. Jacobson's equation-of-state derivation [1] is the direct antecedent used here; Verlinde's entropic-force account [50] and Padmanabhan's thermodynamic-aspects program [51] reach Einstein-level dynamics from horizon thermodynamics and holographic equipartition, sharing with this framework both the entropic premise and a holographic/boundary counting of degrees of freedom. More recently, Bianconi's *gravity from entropy* [52] derives modified Einstein equations from an action built on the quantum relative entropy between the spacetime metric and a matter-induced metric, reproducing the low-energy Einstein equations and an emergent positive cosmological constant *without* invoking horizon entropy or holography. The existence of this last route is worth stating plainly: "gravity from entropy" does not uniquely select the horizon/holographic architecture used here, so this framework's Jacobson–Bekenstein route is a specific commitment among alternatives rather than the only entropic path to Einstein dynamics. What this framework adds to the landscape is not the entropic premise — which it shares — but the discrete substratum beneath the thermodynamics ([Main]; [SM]) and the quantitative consequences specific to it (the fixed 1/4 coefficient, $\epsilon = 2\,l_p$, and the running-vacuum coefficient of §7).

A qualitative convergence is worth noting without overstating it: entropic and emergent-gravity programs generically yield a *dynamical* dark-energy sector rather than a bare constant Λ — the running vacuum of §7.1 here, and the always-positive G-field term of [52]. This convergence is at the level of the qualitative form (dark energy as a derived, non-constant feature) only; it does not transfer magnitude, sign, or mechanism between programs, which differ substantially. The sign and $\sim 10^{-32}$ magnitude of the running-vacuum coefficient predicted here (§7.1) are specific to this construction — not corroborated by, nor inherited from, the neighbouring programs.

8.7 Kinematic emergence of geometry from the substratum

The derivations of §3–§5, and the four convergent necessity routes of §3.2 (*An independent necessity statement*) (thermodynamic, gravitational closure, Lovelock, spin-2 bootstrap), all take a smooth Lorentzian metric as given and fix its *dynamics*. A logically prior question is *kinematic*: does the discrete substratum ([SM]) actually yield a smooth pseudo-Riemannian geometry, with the right dimension and curvature, in the first place? §8.2 treats the continuum limit as a posit. This subsection reports a research program that makes that posit concrete at the weak-field level, by adapting standard discrete-geometry tools — the spectral dimension [53] and the Ollivier–Ricci curvature [54] — to the substratum graph. The results are consistency checks reproducing known Newtonian physics; they add no new empirical content, and the program's honest boundary is stated at the end.

Dimension and curvature tool. The cubic substratum has spectral dimension $d_s \to 3$ (heat-kernel return probability), which with the emergent time direction is the target $3+1$. Encoding a chosen emergent metric into the graph through the framework's own dispersion relation (bond weights fixed by the local light-cone speed) and computing curvature two independent ways — combinatorially (Ollivier–Ricci) and analytically (the Riemannian curvature of that metric) — the two agree (correlation ≈ 0.97; $\kappa \approx \frac{1}{2}K$ as expected in two dimensions). So the discrete-curvature tool faithfully computes the framework's *own* emergent-metric geometry rather than an unrelated combinatorial quantity.

Weak-field gravity at the fixed G. In the weak field the Newtonian force is carried by the emergent-time-rate component g_{00} (the local substratum clock rate), not by spatial curvature; g_{00}-only geodesics reproduce closed Keplerian orbits obeying Kepler's third law. The Newtonian field itself follows from holographic equipartition using the framework's own constants ($\eta = 1/\epsilon^2$; $\hbar = c^3\epsilon^2/4G$) together with the classical horizon temperature of §3.1: one obtains $a = GM/r^2$ with the framework's *fixed* G — fixed through $\epsilon = 2\,l_p$, so inheriting the conditions of §3.2 — and the Poisson equation $\nabla^2\Phi = 4\pi G\rho$ for extended sources. The two ingredients that are heuristic in the usual entropic-gravity argument are, here, consequences of the framework's structure: equipartition is the equipartition *theorem* because the emergent boundary modes are harmonic (the [SM] wave equation), and the entropic force $F = T\,dS/dx$ is a theorem of the reversible, energy-conserving microcanonical dynamics (Lemma 3). Notably the *field* derivation requires no entropy-displacement (Bekenstein) input; the force on a test mass is field $\times$ emergent inertia.

Honest boundary. These are weak-field, consistency-level results: they reproduce Newtonian gravity at the framework's own G, not a new prediction. Three items remain genuinely open. (i) The equipartition degree-of-freedom normalization $N_{\mathrm{dof}} = A/l_p^2 = 4\,S_{\mathrm{BH}}$ — the standard relation in all entropic gravity — is here *adopted*, not derived from the substratum's microscopic component count ($K = 2d = 6$ per site; one boundary-crossing component per cell); reconciling these counts is unsolved. (ii) The full nonlinear $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$, and the Lorentzian/3+1 generalization of the (Euclidean, conformal) curvature check, are not established. (iii) For the optional stronger claim that inertia itself is entropic, the entropy/Compton-phase identification remains an assumption. The program's value is thus (a) showing the discrete-curvature approach targets the framework's real geometry and (b) turning §8.2's posited continuum limit into a computation that reaches the Newtonian checkpoint with the remaining steps explicitly marked — not a claim to have derived general relativity from the substratum.

9. Conclusion

The cosmological horizon realizes the embedded-observation framework of [Main], and the consequences for the gravitational sector are sharp. The emergent action scale $\hbar = c^3(2\,l_p)^2/(4G)$ is determined by thermal self-consistency between the classical horizon temperature and the Euclidean periodicity of the emergent QFT, as a gap equation with ϵ the one free geometric input (§4), conditional on H-slope (§3.2). The Bekenstein-Hawking entropy $S = A/(4l_p^2)$ follows by direct mode counting, with the 1/4 coefficient fixed as the ratio between the Euclidean KMS period and the Jacobson prefactor — a structural constant rather than an empirical fit, carrying the same condition together with the horizon and frame conditions of §2 and §8.5. The GW250114 observation of September 2025 confirms Hawking's area theorem at 99.999% confidence — classical horizon dynamics the framework preserves; the 1/4 coefficient has no direct empirical test. The cosmological constant problem is addressed at Level G1 — dissolved at the descriptive-levels register, with the covariant Level-G3 target equation (§7) the decisive open: the 10^{122} ratio between the quantum vacuum energy and the effective Λ is the compression ratio between two levels of description that refer to incommensurable objects, not a failure of cancellation.

The empirical payoff is the set of predictions in §7. The dark-energy content is geometry-dependent bookkeeping whose smooth late-time expansion contains an $H^2 - H_0^2$ term shared by every smooth model, read in the Type II running-vacuum channel — a structural reading, not a derived law (§7.1): consistent with the Bertini et al. (2025) direct fit to DESI DR2 + DES-Y5 + Planck 2018 [15] (which reports ν consistent with zero at 2σ), and decisively testable by DESI Year 5 and subsequent stage-IV surveys. The coefficient's magnitude is the self-consistent emergent-QFT value $|\nu_{\mathrm{QFT}}| \sim 10^{-32}$ (conditional on the §7.3 split, not yet established at theorem level), observationally indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3.1 and §5 — spatial location alone, with no inherent frequency assignment — to be well below $|\nu_{\mathrm{QFT}}|$ under the natural 2D distribution. The framework's structural prediction is therefore the Type II channel with $|\nu| \approx |\nu_{\mathrm{QFT}}|$ and $w(z) \approx -1$. The dark-sector corollary explains ~95% of the gravitational budget of the universe without new fundamental particles, and the absence of direct dark-matter detection at the LZ and XENONnT experiments is interpreted within the framework as a structural consequence rather than a null result to be explained away. The MOND phenomenology emerges — at the G1 register, with the covariant completion (§7) the standing conditional — from entropy displacement at the cosmological boundary with $a_0 = cH/6$ (the factor 1/6 imported from Verlinde), reproducing the baryonic Tully-Fisher relation and predicting Keplerian decline at $r > r_M$ — the latter confirmed by the Jiao et al. (2023) detection of a Keplerian decline in the Milky Way's own rotation curve at ~ 17 kpc. The $H(z)$ scaling of a_0 predicts systematically more baryon-dominated rotation curves at high redshift, consistent with Genzel et al.'s detection of declining outer velocities at $z = 0.9$–2.4 and testable at higher precision by JWST and ELT kinematics.

The foundational framework developed in [Main] and the cubic-lattice realization developed in [SM] are complementary to the present work: [Main] establishes the equivalence between embedded observation and quantum mechanics, the present paper applies it to the cosmological horizon, and the SM paper applies it to the microscopic lattice substratum. The reconstruction theorem and the synthesis claim — that quantum mechanics, general relativity, and the arrow of time descend as three projections of the same deterministic (S, φ) — are developed in [Substratum]. Together, this four-paper synthesis ([Main], [SM], [GR], [Substratum]) establishes that the Standard Model, general relativity, and the arrow of time are three faces of a single finite deterministic construction, with quantum mechanics emerging as the unique self-consistent description for any observer bounded by a causal horizon. The framework's hierarchical structural realism and its relationship to other unification programs are developed in the fifth core paper [Structure].

Appendix A: Nested Partitions

This appendix develops the nested-partition formalism for the framework, establishing the self-consistency of multiple coexisting causal boundaries (typically a black hole horizon inside a cosmological horizon) and deriving the generalized second law and the Page curve as consequences.

A.1 Setup

Let $\mathcal{C}_V$, $\mathcal{C}_B$, $\mathcal{C}_D$ be finite configuration spaces with $|\mathcal{C}_V|, |\mathcal{C}_B| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D \to \mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D$ be a bijection. The physical picture: V is the region between a black hole horizon and the cosmological horizon, B is the black hole interior, D is the trans-cosmological region. Define $W = V \cup B$ (the cosmological interior).

Two procedures produce a quantum description on V:

Direct: Trace out $B \cup D$ in one step, marginalizing over $\mathcal{C}_B \times \mathcal{C}_D$.

Sequential: (Stage 1) Trace out D, producing emergent QM on W. (Stage 2) Within the emergent QM on W, trace out B, producing a further-reduced description on V.

A.2 Classical Associativity

Theorem A.1. *The direct and sequential procedures produce the same transition probabilities on $\mathcal{C}_V$.*

Proof. The direct transition matrix is:

$$T_{ij}^{\mathrm{dir}}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{b \in \mathcal{C}_B} \sum_{d \in \mathcal{C}_D} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))]$$

The sequential procedure marginalizes over D (Stage 1) then over B (Stage 2). Composing:

$$T_{ij}^{\mathrm{seq}}(t) = \frac{1}{|\mathcal{C}_B|} \sum_{b} \frac{1}{|\mathcal{C}_D|} \sum_{d} \delta_{x_j}[\pi_V(\varphi_t(x_i, b, d))] = T_{ij}^{\mathrm{dir}}(t)$$

The sums commute (Fubini on a finite product). □

A.3 Quantum-Classical Consistency

Theorem A.2 (Quantum-classical consistency of nested trace-outs). *Let Φ_V^{dir} and Φ_V^{seq} be the CPTP channels on $\mathcal{H}_V$ produced by the direct and sequential procedures respectively. Then $\Phi_V^{dir} = \Phi_V^{seq}$.*

Proof. By the Stinespring dilation theorem [30, 31], φ defines a unitary U_φ on $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$. The direct channel is $\Phi_V^{\mathrm{dir}}(\rho_V) = \mathrm{Tr}_{BD}[U_\varphi\,(\rho_V \otimes \rho_B \otimes \rho_D)\,U_\varphi^\dagger]$. The sequential channel is $\Phi_V^{\mathrm{seq}}(\rho_V) = \mathrm{Tr}_B[\mathrm{Tr}_D[U_\varphi\,(\rho_V \otimes \rho_B \otimes \rho_D)\,U_\varphi^\dagger]]$. On finite-dimensional tensor products, $\mathrm{Tr}_B \circ \mathrm{Tr}_D = \mathrm{Tr}_{BD}$, giving $\Phi_V^{\mathrm{seq}} = \Phi_V^{\mathrm{dir}}$. □

A.4 Hamiltonian and $\hbar$ Consistency

Corollary A.3 (Hamiltonian consistency). *The direct and sequential procedures yield the same emergent dynamics in two regimes. At the reconstruction level, with no assumption beyond the §3.3 hypotheses: Theorem A.2 gives equal transition data, so under the §3.3 hypotheses $\hat{H}_V^{dir} = D\,\hat{H}_V^{seq}\,D^\dagger + E_0$ or $\hat{H}_V^{dir} = -D\,\bar{\hat{H}}_V^{seq}\,D^\dagger + E_0$, for a time-independent diagonal unitary D and a real E_0. If moreover the common channel family of Theorem A.2 is exactly unitary, $\Phi_t = \mathrm{Ad}_{U(t)}$ for both procedures, then $\mathrm{Ad}_{U_{dir}(t)} = \mathrm{Ad}_{U_{seq}(t)}$ forces $U_{dir}(t) = e^{i\theta(t)}\,U_{seq}(t)$ with $\theta(t) = -E_0 t$, and $\hat{H}_V^{dir} = \hat{H}_V^{seq} + E_0$ — no antiunitary branch and no nontrivial D.*

Proof. Theorem A.2 gives $\Phi_V^{\mathrm{dir}} = \Phi_V^{\mathrm{seq}}$, hence $T_{ij}^{\mathrm{dir}}(t) = T_{ij}^{\mathrm{seq}}(t)$ for all t (Born rule applied to diagonal elements), and the two-branch theorem (§3.3) gives the reconstructed form. In the exact-unitary regime, equality of the conjugation actions at each time makes $U_{\mathrm{seq}}(t)^\dagger U_{\mathrm{dir}}(t)$ commute with every visible operator, hence a unimodular scalar (kernel: `ad_eq_scalar`); the resulting identity of spectral phase families $e^{-iE_a t} = c(t)\,e^{-iE'_a t}$ forces the eigenvalue differences $E_a - E'_a$ to be mode-independent (kernel: `phase_families_shift`), so $\theta(t) = -E_0 t$ and the generators differ by the scalar shift alone. Exact channel equality in the same visible algebra carries the coherence information that transition probabilities discard, which is why the antiunitary branch and the diagonal gauge both disappear; a general CPTP reduced dynamics, by contrast, has no uniquely defined Hamiltonian part, and the reconstruction-level clause claims none. □

Corollary A.4 ($\hbar$ universality). *Both procedures yield the same emergent action scale $\hbar = c^3\epsilon^2/(4G)$.*

Proof. $\hbar$ depends only on local boundary quantities c, G, ϵ (§3.2, Steps 2–3), not on which horizon defines the partition or how many trace-outs are performed. □

A.5 Additive Dissipators

Theorem A.5 (Additive dissipators). *Let the classical Hamiltonian be spatially local with coupling chain $V \leftrightarrow B \leftrightarrow D$ (no direct V-D coupling). Let τ_B^D be the D-sector timescale and τ_B^B the B-sector timescale. On observation timescales $t \ll \min(\tau_B^D, \tau_B^B)$, the correction to the frozen-limit description on $\mathcal{H}_V$ — an absorbable Hermitian drift at first order in each timescale ratio, with genuine dissipation entering at second order ([Main, §3.2]) — decomposes as:*

$$\mathcal{D}_V = \mathcal{D}_V^{(B)} + \mathcal{D}_V^{(D)} + \mathcal{R}$$

where $\|\mathcal{D}_V^{(B)}\| = \mathcal{O}(\tau_S/\tau_B^B)$, $\|\mathcal{D}_V^{(D)}\| = \mathcal{O}(\tau_S/\tau_B^D)$, and the cross term satisfies $\|\mathcal{R}\| = \mathcal{O}(\tau_S^2/(\tau_B^B \cdot \tau_B^D))$.

Proof. The proof applies the boundary-only dependence lemma (§3.2) twice, tracking the cross term at each stage.

(i) *First application: freeze D.* Apply the boundary-only dependence lemma to the partition (W, D) where $W = V \cup B$. By spatial locality, $H_{\text{tot}} = H_V + H_B + H_D + H_{VB} + H_{BD}$ with no direct V-D coupling. The D-sector is frozen on timescales $t \ll \tau_B^D$, giving:

$$T_{ij}^V(t) = \frac{1}{|\mathcal{C}_B|} \sum_{b \in \mathcal{C}_B} \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i, b))] + \mathcal{O}(t/\tau_B^D)$$

where φ_t^{VB} is the flow restricted to $V \times B$ with D frozen. The $\mathcal{O}(t/\tau_B^D)$ correction arises from D's back-reaction on B through H_{BD}.

(ii) *Dissipator from the frozen-D dynamics.* The frozen-D flow φ_t^{VB} is a bijection on $\mathcal{C}_V \times \mathcal{C}_B$ — the standard Stinespring setup with B as the hidden sector. The emergent dynamics on V is approximately unitary, with total departure from the frozen-B description $\mathcal{D}_V^{(B)} \sim \mathcal{O}(\tau_S/\tau_B^B)$, of which the non-absorbable (dissipative) part enters at second order ([Main, §3.2]).

(iii) *Decomposition of the D-correction.* The $\mathcal{O}(t/\tau_B^D)$ correction from step (i) has two components:

Direct: D's back-reaction shifts the boundary modes of B by $\delta b \sim \mathcal{O}(t/\tau_B^D)$. This propagates through H_{VB} to the visible sector, contributing $\mathcal{D}_V^{(D)} \sim \mathcal{O}(\tau_S/\tau_B^D)$ to the total correction (same first-order/second-order split).

Cross: The shift δb also modifies the effective B-state against which $\mathcal{D}_V^{(B)}$ is computed. The dissipator $\mathcal{D}_V^{(B)}$ depends on the B-state through the Liouville average over b. The D-induced shift changes this average by $\mathcal{O}(t/\tau_B^D)$, so:

$$\|\mathcal{R}\| = \left\|\frac{\partial \mathcal{D}_V^{(B)}}{\partial b}\right\| \cdot \|\delta b\| \leq \|\mathcal{D}_V^{(B)}\| \cdot \mathcal{O}(t/\tau_B^D) = \mathcal{O}\left(\frac{\tau_S}{\tau_B^B} \cdot \frac{\tau_S}{\tau_B^D}\right) = \mathcal{O}\left(\frac{\tau_S^2}{\tau_B^B \cdot \tau_B^D}\right)$$

The inequality uses the fact that $\mathcal{D}_V^{(B)}$ is a continuous function of the B-state distribution and the shift is first-order in t/τ_B^D. $\square$

Bound for the cosmological + stellar black hole case. For a $30M_\odot$ black hole inside the cosmological horizon: $\tau_S \sim 10^{-15}$ s, $\tau_B^B \sim \tau_{\text{scr}} \sim \beta \log S_{\text{BH}} \sim 10^{-3}$ s, $\tau_B^D \sim 1/H \sim 10^{17}$ s. The three terms:

$$\|\mathcal{D}_V^{(B)}\| \sim 10^{-12}, \qquad \|\mathcal{D}_V^{(D)}\| \sim 10^{-32}, \qquad \|\mathcal{R}\| \sim 10^{-44}$$

The cross term is negligible by 12 orders of magnitude relative to the smaller of the two primary corrections.

A.6 The Generalized Second Law

The Bekenstein-Hawking entropy of a horizon is the information content of the corresponding partition boundary (§5): $S_{\text{BH}} = A_{\text{BH}}/(4\,l_p^2)$ modes at the black hole horizon, $S_{\text{dS}} = A_{\text{dS}}/(4\,l_p^2)$ modes at the cosmological horizon. The matter entropy is the von Neumann entropy of the visible-sector state: $S_{\text{matter}} = -\text{Tr}[\rho_V \log \rho_V]$.

Theorem A.6 (Generalized second law). *For any process described by the CPTP channel Φ_V on $\mathcal{H}_V$:*

$$S_{\text{matter}}(t) + S_{\text{BH}}(t) + S_{\text{dS}}(t) \geq S_{\text{matter}}(0) + S_{\text{BH}}(0) + S_{\text{dS}}(0)$$

Proof. The total system (S, φ) evolves unitarily at the fundamental level, so the total fine-grained entropy $S_{\text{total}} = \log|S|$ is constant. The three terms S_{matter}, S_{BH}, S_{dS} are the observer's ignorance about three sectors: V, B, D respectively.

(i) *Subadditivity.* For the tripartite system $\mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$ in a pure state $|\Psi\rangle$ (the total state is deterministic, hence pure conditioned on the initial state), the strong subadditivity of von Neumann entropy [31] gives:

$$S(\rho_{VB}) + S(\rho_{BD}) \geq S(\rho_B) + S(\rho_{VBD})$$

Since $\rho_{VBD} = |\Psi\rangle\langle\Psi|$ is pure, $S(\rho_{VBD}) = 0$. Thus $S(\rho_{VB}) + S(\rho_{BD}) \geq S(\rho_B)$.

(ii) *Identification.* The observer's matter entropy is $S_{\text{matter}} = S(\rho_V)$. By purification on the tripartite system: $S(\rho_V) = S(\rho_{BD})$. The black hole entropy is $S_{\text{BH}} = S(\rho_B)$ restricted to the boundary modes (the number of B-modes coupled across the black hole horizon). The cosmological entropy is $S_{\text{dS}} = S(\rho_D)$ restricted to the boundary modes.

(iii) *Monotonicity and compensation.* The hidden-sector prior is uniform (Liouville marginalization), so the marginal channel is unital: $\Phi_V(I_V) = \text{Tr}_{BD}[U_\varphi\,(I_V \otimes \rho_B \otimes \rho_D)\,U_\varphi^\dagger] = I_V$ for maximally mixed ρ_B, ρ_D. Unital channels are entropy-non-decreasing (their outputs are majorized by their inputs [31]), so $S(\rho_V)$ is non-decreasing under the emergent dynamics. (Relative-entropy contraction alone would not suffice — general CPTP maps can decrease entropy; the uniform prior is load-bearing here.) Any process that decreases S_{BH} (shrinking the black hole) transfers information from B to $V \cup D$. By the unitarity of φ, the total mutual information $I(V : B : D)$ is conserved, so the decrease in S_{BH} is compensated by an increase in $S_{\text{matter}} + S_{\text{dS}}$. $\square$

A.7 The Page Curve

Setup. A black hole with initial horizon area A_0 evaporates. The partition boundary at the black hole horizon moves: modes transition from B (interior) to R (radiation, a subsystem of V) as the horizon shrinks. Define $n_B(t) = A_{\text{BH}}(t)/(4\,l_p^2)$ (the number of boundary modes at the BH horizon) and $n_R(t) = n_B(0) - n_B(t)$ (the number of emitted radiation modes). Conservation: $n_B(t) + n_R(t) = n_B(0) = S_{\text{BH}}(0)$.

The trans-cosmological sector D is frozen on evaporation timescales ($\tau_{\text{evap}} \ll \tau_B^D$), so by Theorem A.5 the relevant dynamics is the bipartite system $B \cup R$ with D decoupled at leading order.

Theorem A.7 (Effective bipartite pure state). *On timescales $t \ll \tau_B^D$, the joint state of B and R is effectively pure: $\rho_{BR} = |\psi(t)\rangle\langle\psi(t)| + \mathcal{O}(\tau_S/\tau_B^D)$, where $|\psi(t)\rangle \in \mathcal{H}_B(t) \otimes \mathcal{H}_R(t)$.*

Proof. The total system evolves unitarily under U_φ (§A.3). The total state $|\Psi\rangle \in \mathcal{H}_V \otimes \mathcal{H}_B \otimes \mathcal{H}_D$ is pure (deterministic evolution from a definite initial state). By the boundary-only dependence lemma, the D-sector is frozen: ρ_D is approximately constant, and $|\Psi\rangle \approx |\psi_{VB}\rangle \otimes |\chi_D\rangle + \mathcal{O}(\tau_S/\tau_B^D)$. Restricting to the $B \cup R$ subsystem (where $R \subset V$ is the radiation): $\rho_{BR} \approx |\psi\rangle\langle\psi|$ is pure at leading order. The correction is $\mathcal{O}(\tau_S/\tau_B^D) \sim 10^{-32}$. $\square$

Theorem A.8 (Cycle typicality, conditional on H-scramble). *For the dynamics φ on the energy shell $\Sigma_E \subset \mathcal{H}_B \otimes \mathcal{H}_R$, the time-averaged entanglement entropy* *of the radiation equals the Page value [32]:*

$$\overline{S(\rho_R)} = S_{\mathrm{Page}}(d_B, d_R) + \mathcal{O}(\epsilon)$$

for all initial conditions except a fraction $\leq (\ln d_{\mathrm{min}}/\epsilon)\, e^{-c\, d_{\mathrm{min}}\, \epsilon^2}$*, where* $d_B = |\mathcal{C}_B(t)|$*,* $d_R = |\mathcal{C}_R(t)|$*,* $d_{\mathrm{min}} = \min(d_B, d_R)$*, and:*

$$S_{\mathrm{Page}}(d_B, d_R) = \ln(\min(d_B, d_R)) - \frac{\min(d_B, d_R)}{2\max(d_B, d_R)}$$

Proof. The bijection φ conserves energy (the wave equation is Hamiltonian), so φ restricts to a bijection on each energy shell Σ_E. Since Σ_E is finite, this restricted bijection decomposes into disjoint cycles $C_1, \ldots, C_k$ with lengths $L_1, \ldots, L_k$ summing to $|\Sigma_E|$. On each cycle C_j, the time average of any observable f is exact:

$$\overline{f}_{C_j} = \frac{1}{L_j} \sum_{s \in C_j} f(s)$$

By Popescu-Short-Winter [29], the set of atypical states — those with $|S(\mathrm{Tr}_B |\psi\rangle\langle\psi|) - S_{\mathrm{Page}}| > \epsilon$ — has measure at most $\eta = e^{-c\, d_{\mathrm{min}}\, \epsilon^2}$, where $d_{\mathrm{min}} = \min(d_B, d_R)$ and $c > 0$ is a universal constant.

The measures are different, and the step below requires a hypothesis (H-scramble). [29] bounds a Haar fraction on the unit sphere of a constrained Hilbert subspace. It does not bound the fraction of an arbitrary discrete set of substratum configurations, nor of a permutation orbit, so "the total number of atypical states is at most $\eta|\Sigma_E|$" does not follow from [29] alone. The gap is not technical. Let the shell be the computational product basis $\{|b\rangle \otimes |r\rangle\}$ at a common energy and let φ be one long cycle permuting it: φ is finite and bijective, conserves the shell, its orbit can contain every basis configuration, its counting measure is exactly uniform, and yet every visited state is a product state with $S(\rho_R) = 0$, while the Page value is $\ln d_{\mathrm{min}} - d_{\mathrm{min}}/(2d_{\mathrm{max}}) > 0$ for $d_B, d_R > 1$ — a gap of 0.886 nats already at $d_B = d_R = 4$. So finite bijectivity, energy conservation and uniform cycle counting together do not give Page typicality.

What the argument needs is a separate condition, **H-scramble**: the orbit ensemble equidistributes over quantum pure states on the shell, in the sense of an approximate unitary design or comparable scrambling property. Under H-scramble the counting step below goes through with $\eta|\Sigma_E|$ as written. H-scramble is not established here, and the lesson generalizes: uniformity over classical microstates is not uniformity over quantum pure states. The same distinction governs H-state in §2.

Call a cycle *bad* if its time-averaged entropy deviates from S_{Page} by more than 2ϵ. Every state's deviation is bounded by $\ln d_{\mathrm{min}}$, so a cycle with atypical fraction f_j satisfies $|\overline{S}_{C_j} - S_{\text{Page}}| \le \epsilon + f_j \ln d_{\min}$; a bad cycle therefore requires $f_j \ge \epsilon / \ln d_{\min}$, i.e. $L_j \le (\ln d_{\min}/\epsilon)\,|\{s \in C_j : s \text{ atypical}\}|$. Summing over all bad cycles:

$$\sum_{j\,\text{bad}} L_j \le \frac{\ln d_{\min}}{\epsilon}\,\eta\,|\Sigma_E|$$

The fraction of initial conditions on Σ_E that land on a bad cycle is therefore at most $(\ln d_{\min}/\epsilon)\,\eta = (\ln d_{\min}/\epsilon)\,e^{-c\,d_{\min}\,\epsilon^2}$ — the polynomial prefactor is negligible against the exponential concentration. For a $30M_\odot$ black hole, $d_{\min} \sim e^{S_{\text{BH}}} \sim e^{10^{77}}$, so $\eta \sim e^{-10^{77}}$. For all but an exponentially negligible fraction of initial conditions, the time-averaged entanglement entropy equals the Page value. □

Theorem A.9 (conditional on A.8, hence on H-scramble) (The Page curve). *The entanglement entropy of the radiation follows:*

$$S_{\text{rad}}(t) = \begin{cases} n_R(t)\cdot \ln q - \frac{q^{n_R(t)}}{2\,q^{n_B(t)}} & \text{for } n_R(t) < n_B(t) \quad \text{(early: before Page time)} \\ n_B(t)\cdot \ln q - \frac{q^{n_B(t)}}{2\,q^{n_R(t)}} & \text{for } n_R(t) > n_B(t) \quad \text{(late: after Page time)} \end{cases}$$

where q is the alphabet size (gauge — it cancels in all ratios), $n_B(t) + n_R(t) = n_B(0)$, and the Page time t_P is defined by $n_R(t_P) = n_B(t_P) = n_B(0)/2$.

Proof. Direct substitution of $d_B = q^{n_B(t)}$ and $d_R = q^{n_R(t)}$ into the Page entropy formula (Theorem A.8), using $\ln(q^n) = n \ln q$ and the conservation law $n_B + n_R = n_B(0)$. The turnover at $n_R = n_B$ follows from $\min(d_B, d_R)$ switching from d_R to d_B. □

The evaporation trajectory. The Hawking temperature (§3.2, Step 4) gives luminosity $L \sim 1/M^2$, hence $dM/dt = -\alpha/M^2$ where $\alpha = \hbar c^4 \Gamma/(15360\pi G^2)$ and Γ is the species-dependent greybody factor (solution-specific). Integrating:

$$M(t) = M_0\big(1 - t/t_{\text{evap}}\big)^{1/3}, \qquad t_{\text{evap}} = \frac{M_0^3}{3\alpha}$$

The boundary mode count $n_B(t) = 4\pi G M(t)^2/(\hbar c) = n_B(0)\cdot(1 - t/t_{\text{evap}})^{2/3}$ and $n_R(t) = n_B(0) - n_B(t)$. The Page time t_P satisfies $n_B(t_P) = n_B(0)/2$:

$$t_P = t_{\text{evap}}\big(1 - 2^{-3/2}\big) \approx 0.646\,t_{\text{evap}}$$

The entropy of the radiation at the Page time is $S_{\text{rad}}(t_P) = n_B(0) \ln q/2 = S_{\text{BH}}(0) \ln q/2$ — half the initial black hole entropy, as expected.

Remark (Comparison with QES + islands). The Page-curve derivation above quantitatively matches the prediction obtained via the quantum extremal surface formula and the entanglement-island construction [41, 42, 43, 44]. In the QES approach, the Page curve emerges through an extremization principle that selects the dominant saddle in a replica-trick computation, with the post-Page-time saddle including an "island" — a region of bulk spacetime that contributes to the radiation's entanglement entropy despite being topologically disconnected. The replica wormhole geometries that connect the replicas in the gravitational path integral are essential to obtaining the saturating Page curve at late times.

The framework here arrives at the same Page time $t_P = (1 - 2^{-3/2})t_{\text{evap}}$ and the same Page entropy $S_{\text{rad}}(t_P) = S_{\text{BH}}(0) \ln q/2$ through a structurally cleaner mechanism: the cycle-typicality argument of Theorem A.8 invokes only Popescu-Short-Winter [29] applied to the deterministic substratum dynamics on the energy shell, with no replica trick, no ergodicity assumption, and no gravitational path integral. The "island" in the QES formulation corresponds, in the present construction, to the post-Page-time regime in which the BH boundary modes form the smaller subsystem; the "replica wormhole" geometries appear to correspond to the substratum's cycle structure on the energy shell. This suggested correspondence — replica wormholes $\leftrightarrow$ substratum cycles — is conjectural at this stage; making it precise would require constructing the gravitational path integral as an effective description of the substratum dynamics in the appropriate limit. What the present derivation establishes — under Theorem A.8's typicality and horizon-factorization hypotheses, at the G1 register — is that the Page curve's quantitative content is reproducible from a finite deterministic substratum without invoking the replica machinery.

The convergence is structural: both approaches require *something* to play the role of typicality (PSW measure-concentration here, replica saddles in QES + islands), and both extract the same Page time and Page entropy from that ingredient. The framework here establishes that the typicality-providing ingredient need not be gravitational at all — it can be combinatorial, on the substratum side of the trace-out — which constitutes a microscopic realization of the entanglement-island construction.

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